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# 261 Aortic Stenosis

**Patrick T. O'Gara, Joseph Loscalzo**

## GLOBAL BURDEN OF VALVULAR HEART DISEASE

Primary valvular heart disease ranks well below coronary heart disease, stroke, hypertension, obesity, and diabetes as a major threat to the public health. Nevertheless, it can cause significant morbidity and lead to premature death. Rheumatic fever **(Chap. 359)** is the dominant cause of valvular heart disease in low- and middle-income countries. Its prevalence has been estimated to range from as low as 1 per 100,000 school-age children in Costa Rica to as high as 150 per 100,000 in China **(Fig. 261-1)**. Rheumatic heart disease accounts for 12–65% of hospital admissions related to cardiovascular disease and 2–10% of hospital discharges in some endemic countries. Prevalence and mortality rates vary among communities even within the same country as a function of overcrowding and the availability of medical resources and population-wide programs for detection and treatment of group A streptococcal pharyngitis. In economically deprived areas, tropical and subtropical climates (particularly on the Indian subcontinent and in Southeast Asia), Central America, and the Middle East, rheumatic valvular disease progresses more rapidly than in more developed nations and frequently causes serious symptoms in patients aged <20 years. This accelerated natural history may be due to repeated infections with more virulent strains of rheumatogenic streptococci. Approximately 30–35 million people live with rheumatic heart disease worldwide, an estimated prevalence characterized by 300,000 new cases and 233,000 case fatalities per year, with the highest prevalence and age-adjusted mortality rates in sub-Saharan Africa, South Asia, and Oceania. The major drivers of untreated streptococcal pharyngitis in these endemic areas include reduced access to high-quality health care and social determinants of health. In the United States, rheumatic heart disease accounted for 3320 deaths in 2017. Although globally the age standardized mortality rate from rheumatic heart disease declined by nearly 50% between 1990 and 2015, the prevalence of heart failure attributable to rheumatic heart disease increased by nearly 90% over the same time interval.

Although there have been recent reports of isolated outbreaks of streptococcal infection in North America, valve disease in high-income countries is dominated by degenerative or inflammatory processes that lead to valve thickening, fibrosis, calcification, and dysfunction. The prevalence of valvular heart disease increases significantly with age. The prevalence of undiagnosed moderate or severe valvular heart disease is ~6% in individuals aged >65 years. Significant left-sided valve disease may affect as many as 12–13% of adults aged >75 years **(Fig. 261-2)**. Severe aortic stenosis (AS) is estimated to affect 3.5% of the population aged >75 years. A Swedish epidemiologic study estimated the incidence of newly diagnosed valvular heart disease at 64 per 100,000 person-years. AS and mitral regurgitation contributed approximately one-half and one-quarter, respectively, of the valvular heart disease diagnoses in this study.

The incidence of infective endocarditis **(Chap. 128)** has increased with the aging of the population, the more widespread prevalence of vascular grafts and intracardiac devices, the emergence of more virulent multidrug-resistant microorganisms, the growing epidemic of diabetes mellitus, and the opioid crisis. The more restricted use of antibiotic prophylaxis since 2007 has not been convincingly associated with an increase in incidence rates for infective endocarditis cases attributable to oropharyngeal pathogens. Infective endocarditis has become a relatively more frequent cause of acute valvular regurgitation. Valve surgery during the acute phase of infective endocarditis is performed in ~50% of hospitalized patients. Duration of intravenous antibiotic use may be shortened in selective cases.

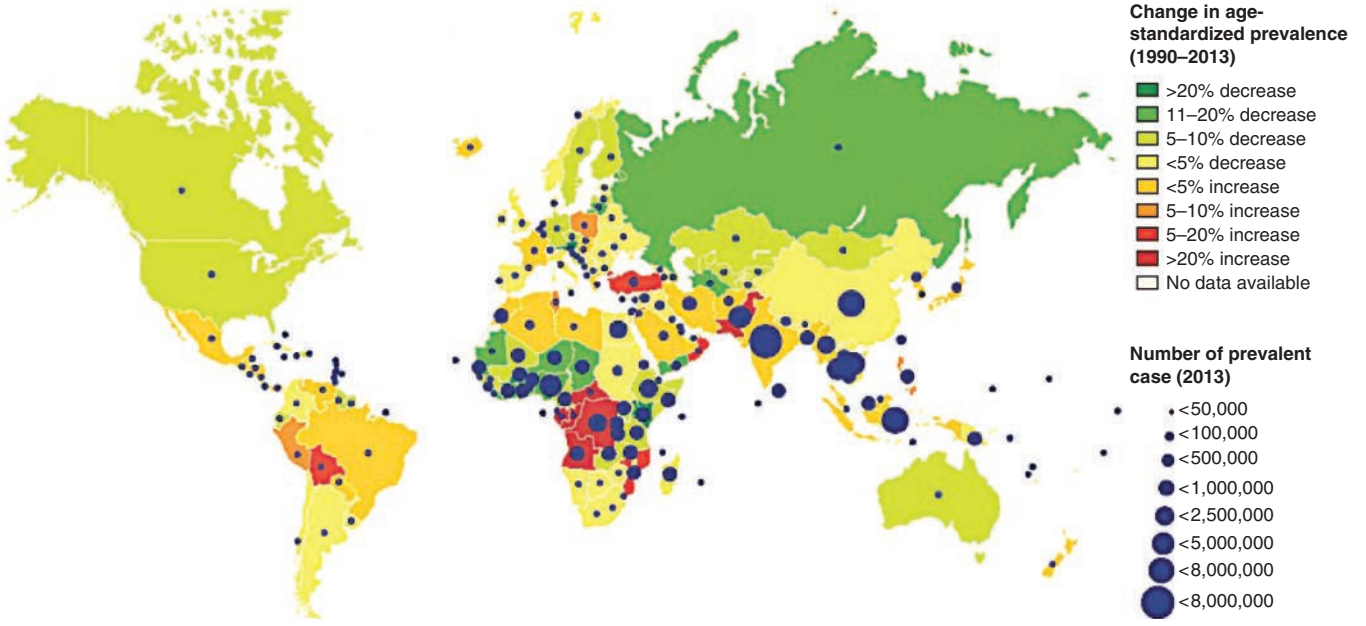



**FIGURE 261-1 The global burden of rheumatic heart disease.** This world map provides a snapshot of both the change in prevalence of rheumatic heart disease cases between 1990 and 2013 (upper right legend) and the estimated number of rheumatic heart disease cases per country (lower right legend). Regions in which the disease is highly prevalent include sub-Saharan Africa, India, China, and Southeast Asia. *(Reproduced with permission from JR Carapetis et al: Acute rheumatic fever and rheumatic heart disease. Nat Rev Dis Primers 2:15084, 2016.)*

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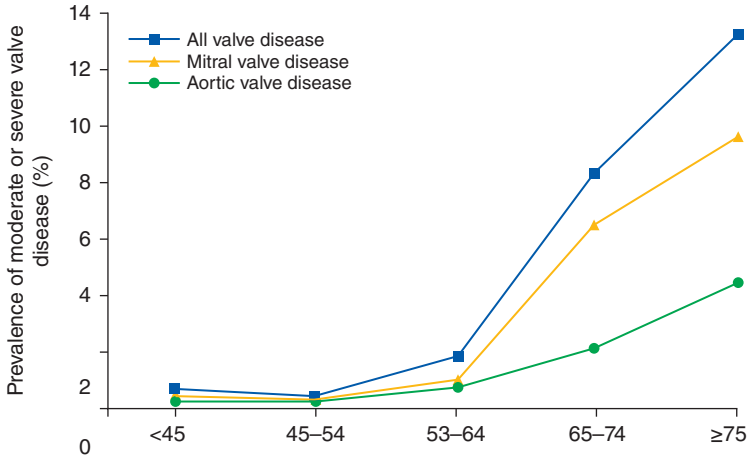



**FIGURE 261-2 The burden of moderate or severe mitral and aortic valve disease in the United States.** Prevalence estimates are derived from three population-based studies comprising a total of 11,911 individuals: The Coronary Artery Risk Development in Young Adults (CARDIA), the Atherosclerosis Risk in Communities (ARIC), and the Cardiovascular Health Study (CHS). *(Reproduced with permission from VT Nkomo et al: Burden of valvular heart diseases: A population-based study. Lancet 368:1005, 2006.)*

Bicuspid aortic valve (BAV) disease affects as many as 0.5–1.4% of the general population and is accompanied by an associated aortopathy in ~30-40% of individuals, a disease process expressed as root or ascending aortic aneurysm formation or descending thoracic aortic coarctation. An increasing number of childhood survivors of congenital heart disease present later in life with valvular dysfunction. The global burden of valvular heart disease will continue to progress.

As is true for many other chronic health conditions, disparities in access to and quality of care for patients with valvular heart disease have been well documented, especially for those patients with rheumatic heart disease in low- and middle-income countries. In the Society for Thoracic Surgeons (STS)/American College of Cardiology (ACC) Transcatheter Valve Therapy (TVT) registry, blacks comprise <5% of patients in the United States who have received a transcatheter valve for AS. Management decisions and outcome differences based on age, sex, race, and geography require intensification of educational efforts and prioritization of resources.

The role of the physical examination in the evaluation of patients with valvular heart disease is also considered in **Chaps. 42 and 239**; of electrocardiography (ECG) in **Chap. 240**; of echocardiography and other noninvasive imaging techniques in **Chap. 241**; and of cardiac catheterization and angiography in **Chap. 242**.

## AORTIC STENOSIS

Aortic stenosis (AS) is the most common valve lesion among adult patients with chronic valvular heart disease; the majority of adult patients with symptomatic, valvular AS are male.

### ETIOLOGY AND PATHOGENESIS

**(Table 261-1)** AS in adults is due to degenerative calcification of the aortic cusps and occurs most commonly on a substrate of congenital disease (BAV), chronic (trileaflet) deterioration, or previous rheumatic inflammation. A pathologic study of specimens removed at the time of aortic valve replacement (AVR) for AS in adults showed that 53% were bicuspid and 4% were unicuspid. The process of aortic valve deterioration and calcification is not a passive one, but rather one that shares many features with vascular atherosclerosis, including endothelial dysfunction, lipid accumulation, inflammatory cell activation, cytokine release, and upregulation of several signaling pathways **(Fig. 261-3)**. Eventually, a fibrocalcific response is established wherein collagen is deposited and valvular myofibroblasts differentiate phenotypically into osteoblasts and actively produce bone matrix proteins that allow for the deposition of calcium hydroxyapatite crystals. Genetic polymorphisms involving the vitamin D receptor, the estrogen receptor in postmenopausal women, interleukin 10, and apolipoprotein E4 have been linked to the development of calcific AS, and a strong familial clustering of cases with trileaflet valves has been reported from western France. Several traditional atherosclerotic risk factors have also been associated with the development and progression of calcific AS, including hypertension, low-density lipoprotein (LDL) cholesterol, lipoprotein a (Lp[a]), diabetes mellitus, smoking, chronic kidney disease, and the metabolic syndrome. In a Canadian observational cohort study, the incidence of severe AS was 144 per 100,000 person-years. Hypertension, diabetes mellitus, and dyslipidemia accounted for approximately one-third of the population-attributable risk for severe AS. The presence of aortic valve sclerosis (focal thickening and calcification of the leaflets not severe enough to cause obstruction) is associated with an excess risk of cardiovascular death and myocardial infarction (MI) among persons aged >65. Approximately 30% of persons aged >65 years exhibit some degree of aortic valve sclerosis. Rate and extent of progression to valve obstruction (stenosis) vary among individual patients.

*Rheumatic disease of the aortic leaflets* produces commissural fusion, sometimes resulting in a bicuspid-appearing valve. This condition, in turn, makes the leaflets more susceptible to trauma and ultimately leads to fibrosis, calcification, and further narrowing. By the time obstruction to left ventricular (LV) outflow causes serious clinical disability, the valve is usually a rigid calcified mass, and careful examination may make it difficult or even impossible to determine the etiology of the underlying process. Rheumatic AS is almost always associated with involvement of the mitral valve and with aortic regurgitation (AR). Mediastinal radiation can also result in late scarring, fibrosis, and calcification of the aortic leaflets.

**TABLE 261-1 Major Causes of Aortic Stenosis**

| VALVE LESION | ETIOLOGIES |
|---|---|
| Aortic stenosis | Congenital (bicuspid, unicuspid) |
| | Degenerative calcific disease |
| | Rheumatic fever |
| | Radiation |

### BICUSPID AORTIC VALVE DISEASE

A bicuspid aortic valve (BAV) is the most common congenital heart valve defect and occurs in 0.5–1.4% of the population with a 2–4:1 male-to-female predominance. The inheritance pattern appears to be autosomal dominant with incomplete penetrance, although some have questioned an X-linked component as suggested by the prevalence of BAV disease among patients with Turner's syndrome. The prevalence of BAV disease among first-degree relatives of an affected individual is ~10%. A single gene defect to explain the majority of cases has not been identified, although mutations in the *NOTCH1*, *GATA5*, and *GATA4* genes have been described in some families. Abnormalities in endothelial nitric oxide synthase and NKX2.5 have been implicated as well. Medial degeneration with ascending aortic aneurysm formation occurs commonly among patients with BAV disease; aortic coarctation is less frequently encountered. Patients with BAV disease have larger aortas than patients with comparable tricuspid aortic valve disease. The aortopathy develops independently of the hemodynamic severity of the valve lesion, but directional shear forces dictated by the anatomic configuration of the valve appear to influence its expression. For example, enlargement of the ascending aorta along its greater curvature is most often associated with right-left cusp fusion, the most common bicuspid variant. Patients with BAV disease are at risk for aneurysm formation and/or dissection. A BAV can be a component of more complex congenital heart disease with or without other left heart obstructing lesions, as seen in Shone's complex.

### OTHER FORMS OF OBSTRUCTION TO LEFT VENTRICULAR OUTFLOW

In addition to valvular AS, three other lesions may be responsible for obstruction to LV outflow: *hypertrophic obstructive cardiomyopathy* **(Chap. 259)**, *discrete fibromuscular/membranous subaortic stenosis*,

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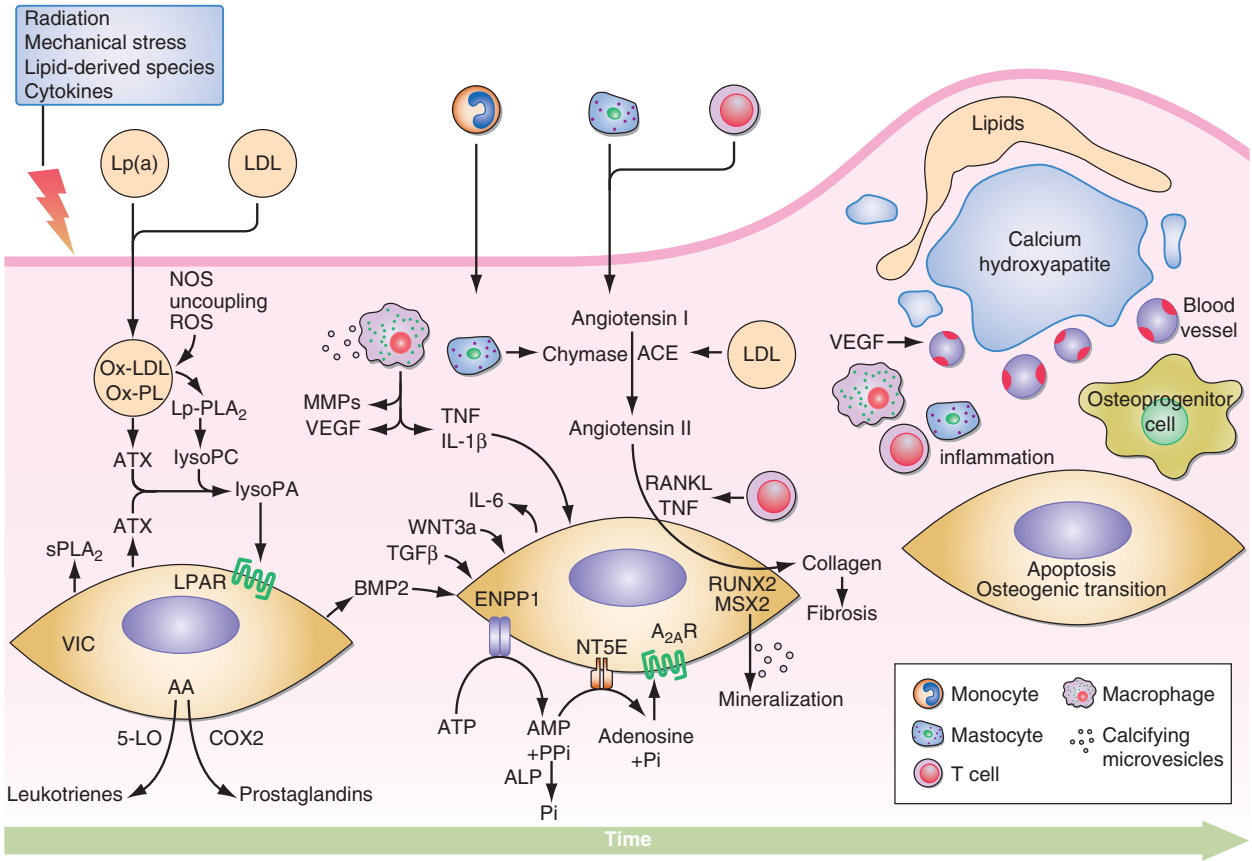



**FIGURE 261-3 Pathogenesis of calcific aortic stenosis.** Lipid and inflammatory cell infiltration occurs across damaged endothelium. A cascade of events follows that leads eventually to formation of disorganized collagen (fibrosis) and calcium hydroxyapatite (bone) deposition. Valvular interstitial cells (VIC) are critical participants in this active process. AA, arachidonic acid; ACE, angiotensin-converting enzyme; ALP, alkaline phosphatase; ApoB, apolipoprotein B; AMP, adenosine monophosphate; ATP, adenosine triphosphate; ATX, autotaxin; A2AR, adenosine A2A receptor; BMP, bone morphogenetic protein; COX2, cyclo-oxygenase 2; ENPP, ectonucleotide pyrophosphatase/phosphodiesterase; IL, interleukin; 5-LO, 5-lipoxygenase; LDL, low-density lipoprotein; Lp(a), lipoprotein(a); LPAR, lysophosphatidic acid receptor; Lp-PLA2, lipoprotein-associated phospholipase A2; lysoPA, lysophosphatidic acid; lysoPC, lysophosphatidylcholine; MMP, matrix metalloproteinase; NOS, nitric oxide synthase; Ox-PL, oxidized phospholipid; Ox-LDL, oxidized LDL; RANKL, receptor activator of nuclear factor-κB ligand; ROS, reactive oxygen species; RUNX2, runt-related transcription factor 2; sPLA2, secreted PLA2; TGFβ , transforming growth factor-β ; TNF, tumor necrosis factor; VEGF, vascular endothelial growth factor; VIC, valvular interstitial cell. *(Reproduced with permission from B Lindman et al: Calcific aortic stenosis. Nat Rev Dis Primers 2:16006, 2016.)*

and *supravalvular AS* (Chap. 269). The causes of LV outflow obstruction can be differentiated on the basis of the cardiac examination and Doppler echocardiographic findings.

## PATHOPHYSIOLOGY

The obstruction to LV outflow produces a systolic pressure gradient between the LV and aorta. When severe obstruction is suddenly produced experimentally, the LV responds by dilation and reduction of stroke volume. However, in some patients, the obstruction may be present at birth and/or increase gradually over the course of many years, and LV contractile performance is maintained by the presence of concentric LV hypertrophy. Initially, this serves as an adaptive mechanism because it reduces toward normal the systolic stress developed by the myocardium, as predicted by the Laplace relation ($S = Pr/h$, where $S$ = systolic wall stress, $P$ = pressure, $r$ = radius, and $h$ = wall thickness). A large transaortic valve pressure gradient may exist for many years without a reduction in cardiac output (CO) or the development of LV dilation. Ultimately, however, excessive hypertrophy becomes maladaptive, LV systolic function declines because of afterload mismatch, abnormalities of diastolic function progress, and irreversible myocardial fibrosis develops.

A mean systolic pressure gradient >40 mmHg with a normal CO or an effective aortic orifice area of ~<1 $cm^2$ (or ~<0.6 $cm^2/m^2$ body surface area in a normal-sized adult)—i.e., less than approximately one-third of the normal orifice area—is generally considered to represent severe obstruction to LV outflow. The elevated LV end-diastolic pressure observed in many patients with severe AS and preserved ejection fraction (EF) signifies the presence of diminished compliance of the hypertrophied LV. Although the CO at rest is within normal limits in most patients with severe AS, it usually fails to rise normally during exercise. Loss of an appropriately timed, vigorous atrial contraction, as occurs in atrial fibrillation (AF) or atrioventricular dissociation, may cause rapid progression of symptoms. Late in the course, contractile function deteriorates because of afterload excess, the CO and LV–aortic pressure gradient decline, and the mean left atrial (LA), pulmonary artery (PA), and right ventricular (RV) pressures rise. LV performance can be further compromised by superimposed epicardial coronary artery disease (CAD). Stroke volume (and thus CO) can also be reduced in patients with significant hypertrophy and a small LV cavity despite a normal EF. Low-flow (defined as a stroke volume index <35 $mL/m^2$), low-gradient (defined as a mean pressure gradient <40 mmHg) AS (with either reduced or normal LV systolic function) is both a diagnostic and therapeutic challenge.

The hypertrophied LV causes an increase in myocardial oxygen requirements. In addition, even in the absence of obstructive CAD, coronary blood flow is impaired to the extent that ischemia can be precipitated under conditions of excess demand. Capillary density is reduced relative to wall thickness, compressive forces are increased, and the elevated LV end-diastolic pressure reduces the coronary driving pressure. The subendocardium is especially vulnerable to ischemia by this mechanism.

## SYMPTOMS

AS is rarely of clinical importance until the valve orifice has narrowed to ~1 $cm^2$. Even severe AS may exist for many years without producing any symptoms because of the ability of the hypertrophied LV to generate the elevated intraventricular pressures required to maintain a

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normal stroke volume. Once symptoms occur, or the LV ejection fraction falls below normal, valve replacement is indicated.

Most patients with pure or predominant AS have gradually increasing obstruction over years but do not become symptomatic until the sixth to eighth decades. Adult patients with BAV disease, however, develop significant valve dysfunction and symptoms one to two decades sooner. Exertional dyspnea, angina pectoris, and syncope are the three cardinal symptoms. Often, there is a history of insidious progression of fatigue and dyspnea associated with gradual curtailment of activities and reduced effort tolerance. *Dyspnea* results primarily from elevation of the pulmonary capillary pressure caused by elevations of LV diastolic pressures secondary to impaired relaxation and reduced LV compliance. *Angina pectoris* usually develops somewhat later and reflects an imbalance between the increased myocardial oxygen requirements and reduced oxygen availability. CAD may or may not be present, although its coexistence is common among AS patients age >65. *Exertional syncope* may result from a decline in arterial pressure caused by vasodilation in exercising muscles and inadequate vasoconstriction in nonexercising muscles in the face of a fixed CO, or from a sudden fall in CO produced by an arrhythmia.

Because the CO at rest is usually well maintained until late in the course, marked fatigability, weakness, peripheral cyanosis, cachexia, and other clinical manifestations of a low CO are usually not prominent until this stage is reached. Orthopnea, paroxysmal nocturnal dyspnea, and pulmonary edema, i.e., symptoms of LV failure, also occur only in the advanced stages of the disease. Severe pulmonary hypertension leading to RV failure and systemic venous hypertension, hepatomegaly, AF, and tricuspid regurgitation (TR) are usually late findings in patients with isolated severe AS.

When AS and mitral stenosis (MS) coexist, the reduction in flow (CO) caused by MS lowers the pressure gradient across the aortic valve and, thereby, masks many of the clinical findings produced by AS. The transaortic pressure gradient can be increased in patients with concomitant AR due to higher aortic valve flow rates.

## PHYSICAL FINDINGS

The heart rhythm is generally regular until late in the course; at other times, AF should suggest the possibility of associated mitral valve disease. Hypertension occurs commonly among older adults with AS. In the late stages, however, when stroke volume declines, the systolic pressure may fall and the pulse pressure narrow. The carotid arterial pulse rises slowly to a delayed peak (*pulsus parvus et tardus*). A thrill or anacrotic "shudder" may be palpable over the carotid arteries, more commonly the left. In the elderly, the stiffening of the arterial wall may mask this important physical sign. In many patients, the *a* wave in the jugular venous pulse is accentuated. This results from the diminished distensibility of the RV cavity caused by the bulging, hypertrophied interventricular septum.

The LV impulse is sometimes displaced laterally in the later stages of the disease. A double apical impulse (with a palpable $S_4$) may be appreciated, particularly with the patient in the left lateral recumbent position. A systolic thrill may be present at the base of the heart to the right of the sternum when leaning forward or in the suprasternal notch.

**Auscultation** An early systolic ejection sound is frequently audible in children, adolescents, and young adults with congenital BAV disease. This sound usually disappears when the valve becomes calcified and rigid. As AS increases in severity, LV systole may become prolonged so that the aortic valve closure sound no longer precedes the pulmonic valve closure sound, and the two components may become synchronous, or aortic valve closure may even follow pulmonic valve closure, causing paradoxical splitting of $S_2$ (Chap. 239). The sound of aortic valve closure can be heard most frequently in patients with AS who have pliable valves, and calcification diminishes the intensity of this sound. Frequently, an $S_4$ is audible at the apex and reflects the presence of LV hypertrophy and an elevated LV end-diastolic pressure; an $S_3$ generally occurs late in the course, when the LV dilates and its systolic function becomes severely compromised.

The murmur of AS is described as an ejection (mid) systolic murmur that commences shortly after the $S_1$, increases in intensity to reach a peak toward the middle of ejection, and ends just before aortic valve closure. It is characteristically low-pitched, rough, and rasping in character, and loudest at the base of the heart, most commonly in the second right intercostal space. It is transmitted upward along the carotid arteries. Occasionally it is transmitted downward and to the apex, where it may be confused with the systolic murmur of mitral regurgitation (MR) (Gallavardin effect). In almost all patients with severe obstruction and preserved CO, the murmur is at least grade III/VI. In patients with mild degrees of obstruction or in those with severe stenosis with heart failure and low CO in whom the stroke volume and, therefore, the transvalvular flow rate are reduced, the murmur may be relatively soft and brief.

## LABORATORY EXAMINATION

**ECG** In most patients with severe AS, there is LV hypertrophy. In advanced cases, ST-segment depression and T-wave inversion (LV "strain") in standard leads I and aVL and in the left precordial leads are evident. However, there is no close correlation between the ECG and the hemodynamic severity of obstruction, and the absence of ECG signs of LV hypertrophy does not exclude severe obstruction. Systemic hypertension can coexist and also contribute to the development of hypertrophy.

**Echocardiogram** The key findings on transthoracic echocardiogram are thickening, calcification, and reduced systolic opening of the valve leaflets and LV hypertrophy. Eccentric closure of the aortic valve cusps is characteristic of congenitally bicuspid valves. Transesophageal echocardiography imaging can display the obstructed orifice extremely well, but it is not routinely required for accurate characterization of AS. The valve gradient and aortic valve area can be estimated by Doppler measurement of the transaortic velocity. Severe AS is defined by a valve area <1 $cm^2$, whereas moderate AS is defined by a valve area of 1–1.5 $cm^2$ and mild AS by a valve area of 1.5–2 $cm^2$. Aortic valve sclerosis, conversely, is accompanied by a jet velocity of <2.5 m/s (peak gradient <25 mmHg). LV dilation and reduced systolic shortening reflect impairment of LV function. There is a robust experience with the use of longitudinal strain to characterize earlier changes in LV systolic function, before a decline in EF can be appreciated. Doppler indices of impaired diastolic function are frequently seen.

Echocardiography is useful for identifying coexisting valvular abnormalities, differentiating valvular AS from other forms of LV outflow obstruction, and measuring the aortic root and proximal ascending aortic dimensions. These aortic measurements are particularly important for patients with BAV disease. Dobutamine stress echocardiography is useful for the evaluation of patients with AS and severe LV systolic dysfunction (low-flow, low-gradient, severe AS with reduced EF), in whom the severity of the AS can often be difficult to judge. Patients with severe AS (i.e., valve area <1 $cm^2$) with a relatively low mean gradient (<40 mmHg) despite a normal EF (low-flow, low-gradient, severe AS with normal EF) are often hypertensive, and efforts to control their systemic blood pressure should be optimized before Doppler echocardiography is repeated. The use of dobutamine stress echocardiography in this setting is generally not advised. When there is continued uncertainty regarding the severity of AS in patients with reduced CO, quantitative analysis of the amount of aortic valve calcium with chest computed tomography (CT) can be helpful. There is increasing use of chest CT angiography to assess aortic valve morphology and function. It has become the imaging method of choice to plan for transcatheter aortic valve implantation (TAVI). Finally, the use of cardiac magnetic resonance (CMR) imaging to screen for the presence of myocardial fibrosis with late gadolinium enhancement in patients with severe AS is an area of active investigation.

**Chest X-Ray** The chest x-ray may show no or little overall cardiac enlargement for many years. Hypertrophy without dilation may produce some rounding of the cardiac apex in the frontal projection and slight backward displacement in the lateral view. A dilated proximal ascending aorta may be seen along the upper right heart border in the frontal view. Aortic valve calcification may be discernible in the lateral view, but it is usually readily apparent on fluoroscopic examination or by echocardiography; the absence of valvular calcification on fluoroscopy in an adult suggests that severe valvular AS is *not* present. In later stages of the disease, as the LV dilates, there is increasing

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roentgenographic evidence of LV enlargement, pulmonary congestion, and enlargement of the LA, PA, and right-sided heart chambers.

**Catheterization** Right- and left-sided heart catheterization for invasive assessment of AS is performed infrequently but can be useful when there is a discrepancy between the clinical and noninvasive findings. Concern has been raised that attempts to cross the aortic valve for measurement of LV pressures are associated with a risk of cerebral embolization. Catheterization can also be useful in three distinct categories of patients: (1) *patients with multivalvular disease*, in whom the role played by each valvular deformity should be defined to aid in the planning of operative treatment; (2) *young, asymptomatic patients with noncalcific congenital AS*, to define the severity of obstruction to LV outflow, because operation or percutaneous aortic balloon valvuloplasty (PABV) may be indicated in these patients if severe AS is present, even in the absence of symptoms; and (3) *patients in whom it is suspected that the obstruction to LV outflow may not be at the level of the aortic valve* but rather at the sub- or supravalvular level.

Coronary angiography is indicated to screen for CAD in appropriate patients with severe AS who are being considered for surgical or transcatheter valve intervention. Angiography can be performed invasively at the time of catheterization for hemodynamic assessment or with noninvasive CT techniques. Decision-making regarding the need for coronary artery revascularization at the time of aortic valve intervention is individualized.

## NATURAL HISTORY

Death in patients with severe AS occurs most commonly in the seventh and eighth decades. Based on data obtained at postmortem examination in patients before surgical treatment became widely available, the average time to death after the onset of various symptoms was as follows: angina pectoris, 3 years; syncope, 3 years; dyspnea, 2 years; heart failure, 1.5–2 years. Moreover, in >80% of patients who died with AS, symptoms had existed for <4 years. Among adults dying with valvular AS, sudden death, which presumably resulted from an arrhythmia, occurred in 10–20%; however, most sudden deaths occurred in patients who had previously been symptomatic. Sudden death as the first manifestation of severe AS is very uncommon (~1% per year) in asymptomatic adult patients. Calcific AS is a progressive disease, with an annual reduction in valve area averaging 0.1 $cm^2$ and annual increases in peak jet velocity and mean valve gradient averaging 0.3 m/s and 7 mmHg, respectively.

# TREATMENT

## Aortic Stenosis (Fig. 261-4)

### MEDICAL TREATMENT

In patients with severe AS (valve area <1 $cm^2$), strenuous physical activity and competitive sports should be avoided, even in the asymptomatic stage. Care must be taken to avoid dehydration and hypovolemia to protect against a significant reduction in CO. Medications used for the treatment of hypertension or CAD, including beta blockers and angiotensin-converting enzyme (ACE) inhibitors, are generally safe for asymptomatic patients with preserved LV systolic function. Nitroglycerin is helpful in relieving angina pectoris in patients with CAD. Neither HMG-CoA reductase inhibitors ("statins") nor inhibitors of the renin-angiotensin-aldosterone system slow the rate of progression of AS. The use of statin medications should be driven by considerations regarding primary and secondary prevention of atherosclerotic cardiovascular disease (ASCVD) events. The need for endocarditis prophylaxis is restricted to AS patients with a prior history of endocarditis.

### SURGICAL TREATMENT

Asymptomatic patients with calcific AS and severe obstruction should be followed carefully for the development of symptoms and by serial echocardiograms for evidence of deteriorating LV function. Operation is indicated in patients with severe AS (valve area <1 $cm^2$ or 0.6 $cm^2/m^2$ body surface area) who are symptomatic, those who exhibit LV systolic dysfunction (EF <50%), and those with AS due to BAV disease and an aneurysmal root or ascending aorta (maximal dimension >5.5 cm). Operation for aneurysm disease is recommended at smaller aortic diameters (4.5–5.0 cm) for patients with a family history of an aortic catastrophe and for patients who exhibit rapid aneurysm growth (>0.5 cm/year). Patients with asymptomatic moderate or severe AS who are referred for coronary artery bypass grafting surgery should also have AVR. In patients without heart failure, the operative risk of surgical AVR (SAVR) (including patients with AS or AR) is ~2% (Table 261-2) but increases as a function of age and the need for concomitant aortic or other heart valve surgery or coronary bypass grafting. The indications for SAVR in the asymptomatic patient have been the subject of intense debate, as surgical outcomes in selected patients have continued to improve. Relative indications for which surgery can be considered include an abnormal response to treadmill exercise; rapid progression of AS, especially when urgent access to medical care might be compromised; very severe AS, defined by an aortic valve jet velocity >5 m/s or mean gradient >60 mmHg and low operative risk; and excessive LV hypertrophy in the absence of systemic hypertension. Exercise testing can be safely performed in asymptomatic patients, as many as one-third of whom will show signs of functional impairment. In a small randomized controlled trial (RCT) of early surgery versus conservative care for asymptomatic patients with very severe AS (defined by a transaortic valve jet velocity ≥4.5 m/s, mean gradient ≥50 mmHg, or aortic valve area ≤0.75 $cm^2$), the rate of operative death or death from cardiovascular causes during follow-up was reduced with early surgery. In the conservative care group, the cumulative incidence of sudden death was 4% at 4 years and 14% at 8 years.

Operation should be carried out promptly (1–3 months) after symptom onset. Clinical decision-making is straightforward for patients with normal flow (>35 mL/$m^2$), high gradient (≥40 mmHg) severe AS. In patients with low-flow, low-gradient severe AS with reduced LVEF, perioperative mortality rates are high (15–20%), and evidence of LV dysfunction usually persists even after a technically successful operation. Long-term postoperative survival correlates with preoperative LV function. Nonetheless, in view of the even worse prognosis of such patients when they are treated medically, there is usually little choice but to advise valve replacement, especially in patients in whom flow reserve can be demonstrated by dobutamine stress echocardiography (defined by a ≥20% increase in stroke volume after dobutamine challenge). Patients in this high surgical risk group are usually treated with TAVI (see below), but robust data from RCTs in this subpopulation of severe AS patients are lacking. The management of patients with low-flow, low-gradient severe AS with normal LVEF is also challenging. Outcomes are improved with surgery or TAVI compared with conservative medical care for symptomatic patients with this type of "paradoxical" low-flow AS, but more research is needed to guide therapeutic decision making for individual patients. In patients in whom severe AS and CAD coexist, relief of the AS and revascularization may sometimes result in striking clinical and hemodynamic improvement.

Because many patients with calcific AS are elderly, particular attention must be directed to the adequacy of hepatic, renal, and pulmonary function before AVR is recommended. Age alone is not a contraindication to SAVR for AS. The perioperative mortality rate depends to a substantial extent on the patient's preoperative clinical and hemodynamic state. Assessment of frailty is a critical component of preprocedural evaluation. Treatment decisions for AS patients who are not at low operative risk are made by a multidisciplinary heart team with representation from general cardiology, interventional cardiology, multimodality imaging, cardiac surgery, and other subspecialties as needed, including geriatrics. The 10-year survival rate of older adult patients with SAVR is ~60%. Recommendations regarding the type of valve prosthesis (biological or mechanical) must weigh the trade-offs between limited bioprosthetic valve durability and the risks of thromboembolism and bleeding with a mechanical valve and are heavily influenced by patient age and preferences. Bioprostheses are generally favored for patients

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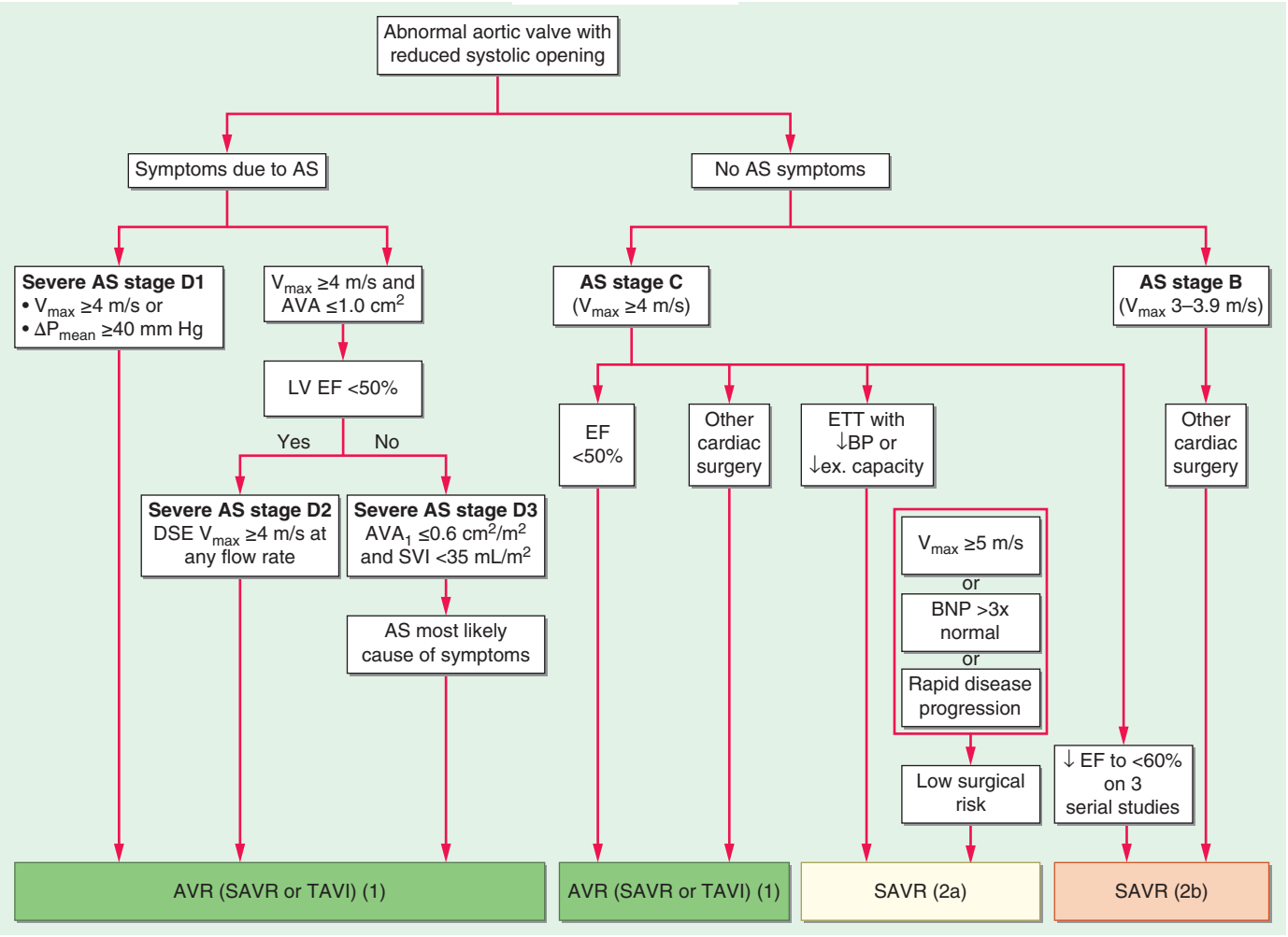



**FIGURE 261-4 Management strategy for patients with aortic stenosis.** Preoperative coronary angiography should be performed routinely as determined by age, symptoms, and coronary risk factors. Cardiac catheterization and angiography may also be helpful when there is a discrepancy between clinical and noninvasive findings. Patients who do not meet criteria for intervention should be monitored with clinical and echocardiographic follow-up. The class designations refer to the American Heart Association/American College of Cardiology methodology for treatment recommendations. Class I recommendations should be performed or are indicated; Class IIa recommendations are considered reasonable to perform; Class IIb recommendations may be considered. The stages refer to the stages of progression of the disease. At disease stage A, risk factors are present for the development of valve dysfunction; stage B refers to progressive, mild-moderate, asymptomatic valve disease; stage C disease is severe in nature but clinically asymptomatic; stage C1 characterizes asymptomatic patients with severe valve disease but compensated ventricular function; stage C2 refers to asymptomatic, severe disease with ventricular decompensation; stage D refers to severe, symptomatic valve disease. With aortic stenosis, stage D1 refers to symptomatic patients with severe aortic stenosis and a high valve gradient (>40 mmHg mean gradient); stage D2 comprises patients with symptomatic, severe, low-flow, low-gradient aortic stenosis and low left ventricular ejection fraction (LVEF); and stage D3 characterizes patients with symptomatic, severe, low-flow, low-gradient aortic stenosis and preserved left ventricular ejection fraction (paradoxical, low-flow, low-gradient severe aortic stenosis). Patients with symptomatic severe AS (left side of the diagram, jet velocity ≥4m/s) should be referred for AVR (SAVR or TAVI). Asymptomatic patients with severe AS (jet velocity ≥4m/s) should be referred for AVR (SAVR or TAVI) for LVEF <50% or when other cardiac surgery is needed (e.g., aneurysm repair). There are several findings for which referral for AVR would be reasonable related to results of exercise testing, the presence of a jet velocity >5 m/s or elevated B-type natriuretic peptide (BNP), provided the patient is considered low risk for complications related to AVR. AS, aortic stenosis; AVA, aortic valve area; AVR, aortic valve replacement; BP, blood pressure; DSE, dobutamine stress echocardiography; EF, ejection fraction; ETT, exercise treadmill test; $\Delta P_{mean}$, mean pressure gradient; SAVR, surgical AVR; TAVI, transcatheter aortic valve implantation; $V_{max}$, maximum velocity. *(Reproduced with permission from CM Otto et al: 2020 AHA/ACC Guideline for management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. Circulation 143:e72, 2021.)*

age >65 years. Shared decision-making with younger patients must be individualized, although increasing numbers of patients age <65 now opt for a biological valve replacement. Approximately 30% of bioprosthetic valves evidence primary valve failure by 10 years, requiring re-replacement (or valve-in-valve TAVI, see below), and an approximately equal percentage of patients with mechanical prostheses develop hemorrhagic complications as a consequence of treatment with vitamin K antagonists. In a large observational study of patients who underwent SAVR in California between 1996 and 2013, receipt of a biological versus a mechanical prosthesis in patients <55 years old was associated with an excess hazard of death over 15 years of follow-up. Homograft AVR is usually reserved for patients with aortic valve endocarditis.

The Ross procedure involves replacement of the diseased aortic valve with the autologous pulmonic valve and implantation of a homograft in the native pulmonic position. It is a technically complex procedure that may be considered in young or middle-aged adult patients when surgical and institutional expertise are available. Late postoperative complications include aortic root dilation, AR, and pulmonary homograft stenosis.

**TABLE 261-2 Mortality Rates After Aortic Valve Surgery[a]**

| OPERATION | NUMBER | UNADJUSTED OPERATIVE MORTALITY (%) |
|---|---|---|
| AVR (isolated) | 25,274 | 1.9 |
| AVR + CAB | 15,855 | 3.6 |

[a]Data are for calendar year 2018 during which 1088 participant groups reported a total of 287,872 procedures.

*Abbreviations:* AVR, aortic valve replacement; CAB, coronary artery bypass.

*Source:* Adapted from ME Bowdish et al: Ann Thorac Surg 109:1646, 2020.

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## PERCUTANEOUS AORTIC BALLOON VALVULOPLASTY

This procedure is preferable to operation in many children and young adults with congenital, noncalcific AS **(Chap. 269)**. It is not recommended as definitive therapy in adults with severe calcific AS because of a very high restenosis rate (80% within 1 year) and the risk of procedural complications, but on occasion, it has been used successfully as a bridge to operation or TAVI in patients with severe LV dysfunction and shock. It is performed routinely as part of the TAVI procedure (see below).

## TRANSCATHETER AORTIC VALVE IMPLANTATION

TAVI surpassed SAVR for treatment of AS in the United States in 2016 and is now available to symptomatic patients across the entire surgical risk spectrum (prohibitive, high, intermediate, and low) on the basis of the favorable results seen in a series of landmark RCTs reported over the past decade **(Fig. 261-5)**. Application of TAVI in asymptomatic AS patients is under active investigation. It is most commonly performed using one of two systems, a balloon-expandable valve or a self-expanding valve, both of which incorporate a pericardial bioprosthesis **(Fig. 261-6)**. TAVI is most frequently undertaken via the transfemoral route, although trans-LV apical, subclavian, carotid, and ascending aortic routes have been used. Aortic balloon valvuloplasty under rapid RV (or LV) pacing is performed as a first step to create an orifice of sufficient size for the prosthesis. Procedural success rates exceed 95% in appropriately selected patients. Valve performance characteristics are excellent over 5 years; longer-term durability assessment is ongoing. Outcomes achieved with this transformative technology have been very favorable and have allowed the extension of AVR to groups of patients previously considered at high or prohibitive risk for conventional surgery. Nevertheless, some prohibitive or high surgical risk patients are not candidates for this procedure because their comorbidity profile and frailty would make its undertaking inappropriate. The heart team is specifically charged with making challenging decisions of this nature. The use of these devices for treatment of patients with structural deterioration of bioprosthetic aortic valves (valve-in-valve TAVI), as an alternative to reoperative valve replacement, has increased sharply over the past 5 years. The technology has also been increasingly applied to BAV patients despite the fact that patients with this anatomy were excluded from the landmark RCTs.

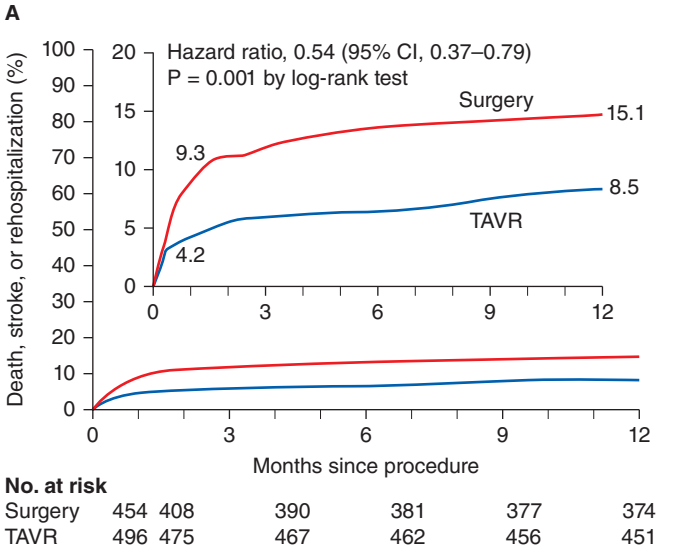



**FIGURE 261-6 Transcatheter aortic valve replacement (TAVR) with a balloon expandable valve versus surgical aortic valve replacement in low surgical risk patients.** Shown are Kaplan-Meier estimates of the rate of the primary composite end point including death from any cause, stroke, or rehospitalization. In this randomized trial, transfemoral TAVR resulted in a marked reduction in the composite endpoint at 1 year, although the individual components did not differ significantly. *(From MJ Mack et al: Transcatheter aortic-valve replacement with a balloon-expandable valve in low-risk patients. N Engl J Med 380:1695, 2019. Copyright © 2019 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.)*

Compared with SAVR, transfemoral TAVI results in fewer periprocedural deaths and confers lower risks of strokes, major bleeding, and AF. Hospital lengths of stay are shorter and return to normal activity more rapid with TAVI. Rates of permanent pacemaker use, perivalvular AR, and vascular complications are lower with SAVR. The choice between TAVI and SAVR for patients with trileaflet AS who prefer a biological prosthesis rests on several clinical, imaging, and technical considerations **(Fig. 261-7 and Table 261-3)**. Because there

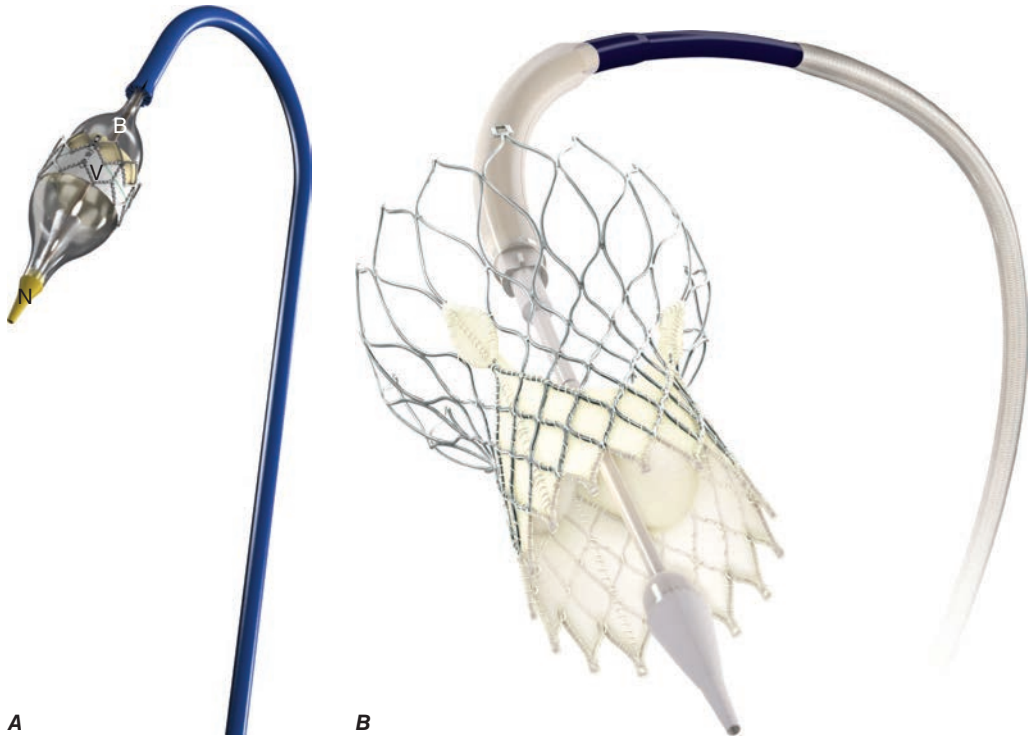



**FIGURE 261-5** Balloon-expandable (***A***) and self-expanding (***B***) valves for transcatheter aortic valve replacement (TAVR). B, inflated balloon; N, nose cone; V, valve. *(Part A, courtesy of Edwards Lifesciences, Irvine, CA; with permission. NovaFlex+ is a trademark of Edwards Lifesciences Corporation. Part B, © Medtronic, Inc. 2015. Medtronic CoreValve Transcatheter Aortic Valve. CoreValve is a registered trademark of Medtronic, Inc.)*

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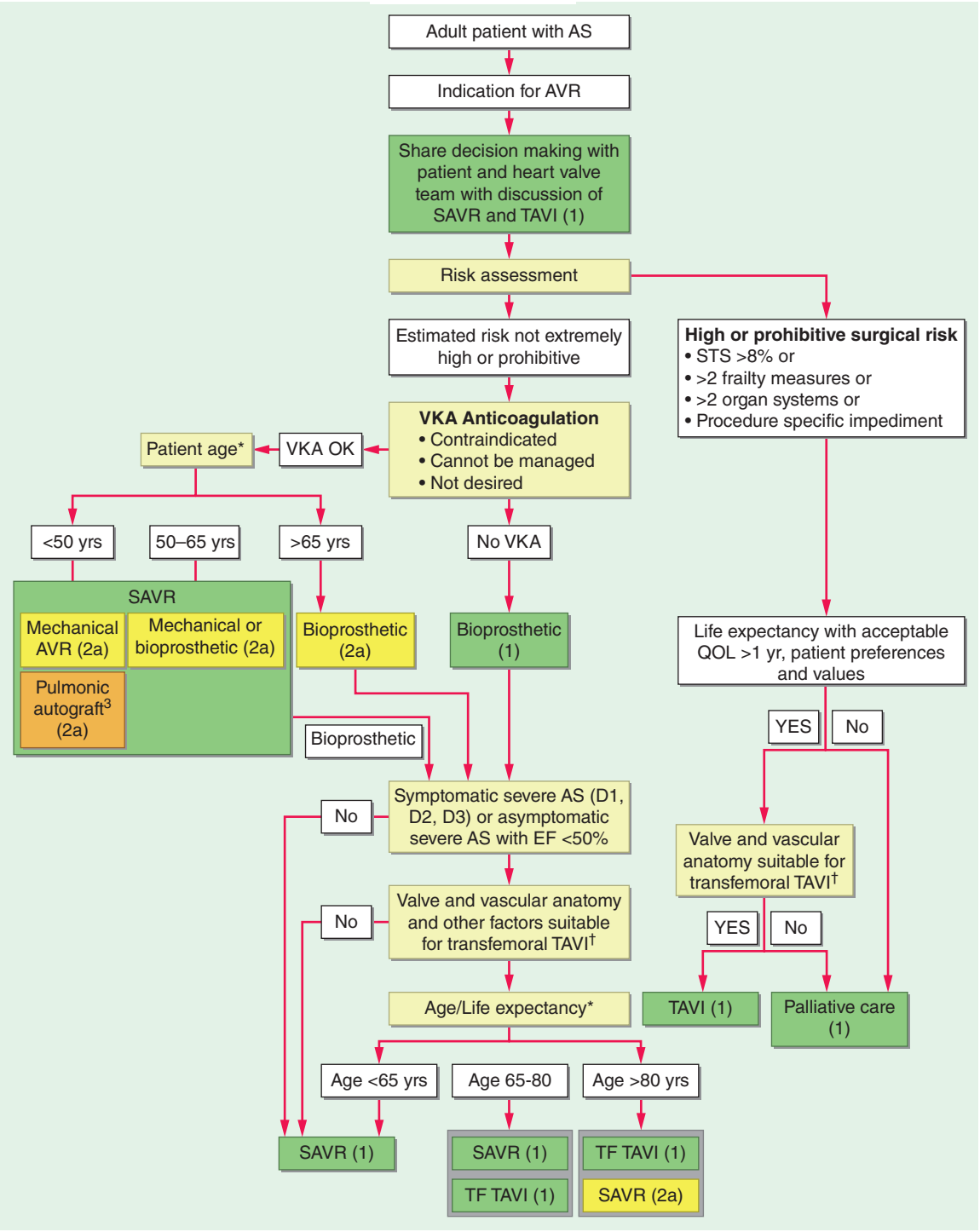



**FIGURE 261-7 Choice of surgical aortic valve replacement (SAVR) versus transfemoral transcatheter aortic valve implantation (TAVI) when indications for aortic valve replacement are met.** For patients who are not prohibitive or high surgical risk candidates, TAVI is not recommended for patients age <65 years (left hand side of flow diagram). For prohibitive or high surgical risk patients, TAVI is preferred over SAVR but is recommended on an individual basis only after multidisciplinary heart team consensus decision-making in collaboration with the patient and family. Palliative care is recommended when TAVI is considered futile (right side of flow diagram). AS, aortic stenosis; AVR, aortic valve replacement; EF, ejection fraction; QOL, quality of life; STS, Society of Thoracic Surgeons; VKA, vitamin K antagonist. *Approximate ages, based on US Actuarial Life Expectancy tables, are provided for guidance. The balance between expected patient longevity and valve durability varies continuously across the age range, with more durable valves preferred for patients with a longer life expectancy. Bioprosthetic valve durability is finite (with shorter durability for younger patients), whereas mechanical valves are very durable but require lifelong anticoagulation. Long-term (20-y) data on outcomes with surgical bioprosthetic valves are available; robust data on transcatheter bioprosthetic valves extend to only 5 years, leading to uncertainty about longer-term outcomes. The decision about valve type should be individualized on the basis of patient-specific factors that might affect longevity. †Placement of a transcatheter valve requires vascular anatomy that allows transfemoral delivery and the absence of aortic root dilation that would require surgical replacement. Valvular anatomy must be suitable for placement on the specific prosthetic valve, including annulus size and shape, leaflet number and calcification, and ostial height. *(Reproduced with permission from CM Otto et al: 2020 AHA/ACC Guideline for management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. Circulation 143:e72, 2021.)*

are scant RCT data on TAVI outcomes in patients <65 years, SAVR is recommended in this age group. Aortic valve/root anatomy, as well as the extent, severity, and distribution of calcium, and the distance of the coronary arteries from the plane of the annulus, may dictate a surgical approach, as could the need to perform a concomitant procedure such as ascending aortic replacement. Lastly, inability to achieve transfemoral access is a relative impediment to TAVI given the higher complication rates observed when this procedure is undertaken from other vascular access sites.

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**TABLE 261-3 Factors Favoring SAVR, TAVI, or Palliative Care in Patients with Aortic Stenosis**

| | FAVORS SAVR | FAVORS TAVI | FAVORS PALLIATION |
|---|---|---|---|
| Age/life expectancy[a] | • Younger age/longer life expectancy | • Older age/fewer expected remaining years of life | • Limited life expectancy |
| Valve anatomy | • Bicuspid aortic valve<br>• Subaortic (LVOT) calcification<br>• Rheumatic valve disease<br>• Small or large aortic annulus[b] | • Calcific trileaflet AS | |
| Prosthetic valve preference | • Mechanical or surgical bioprosthetic valve preferred<br>• Concern for patient-prosthesis mismatch (annular enlargement might be considered) | • Bioprosthetic valve preferred<br>• Favorable ratio of life expectancy to valve durability<br>• TAVI provides larger valve area than same-sized SAVR | |
| Concurrent cardiac conditions | • Aortic dilation[c]<br>• Severe primary MR<br>• Severe CAD requiring bypass grafting<br>• Septal hypertrophy requiring myectomy<br>• Atrial fibrillation | • Severe calcification of the ascending aorta ("porcelain" aorta) | • Irreversible severe LV systolic dysfunction<br>• Severe MR due to annular calcification |
| Noncardiac conditions | | • Severe lung, liver, or renal disease<br>• Mobility issues (high risk for sternotomy) | • Symptoms likely due to noncardiac conditions<br>• Severe dementia<br>• Moderate to severe involvement of 2 or more other organ systems |
| Frailty | • Not frail or few frailty measures | • Frailty likely to improve after TAVI | • Severe frailty unlikely to improve after TAVI |
| Estimated risk of SAVR or TAVI | • SAVR risk low<br>• TAVI risk high | • TAVI risk low to medium<br>• SAVR risk high to prohibitive | • Prohibitive SAVR risk (>15%) or post-TAVI life expectancy <1 year |
| Procedure-specific impediments | • Valve anatomy, annular size, or low coronary ostial height precludes TAVI<br>• Vascular access does not allow transfemoral TAVI | • Previous cardiac surgery with at-risk coronary grafts<br>• Previous chest irradiation | • Valve anatomy, annular size, or coronary ostial height precludes TAVI<br>• Vascular access does not allow transfemoral TAVI |
| Goals of care and patient preferences and values | • Less uncertainty about valve durability<br>• Avoid repeat intervention<br>• Lower risk of permanent pacer<br>• Life prolongation<br>• Symptom relief<br>• Improved long-term exercise capacity and QOL<br>• Avoid vascular complications<br>• Accepts longer hospital stay, pain in recovery period | • Accepts uncertainty about valve durability and possible repeat intervention<br>• Higher risk of permanent pacer<br>• Life prolongation<br>• Symptom relief<br>• Improved exercise capacity and QOL<br>• Prefers shorter hospital stay, less postprocedure pain | • Life prolongation not an important goal<br>• Avoid futile or unnecessary diagnostic or therapeutic procedures<br>• Avoid procedural stroke risk<br>• Avoid possibility of cardiac pacer |

[a]Data on bioprosthetic valve durability are more robust for SAVR valves than for TAVI valves. Mechanical valves are very durable but require lifelong anticoagulation. Choice of prosthesis is a shared decision-making process accounting for individual patient values and preferences. [b]Surgical root enlargement can be performed at time of SAVR to allow a use of a larger prosthesis and reduce the occurrence of prosthesis-patient mismatch. [c]Aortic root or ascending aortic enlargement may require surgical correction at time of SAVR.

*Abbreviations:* AS, aortic stenosis; CAD, coronary artery disease; LV, left ventricular; LVOT, left ventricular outflow tract; MR, mitral regurgitation; QOL, quality of life; SAVR, surgical aortic valve replacement; TAVI, transcatheter aortic valve implantation.

*Source:* Reproduced with permission from CR Burke et al: Goals of care in patients with severe aortic stenosis. Eur Heart J 41:929, 2020.

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# 262 Aortic Regurgitation

**Patrick T. O'Gara, Joseph Loscalzo**

## ETIOLOGY

(Table 262-1) Aortic regurgitation (AR) may be caused by primary valve disease, aortic root disease, or their combination.

**Primary Valve Disease** Rheumatic disease results in thickening, deformity, and shortening of the individual aortic valve cusps, changes that prevent their proper opening during systole and closure during diastole. A rheumatic origin is much less common in patients with isolated AR who do not have associated rheumatic mitral valve disease. Patients with congenital bicuspid aortic valve (BAV) disease may develop predominant AR, and ~20% of these patients will require aortic valve surgery between 10 and 40 years of age. Congenital

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**TABLE 262-1 Major Causes of Aortic Regurgitation**

| VALVE LESION | ETIOLOGIES |
|---|---|
| **Aortic regurgitation** | **Valvular** |
| | Congenital (bicuspid) |
| | Endocarditis |
| | Rheumatic fever |
| | Myxomatous (prolapse) |
| | Radiation |
| | Trauma |
| | Syphilis |
| | Ankylosing spondylitis |
| | **Aortic root disease** |
| | Aortic dissection |
| | Medial degeneration |
| | Marfan syndrome |
| | Bicuspid aortic valve |
| | Nonsyndromic familial aneurysm |
| | Aortitis |
| | Hypertension |

fenestrations of the aortic valve occasionally produce mild AR. Membranous subaortic stenosis results in a high velocity systolic jet that often leads to thickening and scarring of the aortic valve leaflets and secondary AR. Prolapse of an aortic cusp, resulting in progressive chronic AR, occurs in ~15% of patients with ventricular septal defect (Chap. 269), but may also occur as an isolated phenomenon or as a consequence of myxomatous degeneration sometimes associated with mitral and/or tricuspid valve involvement.

AR may result from infective endocarditis (IE), which can develop on a valve previously affected by rheumatic disease, a congenitally deformed valve, or on a normal aortic valve, and may lead to perforation or erosion of one or more leaflets. The aortic valve leaflets may become scarred and retracted during the course of syphilis or ankylosing spondylitis and contribute further to the AR that derives primarily from the associated root dilation. Although traumatic rupture or avulsion of an aortic cusp is an uncommon cause of acute AR, it represents the most frequent serious lesion in patients surviving nonpenetrating cardiac injuries. The coexistence of hemodynamically significant aortic stenosis (AS) with AR usually excludes all the rarer forms of AR because it occurs almost exclusively in patients with rheumatic or congenital AR. In patients with AR due to primary valvular disease, dilation of the aortic annulus may occur secondarily and lead to worsening regurgitation.

**Primary Aortic Root Disease** AR also may be due entirely to marked aortic annular dilation, i.e., aortic root disease, without primary involvement of the valve leaflets; widening of the aortic annulus and lack of diastolic coaptation of the aortic leaflets are responsible for the AR (Chap. 280). Medial degeneration of the ascending aorta, which may or may not be associated with other manifestations of Marfan syndrome; idiopathic dilation of the aorta; annuloaortic ectasia; osteogenesis imperfecta; and severe, chronic hypertension may all widen the aortic annulus and lead to progressive AR. Occasionally AR is caused by retrograde dissection of the aorta involving the aortic annulus. Syphilis and ankylosing spondylitis, both of which may also affect the aortic leaflets, may be associated with cellular infiltration and scarring of the media of the thoracic aorta, leading to aortic dilation, aneurysm formation, and severe regurgitation. In syphilis of the aorta (Chap. 182), now a very rare condition, the involvement of the intima may narrow the coronary ostia, which in turn may be responsible for myocardial ischemia. Takayasu's aortitis and giant cell aortitis can also result in aneurysm formation and secondary AR.

## PATHOPHYSIOLOGY

The total stroke volume ejected by the left ventricle (LV) (i.e., the sum of the effective forward stroke volume and the volume of blood that regurgitates back into the LV) is increased in patients with AR. In patients with severe AR, the volume of regurgitant flow may equal the effective forward stroke volume. In contrast to MR, in which a portion of the LV stroke volume is delivered into the low-pressure left atrium (LA), in AR, the entire LV stroke volume is ejected into a high-pressure zone, the aorta. An increase in the LV end-diastolic volume (increased preload) constitutes the major hemodynamic compensation for AR. The dilation and eccentric hypertrophy of the LV allow this chamber to eject a larger stroke volume without requiring any increase in the relative shortening of each myofibril. Therefore, severe AR may occur with a normal effective forward stroke volume and a normal LV ejection fraction (LVEF, total [forward plus regurgitant] stroke volume/end-diastolic volume), together with an elevated LV end-diastolic pressure and volume. However, through the operation of Laplace's law, LV dilation increases the LV systolic tension required to develop any given level of systolic pressure. Chronic AR is, thus, a state in which LV preload and afterload are both increased. Ultimately, these adaptive measures fail. As LV function deteriorates, the end-diastolic volume rises further and the forward stroke volume and ejection fraction (EF) decline. Deterioration of LV function often precedes the development of symptoms. Considerable thickening of the LV wall also occurs with chronic AR, and at autopsy, the hearts of these patients may be among the largest encountered, sometimes weighing >1000 g.

The reverse diastolic pressure gradient from aorta to LV, which drives the AR flow, falls progressively during diastole, accounting for the typical decrescendo nature of the diastolic murmur. Equilibration between aortic and LV pressures may occur toward the end of diastole in patients with chronic severe AR, particularly when the heart rate is slow. In patients with acute severe AR, the LV is unprepared for the regurgitant volume load. LV compliance is normal or reduced, and LV diastolic pressures rise rapidly, occasionally to levels >40 mmHg. The LV pressure may exceed the LA pressure toward the end of diastole, and this reversed pressure gradient closes the mitral valve prematurely.

In patients with chronic severe AR, the effective forward cardiac output (CO) usually is normal or only slightly reduced at rest, but often it fails to rise normally during exercise. An early sign of LV dysfunction is a reduction in the EF. In advanced stages, there may be considerable elevation of the LA, pulmonary artery (PA) wedge, PA, and right ventricular (RV) pressures and lowering of the forward CO at rest.

*Myocardial ischemia* may occur in patients with AR because myocardial oxygen requirements are elevated by LV dilation, hypertrophy, and elevated LV systolic tension, and coronary blood flow may be compromised. A large fraction of coronary blood flow occurs during diastole, when aortic pressure is low, thereby reducing coronary perfusion or driving pressure. This combination of increased oxygen demand and reduced supply may cause myocardial ischemia, particularly of the subendocardium, even in the absence of epicardial coronary artery disease (CAD).

## HISTORY

Approximately three-fourths of patients with pure or predominant valvular AR are men; women predominate among patients with primary valvular AR who have associated rheumatic mitral valve disease. A history compatible with IE may sometimes be elicited from patients with rheumatic or congenital involvement of the aortic valve, and the infection often precipitates or seriously aggravates preexisting symptoms.

In patients with *acute severe AR*, as may occur in IE, aortic dissection, or trauma, the LV cannot dilate sufficiently to maintain stroke volume, and LV diastolic pressure rises rapidly with associated marked elevations of LA and PA wedge pressures. Pulmonary edema and/or cardiogenic shock may develop rapidly.

*Chronic severe AR* may have a long latent period, and patients may remain relatively asymptomatic for as long as 10–15 years. Uncomfortable awareness of the heartbeat, especially on lying down, may be an early complaint. Sinus tachycardia, during exertion or with emotion, or premature ventricular contractions may produce particularly uncomfortable palpitations as well as head pounding. These complaints may persist for many years before the development of exertional dyspnea, usually the first symptom of diminished cardiac reserve. The dyspnea is followed by orthopnea, paroxysmal nocturnal dyspnea, and excessive

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diaphoresis. Anginal chest pain even in the absence of CAD may occur in patients with severe AR, even in younger patients. Anginal pain may develop at rest as well as during exertion. Nocturnal angina may be a particularly troublesome symptom, and it may be accompanied by marked diaphoresis. The anginal episodes can be prolonged and often do not respond satisfactorily to sublingual nitroglycerin. Systemic fluid accumulation, including congestive hepatomegaly and ankle edema, may develop late in the course of the disease.

## PHYSICAL FINDINGS

In chronic severe AR, the jarring of the entire body and the bobbing motion of the head with each systole can be appreciated, and the abrupt distention and collapse of the larger arteries are easily visible. The examination should be directed toward the detection of conditions predisposing to AR, such as bicuspid valve, IE, Marfan syndrome, or ankylosing spondylitis.

**Arterial Pulse** A rapidly rising "water-hammer" pulse, which collapses suddenly as arterial pressure falls rapidly during late systole and diastole (Corrigan's pulse), and capillary pulsations, an alternate flushing and paling of the skin at the root of the nail while pressure is applied to the tip of the nail (Quincke's pulse), are characteristic of chronic severe AR. A booming "pistol-shot" sound can be heard over the femoral arteries (Traube's sign), and a to-and-fro murmur (Duroziez's sign) is audible if the femoral artery is lightly compressed with a stethoscope.

The arterial pulse pressure is widened as a result of both systolic hypertension and a lowering of the diastolic pressure. The measurement of arterial diastolic pressure with a sphygmomanometer may be complicated by the fact that systolic sounds are frequently heard with the cuff completely deflated. However, the level of cuff pressure at the time of muffling of the Korotkoff sounds (phase IV) generally corresponds fairly closely to the true intra-arterial diastolic pressure. As the disease progresses and the LV end-diastolic pressure rises, the arterial diastolic pressure may actually rise as well, because the aortic diastolic pressure cannot fall below the LV end-diastolic pressure. For the same reason, acute severe AR may also be accompanied by only a slight widening of the pulse pressure. Such patients are invariably tachycardic as the heart rate increases in an attempt to preserve the CO.

**Palpation** In patients with chronic severe AR, the LV impulse is heaving and displaced laterally and inferiorly. The systolic expansion and diastolic retraction of the apex are prominent. A diastolic thrill may be palpable along the left sternal border in thin-chested individuals, and a prominent systolic thrill may be palpable in the suprasternal notch and transmitted upward along the carotid arteries. This systolic thrill and the accompanying murmur do not necessarily signify the coexistence of AS. In some patients with AR or with combined AS and AR, the carotid arterial pulse may be bisferiens, i.e., with two systolic waves separated by a trough (see Fig. 239-2*C*).

**Auscultation** In patients with severe AR, the aortic valve closure sound ($A_2$) is usually absent. A systolic ejection sound is audible in patients with BAV disease, and occasionally an $S_4$ also may be heard. The murmur of chronic AR is typically a high-pitched, blowing, decrescendo diastolic murmur, heard best in the third intercostal space along the left sternal border (see Fig. 239-5*B*). In patients with mild AR, this murmur is brief, but as the severity increases, it generally becomes louder and longer, indeed holodiastolic. When the murmur is soft, it can be heard best with the diaphragm of the stethoscope and with the patient sitting up, leaning forward, and with the breath held in forced expiration. In patients in whom the AR is caused by primary valvular disease, the diastolic murmur is usually louder along the left than the right sternal border. However, when the murmur is louder along the right sternal border, it suggests that the AR is caused by aneurysmal dilation of the aortic root. "Cooing" or musical diastolic murmurs suggest eversion of an aortic cusp vibrating in the regurgitant stream.

A mid-systolic ejection murmur is frequently audible in isolated AR. It is generally heard best at the base of the heart and is transmitted along the carotid arteries. This murmur may be quite loud without signifying aortic valve obstruction. A third murmur sometimes heard in patients with severe AR is the *Austin Flint murmur*, a soft, low-pitched, rumbling mid-to-late diastolic murmur. It is probably produced by the diastolic displacement of the anterior leaflet of the mitral valve by the AR stream and is not associated with hemodynamically significant mitral valve obstruction. The auscultatory features of AR are intensified by strenuous and sustained handgrip, which augments systemic vascular resistance and increases LV afterload.

In acute severe AR, the elevation of LV end-diastolic pressure may lead to early closure of the mitral valve, a soft $S_1$, a pulse pressure that is not particularly wide, and a soft, short, early diastolic murmur of AR.

## LABORATORY EXAMINATION

**ECG** In patients with chronic severe AR, ECG signs of LV hypertrophy are common (Chap. 240). In addition, these patients frequently exhibit ST-segment depression and T-wave inversion in leads I, aVL, $V_5$, and $V_6$ ("LV strain"). Left axis deviation and/or QRS prolongation may also be present.

**Echocardiogram** LV size is increased in chronic AR, and systolic function is normal or even supernormal until myocardial contractility declines, as signaled by a decrease in EF or increase in the end-systolic dimension. A rapid, high-frequency diastolic fluttering of the anterior mitral leaflet produced by the impact of the regurgitant jet is a characteristic finding. The echocardiogram is also useful in determining the cause of AR, by detecting dilation of the aortic annulus and root, aortic dissection (see Fig. 241-5), or primary leaflet pathology. With severe AR, the central jet width assessed by color flow Doppler imaging exceeds 65% of the width of the LV outflow tract, the regurgitant volume is ≥60 mL/beat, the regurgitant fraction is ≥50%, and there is diastolic flow reversal in the proximal portion of the descending thoracic aorta. The continuous-wave Doppler profile of the AR jet shows a rapid deceleration time in patients with acute severe AR, due to the rapid increase in LV diastolic pressure. Surveillance transthoracic echocardiography (TTE) forms the cornerstone of longitudinal follow-up and allows for the early detection of changes in LV size and/or function. For patients in whom TTE is limited by poor acoustical windows or inadequate characterization of LV function or the cause or severity of the regurgitation, cardiac magnetic resonance (CMR) imaging can be performed. This modality also allows for accurate assessment of aortic size and contour. It can also be utilized to screen for LV fibrosis as assessed with late gadolinium enhancement. Both CMR imaging and cardiac computed tomography (CT) can provide detailed assessment of aortic valve and root anatomy. Transesophageal echocardiography (TEE) can also provide detailed anatomic assessment of the valve, root, and portions of the aorta. There is increasing experience with the use of three-dimensional echocardiography to measure LV volumes.

**Chest X-Ray** In chronic severe AR, the apex is displaced downward and to the left in the frontal projection. In the left anterior oblique and lateral projections, the LV is displaced posteriorly and encroaches on the spine. When AR is caused by primary disease of the aortic root, aneurysmal dilation of the aorta may be noted, and the aorta may fill the retrosternal space in the lateral view. Echocardiography, CMR imaging, and chest CT angiography are more sensitive than the chest x-ray for the detection of root and ascending aortic enlargement.

**Cardiac Catheterization and Angiography** When needed, right and left heart catheterization with contrast aortography can provide confirmation of the magnitude of regurgitation and the status of LV function. Coronary angiography is performed routinely in appropriate patients prior to surgery, although this anatomic information can also be gained with coronary CT angiographic techniques.

## TREATMENT

### Aortic Regurgitation

#### ACUTE AORTIC REGURGITATION (FIG. 262-1)

Patients with acute severe AR may respond to intravenous diuretics and vasodilators (such as sodium nitroprusside), but stabilization is

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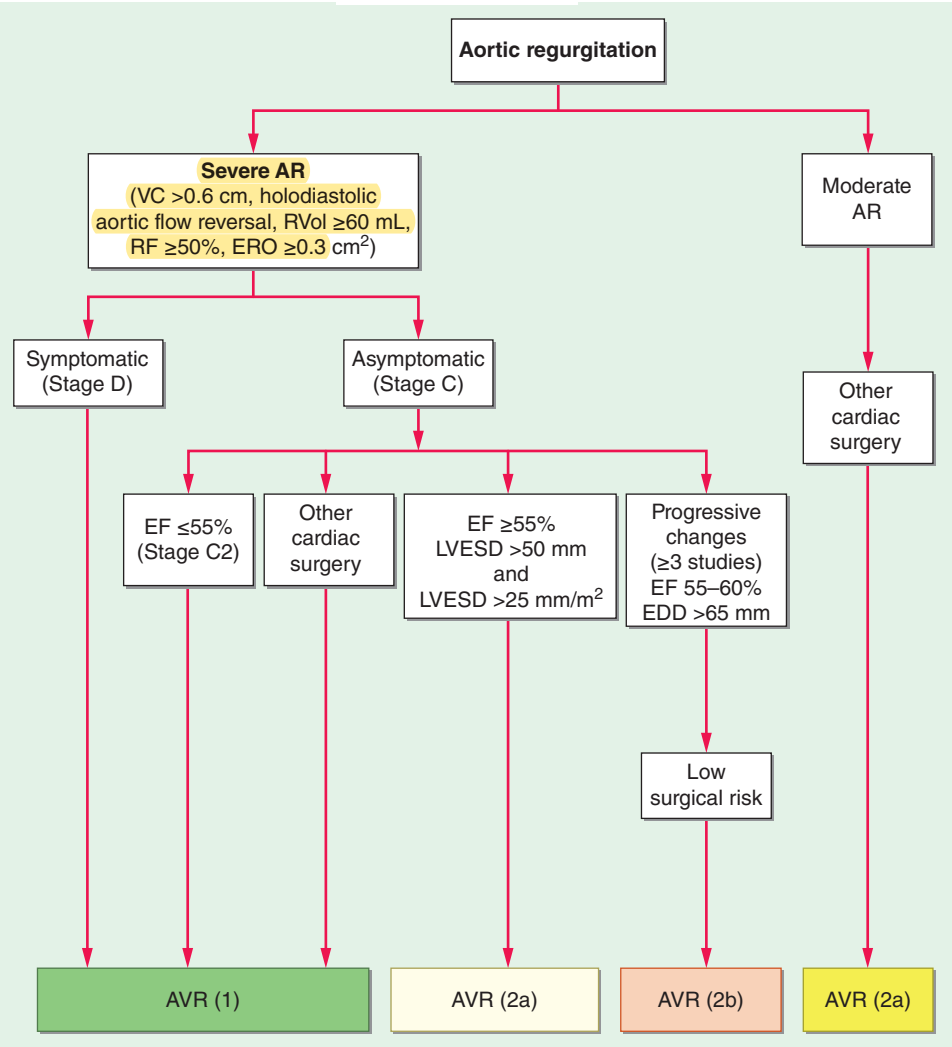



**FIGURE 262-1 Management of patients with aortic regurgitation.** See legend for Fig. 261-4 for explanation of treatment recommendations (Class I, IIa, and IIb) and disease stages (B, C1, C2, and D). Preoperative coronary angiography should be performed routinely as determined by age, symptoms, and coronary risk factors. Cardiac catheterization and angiography may also be helpful when there is a discrepancy between clinical and noninvasive findings. Surgery is indicated for patients with severe AR and symptoms, LV dysfunction, or other indications for operation (e.g., aneurysm disease). Surgery is also reasonable once the LV indexed end-systolic dimension reaches or exceeds 25 mm/m$^2$. Patients who do not meet criteria for intervention should be monitored periodically with clinical and echocardiographic follow-up. AR, aortic regurgitation; AVR, aortic valve replacement (valve repair may be appropriate in selected patients); EDD, end-diastolic dimension; EF, ejection fraction; ERO, effective regurgitant orifice; LV, left ventricular; LVEDD, left ventricular end-diastolic dimension; LVEF, left ventricular ejection fraction; LVESD, left ventricular end-systolic dimension; RF, regurgitant fraction; RVol, regurgitant volume; VC, vena contracta. *(Reproduced with permission from CM Otto et al: 2020 AHA/ACC Guideline for management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. Circulation 143:e72, 2021.)*

usually short-lived and operation is indicated urgently. Intra-aortic balloon counterpulsation is contraindicated. Beta blockers are best avoided so as not to reduce the CO further or slow the heart rate, thus allowing more time for diastolic filling of the LV. Surgery is the treatment of choice and is usually necessary within 24 h of diagnosis.

## CHRONIC AORTIC REGURGITATION

The onset of symptoms, or LV systolic dysfunction, is an indication for surgery. Medical treatment with diuretics and vasodilators (angiotensin-converting enzyme inhibitors, angiotensin receptor blockers [ARBs], dihydropyridine calcium channel blockers, or hydralazine) may be useful as a temporizing measure. Surgery can then be performed in a more controlled setting. The use of vasodilators to extend the compensated phase of chronic severe AR in asymptomatic patients before the onset of symptoms or the development of LV dysfunction is not useful, although these agents should be employed to treat hypertension (systolic blood pressure >140 mm Hg). It is often difficult to achieve adequate blood pressure control because of the increased stroke volume and enhanced LV ejection that accompany severe AR. Cardiac arrhythmias and systemic infections are poorly tolerated in patients with severe AR and must be treated promptly and vigorously. Although nitroglycerin and long-acting nitrates are not as helpful in relieving anginal pain as they are in patients with coronary heart disease, they are worth a trial. Patients with syphilitic aortitis should receive a full course of penicillin therapy (Chap. 182). Beta blockers and the ARB losartan may be useful to retard the rate of aortic root enlargement in young patients with Marfan's syndrome and aortic root dilation. A randomized controlled trial showed no difference in efficacy between atenolol and losartan for this indication. Whether beta blockers or ARBs are useful in retarding the rate of growth of aortic aneurysms in other patient subsets (e.g., BAV disease with aortopathy, Takayasu's disease) has not been demonstrated. Beta blockers in patients with valvular AR were previously considered relatively contraindicated due to concern that slowing of the heart rate would allow more time for diastolic regurgitation. Observational reports, however, have suggested that beta blockers may provide functional benefit in some patients with chronic AR. Beta blockers can sometimes provide incremental blood pressure lowering in patients with chronic AR and hypertension. They can also lessen the sense of forceful heart action that many patients find uncomfortable. Patients with severe AR, particularly those with an associated aortopathy, should avoid isometric exercises.

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## SURGICAL TREATMENT

In deciding on the advisability and proper timing of surgical treatment, two points should be kept in mind: (1) patients with chronic severe AR usually do not become symptomatic until *after* the development of myocardial dysfunction; and (2) when delayed too long (defined as >1 year from onset of symptoms or LV dysfunction), surgical treatment often does not restore normal LV size and function. Therefore, in patients with chronic severe AR, careful clinical follow-up and noninvasive testing with echocardiography at ~6- to 12-month intervals are necessary if operation is to be undertaken at the optimal time, i.e., *after* the onset of LV dysfunction but *prior to* the development of severe symptoms. Exercise testing may be helpful to assess effort tolerance more objectively. Operation can be deferred as long as the patient both remains asymptomatic and retains normal LV function without severe or progressive chamber dilation.

Aortic valve replacement (AVR) is indicated for the treatment of severe AR in symptomatic patients irrespective of LV function. In general, the operation should be carried out in asymptomatic patients with severe AR and progressive LV dysfunction defined by an LVEF <55% on serial studies, an LV end-systolic dimension >50 mm (>25 mm/m$^2$), or an LV diastolic dimension >65 mm. Smaller dimensions may be appropriate thresholds in individuals of smaller stature or when there is evidence of progressively decreasing LV function or increasing LV size on serial studies and the anticipated risks for surgical morbidity and mortality are low. Two case series from surgical referral centers have suggested that surgery should be performed at an even lower threshold for LV end-systolic dimension index (≥20 mm/m$^2$), but data from randomized controlled trials are lacking. Patients with severe AR without indications for operation should be followed by clinical and echocardiographic examination every 6–12 months. Transcatheter aortic valve implantation (TAVI) is not recommended for patients with severe AR who are surgical candidates. Technical success with TAVI in patients with chronic AR is limited by the degree of aortic annular dilation and the relative paucity of valvular and annular calcium.

Surgical options for management of aortic valve and root disease have expanded considerably over the past decade. AVR with a suitable mechanical or tissue (biological) prosthesis is generally necessary in patients with rheumatic AR and in many patients with other causes of valvular AR. Rarely, when a leaflet has been perforated during IE or torn from its attachments to the aortic annulus by thoracic trauma, primary surgical repair may be possible. When AR is due to aneurysmal dilation of the root or proximal ascending aorta rather than to primary valve involvement, it may be possible to reduce or eliminate the regurgitation by narrowing the annulus or by excising a portion of the aortic root without replacing the valve. Elective, valve-sparing aortic root reconstruction generally involves reimplantation of the valve in a contoured graft with reattachment of the coronary artery buttons into the side of the graft and is best undertaken in specialized surgical centers **(Fig. 262-2)**. Resuspension of the native aortic valve leaflets is possible in ~50% of patients with acute AR in the setting of type A aortic dissection.

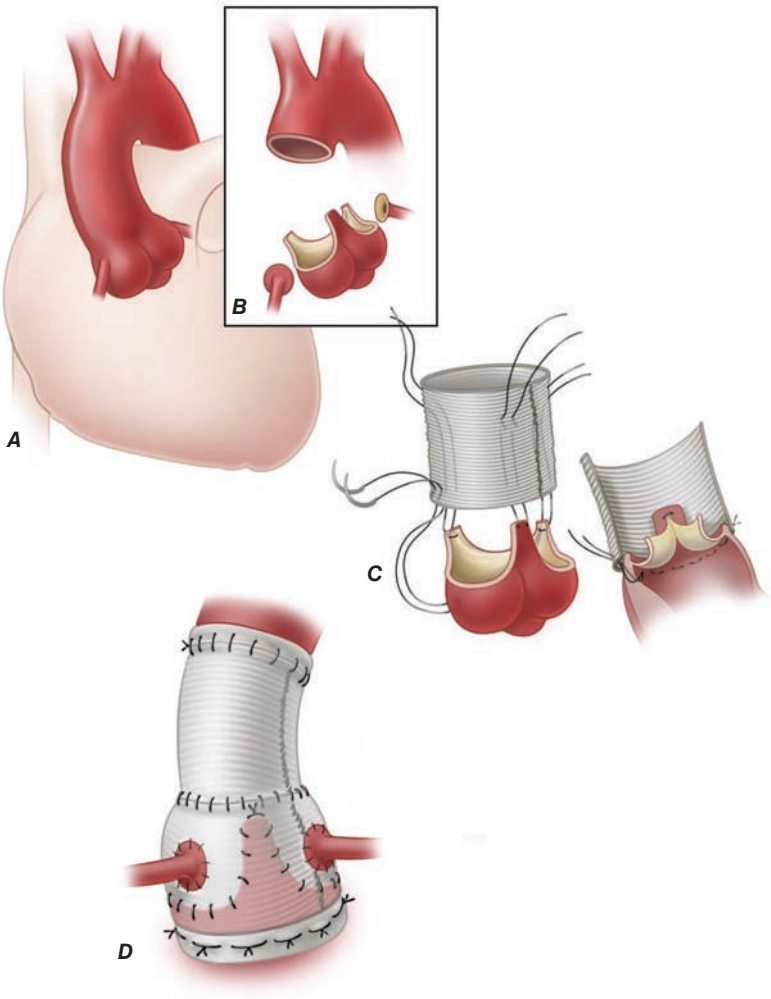



**FIGURE 262-2 Valve-sparing aortic root reconstruction (David procedure).** Aortic root and proximal ascending aorta (***A***) are resected (***B***) with sinuses of Valsalva and mobilized coronary artery buttons remaining. Subannular sutures (***C***) are placed, commissural posts are drawn up inside the valve, and the annular sutures are passed through the proximal end of the graft. The annular sutures are tied (***D***), the valve is reimplanted inside the graft, aortic continuity is reestablished with another graft of appropriate size, and the coronary buttons are attached to the side of the graft. *(From P Steltzer et al [eds]: Valvular Heart Disease: A Companion to Braunwald's Heart Disease, 3rd ed, Fig. 12-27, p. 200.)*

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**TABLE 262-2 Mortality Rates After Aortic Valve Surgery[a]**

| OPERATION | NUMBER | UNADJUSTED OPERATIVE MORTALITY (%) |
|---|---|---|
| AVR (isolated) | 25,274 | 1.9 |
| AVR + CAB | 15,855 | 3.6 |

[a]Data are for calendar year 2018 during which 1088 participant groups reported a total of 287,872 procedures.

*Abbreviations:* AVR, aortic valve replacement; CAB, coronary artery bypass.

*Source:* Adapted from ME Bowdish et al: Ann Thorac Surg 109:1646, 2020.

In other conditions, however, AR can be effectively eliminated only by replacing the aortic valve, as well as the dilated or aneurysmal ascending aorta responsible for the regurgitation, and implanting a composite valve-graft conduit. This formidable procedure entails a higher risk than isolated AVR.

As is true in patients with other valvular heart disease, both operative and late mortality risks are largely dependent on the stage of the disease and myocardial function at the time of operation. The overall operative mortality rate for isolated AVR (performed for either or both AS or AR) is ~2% (Table 262-2). However, patients with AR, marked cardiac enlargement, and established LV dysfunction experience an operative mortality rate of ~10% and a late mortality rate of ~5% per year due to LV failure despite a technically satisfactory operation. Nonetheless, because of the very poor prognosis with medical management, even patients with advanced LV systolic dysfunction should be considered for operation.

Patients with acute severe AR require prompt (24–48 h) surgical treatment, which may be lifesaving.

## FURTHER READING

LACRO RV et al: Atenolol versus losartan in children and young adults with Marfan's syndrome. N Engl J Med 371:2061, 2014.

MALAISRIE SC, MCCARTHY PM: Surgical approach to disease of the aortic valve and the aortic root, in *Valvular Heart Disease: A Companion to Braunwald's Heart Disease*, 5th ed. CM Otto, RO Bonow (eds). Philadelphia, Elsevier Saunders, 2020, pp 267–288.

OTTO CM et al: 2020 ACC/AHA guideline for the management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 143:e72, 2021.

# 263 Mitral Stenosis

**Patrick T. O'Gara, Joseph Loscalzo**

The role of the physical examination in the evaluation of patients with valvular heart disease is also considered in Chaps. 42 and 239; of electrocardiography (ECG) in Chap. 240; of echocardiography and other noninvasive imaging techniques in Chap. 241; and of cardiac catheterization and angiography in Chap. 242.

## MITRAL STENOSIS

### ETIOLOGY AND PATHOLOGY

Rheumatic fever is the leading cause of mitral stenosis (MS) (Table 263-1; see also Chap. 359). Other less common etiologies of obstruction to left ventricular inflow include congenital mitral valve stenosis, cor triatriatum, mitral annular calcification with extension onto the leaflets, systemic lupus erythematosus, rheumatoid arthritis, left atrial myxoma, and infective endocarditis with large vegetations.

**TABLE 263-1 Major Causes of Mitral Stenosis**

| Etiologies |
|---|
| Rheumatic fever |
| Congenital (parachute valve, cor triatriatum) |
| Severe mitral annular calcification with leaflet involvement |
| SLE, RA |
| Myxoma |
| IE with large vegetations |

*Abbreviations:* IE, infective endocarditis; RA, rheumatoid arthritis; SLE, systemic lupus erythematosus.

Pure or predominant MS occurs in ~40% of all patients with rheumatic heart disease and a history of rheumatic fever (Chap. 359). In other patients with rheumatic heart disease, lesser degrees of MS may accompany mitral regurgitation (MR) and aortic valve disease. With reductions in the incidence of acute rheumatic fever, particularly in temperate climates and middle- to high-income countries, the incidence of MS has declined considerably over the past several decades. However, it remains a major problem in low-income countries, especially in sub-Saharan Africa, India, Southeast Asia, and Oceania (Chap. 261).

In rheumatic MS, chronic inflammation leads to diffuse thickening of the valve leaflets with formation of fibrous tissue often with calcific deposits. The mitral commissures fuse, the chordae tendineae fuse and shorten, the valvular cusps become rigid, and the pathologic process eventually leads to narrowing at the apex of the funnel-shaped ("fish-mouth") valve. Although the initial insult to the mitral valve is rheumatic, later changes may be exacerbated by inflammation, fibrosis, and trauma to the valve due to altered flow patterns. Calcification of the stenotic mitral valve immobilizes the leaflets and narrows the orifice further. Thrombus formation and arterial embolization may arise from the calcific valve itself, but in patients with atrial fibrillation (AF), thrombi arise more frequently from the dilated left atrium (LA), particularly from within the LA appendage.

### PATHOPHYSIOLOGY

In normal adults, the area of the mitral valve orifice is 4–6 $cm^2$. In the presence of significant obstruction, i.e., when the orifice area is reduced to <~2 $cm^2$, blood can flow from the LA to the left ventricle (LV) only if propelled by an abnormally elevated left atrioventricular pressure gradient, the hemodynamic hallmark of MS. When the mitral valve opening is reduced to <1.5 $cm^2$, referred to as "severe" MS, an LA pressure of ~25 mmHg is required to maintain a normal cardiac output (CO). The elevated pulmonary venous and pulmonary arterial (PA) wedge pressures reduce pulmonary compliance, contributing to exertional dyspnea. The first bouts of dyspnea are usually precipitated by clinical events that increase the rate of blood flow across the mitral orifice, resulting in further elevation of the LA pressure (see below).

To assess the severity of obstruction hemodynamically, both the transvalvular pressure gradient and the flow rate must be measured (Chap. 242). The latter depends not only on the CO but on the heart rate, as well. An increase in heart rate shortens diastole proportionately more than systole and diminishes the time available for flow across the mitral valve. Therefore, at any given level of CO, tachycardia, including that associated with rapid AF, augments the transvalvular pressure gradient and elevates further the LA pressure. Similar considerations apply to the pathophysiology of tricuspid stenosis (TS).

The LV diastolic pressure and ejection fraction (EF) are normal in isolated MS. In MS and sinus rhythm, the elevated LA and PA wedge pressures exhibit a prominent atrial contraction pattern (*a* wave) and a gradual pressure decline after the *v* wave and mitral valve opening (*y* descent). In severe MS and whenever pulmonary vascular resistance is significantly increased, the PA pressure (PAP) is elevated at rest and rises further during exercise, often causing secondary elevations of right ventricular (RV) end-diastolic pressure and volume.

**Cardiac Output** In patients with severe MS (mitral valve orifice 1–1.5 $cm^2$), the CO is normal or almost so at rest, but rises subnormally

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during exertion. In patients with very severe MS (valve area <1 $cm^2$), particularly those in whom pulmonary vascular resistance is markedly elevated, the CO is subnormal at rest and may fail to rise or may even decline during activity.

**Pulmonary Hypertension** The clinical and hemodynamic features of MS are influenced importantly by the level of the PAP. Pulmonary hypertension results from (1) passive backward transmission of the elevated LA pressure; (2) pulmonary arteriolar constriction (the so-called "second stenosis"), which presumably is triggered by LA and pulmonary venous hypertension (reactive pulmonary hypertension); (3) interstitial edema in the walls of the small pulmonary vessels; and (4) at end stage, organic obliterative changes in the pulmonary vascular bed. Severe pulmonary hypertension results in RV enlargement, secondary tricuspid regurgitation (TR), and pulmonic regurgitation (PR), as well as right-sided heart failure.

## SYMPTOMS

In temperate climates, the latent period between the initial attack of rheumatic carditis (in the increasingly rare circumstances in which a history of one can be elicited) and the development of symptoms due to MS is generally about two decades; most patients begin to experience disability in the fourth decade of life. Studies carried out before the development of surgical mitral valvotomy revealed that once a patient with MS became seriously symptomatic, the disease progressed inexorably to death within 2–5 years.

In patients whose mitral orifices are large enough to accommodate a normal blood flow with only mild elevations of LA pressure, marked elevations of this pressure leading to dyspnea and cough may be precipitated by sudden changes in the heart rate, volume status, or CO, as, for example, with severe exertion, excitement, fever, severe anemia, paroxysmal AF and other tachycardias, sexual intercourse, pregnancy, and thyrotoxicosis. As MS progresses, lesser degrees of stress precipitate dyspnea, the patient becomes limited in daily activities, and orthopnea and paroxysmal nocturnal dyspnea develop. The development of persistent AF often marks a turning point in the patient's course and is generally associated with acceleration of the rate at which symptoms progress. *Hemoptysis* **(Chap. 39)** results from rupture of pulmonary-bronchial venous connections secondary to pulmonary venous hypertension. It occurs most frequently in patients who have elevated LA pressures without markedly elevated pulmonary vascular resistances and is rarely fatal. *Recurrent pulmonary emboli* **(Chap. 279)**, sometimes with infarction, are an important cause of morbidity and mortality late in the course of MS. *Pulmonary infections*, i.e., bronchitis, bronchopneumonia, and lobar pneumonia, commonly complicate untreated MS, especially during the winter months.

**Pulmonary Changes** In addition to the aforementioned changes in the pulmonary vascular bed, fibrous thickening of the walls of the alveoli and pulmonary capillaries occurs commonly in MS. The vital capacity, total lung capacity, maximal breathing capacity, and oxygen uptake per unit of ventilation are reduced **(Chap. 285)**. Pulmonary compliance falls further as pulmonary capillary pressure rises during exercise.

**Thrombi and Emboli** *Thrombi* may form in the left atria, particularly within the enlarged atrial appendages of patients with MS. Systemic embolization, the incidence of which is 10–20%, occurs more frequently in patients with AF, in patients >65 years of age, and in those with a reduced CO. However, systemic embolization may be the presenting feature in otherwise asymptomatic patients with only mild MS.

## PHYSICAL FINDINGS

**(See also Chaps. 42 and 239)**

**Inspection and Palpation** In patients with severe MS, there may be a malar flush with pinched and blue facies. In patients with sinus rhythm and severe pulmonary hypertension or associated TS, the jugular venous pulse reveals prominent *a* waves due to vigorous right atrial systole. The systemic arterial pressure is usually normal or slightly low. A parasternal lift signifies an enlarged RV. A diastolic thrill may rarely be present at the cardiac apex, with the patient in the left lateral recumbent position.

**Auscultation** The first heart sound ($S_1$) is usually accentuated in the early stages of the disease and slightly delayed. The pulmonic component of the second heart sound ($P_2$) also is often accentuated with elevated PAPs, and the two components of the second heart sound ($S_2$) are closely split. The opening snap (OS) of the mitral valve is most readily audible in expiration at, or just medial to, the cardiac apex. This sound generally follows the sound of aortic valve closure ($A_2$) by 0.05–0.12 s. The time interval between $A_2$ and OS varies inversely with the severity of the MS. The OS is followed by a low-pitched, rumbling, diastolic murmur, heard best at the apex with the patient in the left lateral recumbent position **(see Fig. 239-5)**; it is accentuated by mild exercise (e.g., a few rapid sit-ups) carried out just before auscultation. In general, the duration of this murmur correlates with the severity of the stenosis in patients with preserved CO. In patients with sinus rhythm, the murmur often reappears or becomes louder during atrial systole (presystolic accentuation). Soft, grade I or II/VI systolic murmurs may be heard at or medial to the apex and may signify mixed mitral valve disease with regurgitation. Hepatomegaly, ankle edema, ascites, and pleural effusion, particularly in the right pleural cavity, may occur in patients with MS and RV failure.

**Associated Lesions** With severe pulmonary hypertension, a pansystolic murmur produced by functional TR may be audible along the left sternal border. This murmur is usually louder during inspiration and diminishes during forced expiration (Carvallo's sign). When the CO is markedly reduced in MS, the typical auscultatory findings, including the diastolic rumbling murmur, may not be detectable (silent MS), but they may reappear as compensation is restored. The *Graham Steell murmur* of PR, a high-pitched, diastolic, decrescendo blowing murmur along the left sternal border, results from dilation of the pulmonary valve ring and occurs in patients with mitral valve disease and severe pulmonary hypertension. This murmur may be indistinguishable from the more common murmur produced by aortic regurgitation (AR), although it may increase in intensity with inspiration and is accompanied by a loud and often palpable $P_2$.

## LABORATORY EXAMINATION

**ECG** In MS and sinus rhythm, the P wave usually suggests LA enlargement **(see Fig. 240-8)**. It may become tall and peaked in lead II and upright in lead $V_1$ when severe pulmonary hypertension or TS complicates MS and right atrial (RA) enlargement develops. The QRS complex is usually normal. However, with severe pulmonary hypertension, right axis deviation and RV hypertrophy are often present.

**Echocardiogram (See also Chap. 241)** Transthoracic echocardiography (TTE) with color flow and spectral Doppler imaging provides critical information, including measurements of mitral inflow velocity during early (E wave) and late (A wave in patients in sinus rhythm) diastolic filling, estimates of the transvalvular peak and mean gradients and mitral orifice area, the presence and severity of any associated MR, the extent of leaflet calcification and restriction, the degree of distortion of the subvalvular apparatus, and the anatomic suitability for percutaneous mitral balloon commissurotomy (PMBC; see below). In addition, TTE provides an assessment of LV and RV function, chamber sizes, an estimation of the PA systolic pressure based on the tricuspid regurgitant jet velocity, and an indication of the presence and severity of any associated valvular lesions, such as aortic stenosis (AS) and/or regurgitation. Transesophageal echocardiography (TEE) provides superior images and should be used when TTE is inadequate for guiding management decisions. TEE is especially indicated to exclude the presence of LA thrombus prior to PMBC. The performance of TTE with exercise to evaluate the mean mitral diastolic gradient and PAPs can be very helpful in the evaluation of patients with MS when there is a discrepancy between the clinical findings and the resting hemodynamics.

**Chest X-Ray** The earliest changes are straightening of the upper left border of the cardiac silhouette, prominence of the main PAs, dilation

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of the upper lobe pulmonary veins, and posterior displacement of the esophagus by an enlarged LA. Kerley B lines are fine, dense, opaque, horizontal lines that are most prominent in the lower and mid-lung fields that result from distention of interlobular septae and lymphatics with edema when the resting mean LA pressure exceeds ~20 mmHg.

## DIFFERENTIAL DIAGNOSIS

Like MS, significant MR may also be associated with a prominent diastolic murmur at the apex due to increased antegrade transmitral flow, but in patients with isolated MR, this diastolic murmur commences slightly later than in patients with MS, and there is often clear-cut evidence of LV enlargement. An OS and increased $P_2$ are absent, and $S_1$ is soft or absent. An apical pansystolic murmur of at least grade III/VI intensity as well as an $S_3$ suggests significant MR. Similarly, the apical mid-diastolic murmur associated with severe AR (*Austin Flint murmur*) may be mistaken for MS but can be differentiated from it because it is not intensified in pre-systole and becomes softer with administration of amyl nitrite or other arterial vasodilators. TS, which occurs rarely in the absence of MS, may mask many of the clinical features of MS or be clinically silent; when present, the diastolic murmur of TS increases with inspiration and the *y* descent in the jugular venous pulse is delayed.

*Atrial septal defect* **(Chap. 269)** may be mistaken for MS; in both conditions, there is often clinical, ECG, and chest x-ray evidence of RV enlargement and accentuation of pulmonary vascularity. However, the absence of LA enlargement and of Kerley B lines and the demonstration of fixed splitting of $S_2$ with a grade II or III mid-systolic murmur at the mid to upper left sternal border all favor atrial septal defect over MS. Atrial septal defects with large left-to-right shunts may result in functional TS because of the enhanced diastolic flow.

*Left atrial myxoma* **(Chap. 271)** may obstruct LA emptying, causing dyspnea, a diastolic murmur, and hemodynamic changes resembling those of MS. However, patients with an LA myxoma often have features suggestive of a systemic disease, such as weight loss, fever, anemia, systemic emboli, and elevated serum IgG and interleukin 6 (IL-6) concentrations. The auscultatory findings may change markedly with body position. The diagnosis can be established by the demonstration of a characteristic echo-producing mass in the LA with TTE.

## CARDIAC CATHETERIZATION

Left and right heart catheterization can be useful when there is a discrepancy between the clinical and noninvasive findings, including those from TEE and exercise echocardiographic testing when appropriate. Catheterization can also be helpful in assessing associated lesions, such as AS and AR, and in patients with recurring or worsening symptoms later after mitral valve intervention. Computed tomographic coronary angiography is increasingly used to screen preoperatively for the presence of coronary artery disease in appropriate patients prior to heart valve surgery or transcatheter treatment.

# TREATMENT

## Mitral Stenosis (Fig. 263-1)

Penicillin prophylaxis of group A β-hemolytic streptococcal infections **(Chap. 359)** for secondary prevention of rheumatic fever is

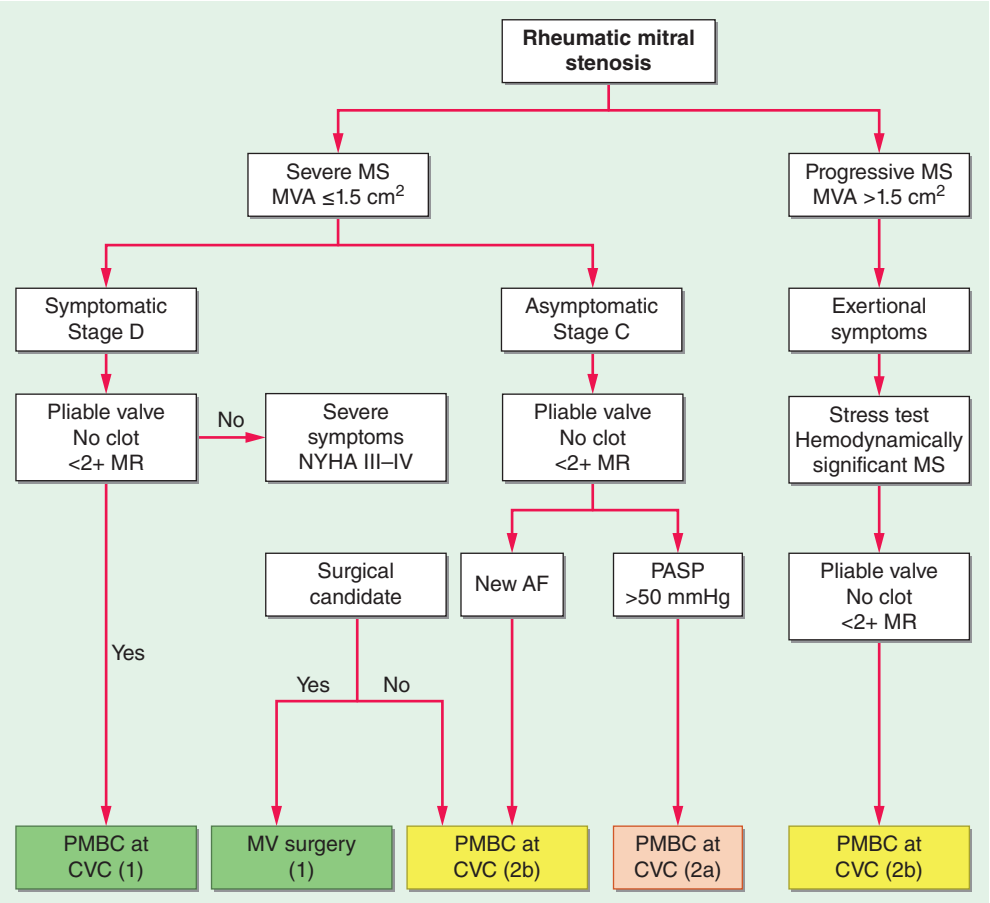



**FIGURE 263-1 Management of rheumatic mitral stenosis.** See legend for **Fig. 261-4** for explanation of treatment recommendations (Class I, IIa, IIb) and disease stages (C, D). Preoperative coronary angiography should be performed routinely as determined by age, symptoms, and coronary risk factors. Cardiac catheterization and angiography may also be helpful when there is a discrepancy between clinical and noninvasive findings. AF, atrial fibrillation; CVC, comprehensive valve center; MR, mitral regurgitation; MS, mitral stenosis; MV, mitral valve; MVA, mitral valve area; MVR, mitral valve surgery (repair or replacement); NYHA, New York Heart Association; PASP, pulmonary arterial systolic pressure; PMBC, percutaneous mitral balloon commissurotomy. *(Reproduced with permission from CM Otto et al: ACC/AHA guideline for the management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 143:e72, 2021.)*

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important for at-risk patients with rheumatic MS. Recommendations for infective endocarditis prophylaxis are similar to those for other valve lesions and are restricted to patients at high risk for complications from infection, including patients with a history of endocarditis. In symptomatic patients, some improvement usually occurs with restriction of sodium intake and small doses of oral diuretics. Beta blockers, nondihydropyridine calcium channel blockers (e.g., verapamil or diltiazem), and digitalis glycosides are useful in slowing the ventricular rate of patients with AF. Vitamin K antagonist therapy (such as warfarin) targeted to an international normalized ratio (INR) of 2–3 should be administered indefinitely to patients with MS who have AF, a history of thromboembolism, or demonstrated LA thrombus. The routine use of a vitamin K antagonist in patients in sinus rhythm with LA enlargement (maximal dimension >5.5 cm) with or without spontaneous echo contrast is more controversial. As of this writing, non–vitamin K oral anticoagulants (e.g., apixaban, rivaroxaban) have not been adequately studied in patients with moderate or severe rheumatic MS and, thus, are not recommended.

If AF is of relatively recent onset in a patient whose MS is not severe enough to warrant PMBC or surgical intervention, reversion to sinus rhythm pharmacologically or by means of electrical countershock is indicated. Usually, cardioversion should be undertaken after the patient has had at least 3 consecutive weeks of anticoagulant treatment to a therapeutic INR. If cardioversion is indicated more urgently, then intravenous heparin should be provided and TEE performed to exclude the presence of LA thrombus before the procedure. Conversion to sinus rhythm is rarely successful or sustained in patients with severe MS, particularly those in whom the LA is significantly enlarged or in whom AF has been present for >1 year, conditions that favor the development of an LA myopathy.

## MITRAL COMMISSUROTOMY

Unless there is a contraindication, mitral commissurotomy is indicated in symptomatic (New York Heart Association [NYHA] Functional Class II–IV) patients with isolated severe MS, whose effective orifice (valve area) is <~1 $cm^2/m^2$ body surface area, or <1.5 $cm^2$ in normal-sized adults. Mitral commissurotomy can be carried out either percutaneously or surgically. In PMBC **(Figs. 263-2 and 263-3)**, a catheter is directed into the LA after transseptal puncture, and a single balloon is directed across the valve and inflated in the valvular orifice. Ideal patients have relatively pliable leaflets with little or no commissural calcium. In addition, the subvalvular structures should not be significantly scarred or thickened, and there should be no LA thrombus. Any associated MR should be of ≤2+/4+ severity. The short- and long-term results of this procedure in appropriate patients are similar to those of surgical commissurotomy, but with less morbidity and a lower periprocedural mortality rate. Event-free survival in younger (<45 years) patients with pliable valves is excellent, with rates as high as 80–90% over 3–7 years. Therefore, PMBC is the procedure of choice for such patients when it can be performed by a skilled operator in a high-volume center.

TTE is helpful in identifying patients for the percutaneous procedure; TEE is performed routinely to exclude LA thrombus and to assess the degree of MR at the time of the scheduled procedure. An "echo score" has been developed to help guide decision-making. The score accounts for the degree of leaflet thickening, calcification, and mobility, and for the extent of subvalvular thickening. A lower score predicts a higher likelihood of successful PMBC.

In patients in whom PMBC is not possible or unsuccessful, or in many patients with restenosis after previous surgery, an "open" surgical commissurotomy using cardiopulmonary bypass is necessary. In addition to opening the valve commissures, it is important to loosen any subvalvular fusion of papillary muscles and chordae tendineae; to remove large deposits of calcium, thereby improving valvular function; and to remove atrial thrombi. The perioperative mortality rate for this type of mitral valve repair procedure is ~2%.

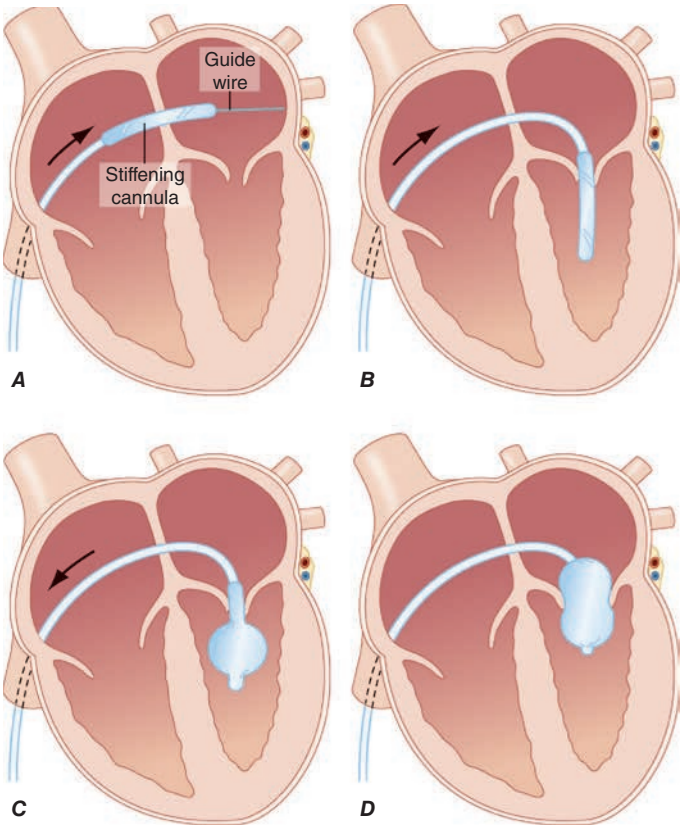



**FIGURE 263-2 Inoue balloon technique for percutaneous mitral balloon commissurotomy.** ***A.*** After transseptal puncture, the deflated balloon catheter is advanced across the interatrial septum, then across the mitral valve and into the left ventricle. ***B–D.*** The balloon is inflated stepwise within the mitral orifice.

Successful commissurotomy is defined by a 50% reduction in the mean mitral valve gradient and a doubling of the mitral valve area. Successful commissurotomy, whether balloon or surgical, usually results in striking symptomatic and hemodynamic improvement

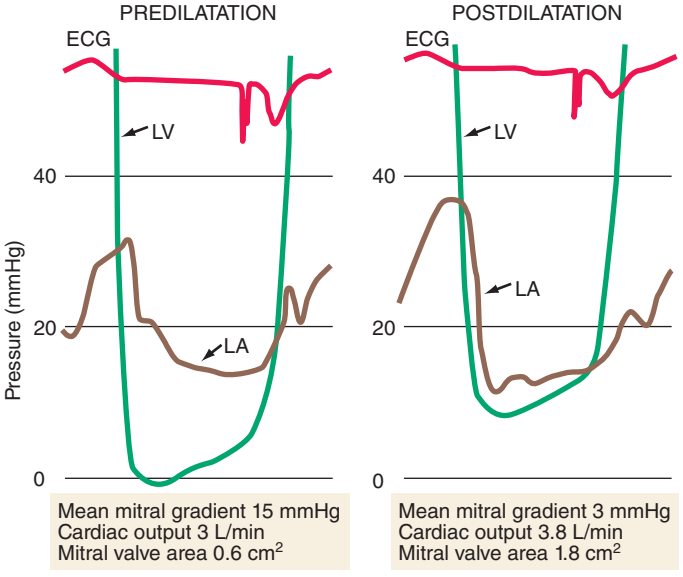



**FIGURE 263-3 Simultaneous left atrial (LA) and left ventricular (LV) pressure** before and after percutaneous mitral balloon commissurotomy (PMBC) in a patient with severe mitral stenosis. ECG, electrocardiogram. *(Courtesy of Raymond G. McKay, MD.)*

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**TABLE 263-2 Mortality Rates after Mitral Valve Surgery[a]**

| OPERATION | NUMBER | UNADJUSTED OPERATIVE MORTALITY (%) |
|---|---|---|
| MVR (isolated) | 10,699 | 4.5 |
| MVR + CAB | 3509 | 9.6 |
| MVRp | 12,424 | 1.2 |
| MVRp + CAB | 4093 | 5.4 |

[a]Data are for calendar year 2018 during which 1088 participant groups reported a total of 287,872 procedures. Surgical mitral valve commissurotomy cases are included in the mitral valve repair procedures.

*Abbreviations:* CAB, coronary artery bypass; MVR, mitral valve replacement; MVRp, mitral valve repair.

*Source:* Adapted from ME Bowdish et al: Ann Thorac Surg 109:1646, 2020.

and prolongs survival. However, there is no evidence that the procedure improves the prognosis of patients with slight or no functional impairment. Therefore, unless recurrent systemic embolization or severe pulmonary hypertension has occurred (PA systolic pressures >50 mmHg at rest or >60 mmHg with exercise), commissurotomy is *not* recommended for patients who are asymptomatic and/or who have mild or moderate stenosis (mitral valve area >1.5 $cm^2$). When there is little symptomatic improvement after commissurotomy, it is likely that the procedure was ineffective, that it induced MR, or that associated valvular or myocardial disease was present. About half of all patients undergoing surgical mitral commissurotomy require reoperation by 10 years. In the pregnant patient with MS, commissurotomy should be carried out if pulmonary congestion occurs despite intensive medical treatment. PMBC is the preferred strategy in this setting and is performed with TEE and no or minimal x-ray exposure.

Mitral valve replacement (MVR) is necessary in patients with MS and significant associated MR, those in whom the valve has been severely distorted by previous transcatheter or operative manipulation, or those in whom the surgeon does not find it possible to improve valve function significantly with commissurotomy. MVR is now routinely performed with preservation of the chordal attachments to optimize LV functional recovery. Perioperative mortality rates with MVR vary with age, LV function, the presence of CAD, and associated comorbidities. They average 5% overall but are lower in young patients and may be twice as high in patients >65 years of age with significant comorbidities (**Table 263-2**). Because there are also long-term complications of valve replacement, patients in whom preoperative evaluation suggests the possibility that MVR may be required should be operated on only if they have severe MS—i.e., an orifice area ≤1.5 $cm^2$—and are in NYHA Class III, i.e., symptomatic with ordinary activity despite optimal medical therapy. The overall 10-year survival of surgical survivors is ~70%. Long-term prognosis is worse in patients >65 years of age and those with marked disability and marked depression of the CO preoperatively. Pulmonary hypertension and RV dysfunction are additional risk factors for poor outcome.

## FURTHER READING

NISHIMURA RA et al: Mitral valve disease: Current management and future challenges. Lancet 387:1324, 2016.

OTTO CM et al: 2020 ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 143:e72, 2021.

# 264 Mitral Regurgitation

**Patrick T. O'Gara, Joseph Loscalzo**

The role of the physical examination in the evaluation of patients with valvular heart disease is also considered in **Chaps. 42 and 239**; of electrocardiography (ECG) in **Chap. 240**; of echocardiography and other noninvasive imaging techniques in **Chap. 241**; and of cardiac catheterization and angiography in **Chap. 242**.

## ETIOLOGY

Mitral regurgitation (MR) may result from an abnormality or disease process that affects any one or more of the five functional components of the mitral valve apparatus (leaflets, annulus, chordae tendineae, papillary muscles, and subjacent myocardium) (**Table 264-1**). Acute MR can occur in the setting of acute myocardial infarction (MI) with papillary muscle rupture (**Chap. 275**), following blunt chest wall trauma, or during the course of infective endocarditis (IE) owing to leaflet perforation or destruction. With acute MI, the posteromedial papillary muscle is involved much more frequently than the anterolateral papillary muscle because of its singular blood supply. Transient, acute MR can occur during periods of active ischemia and bouts of angina pectoris. Rupture of chordae tendineae can result in "acute-on-chronic MR" in patients with myxomatous degeneration of the valve apparatus.

Chronic MR can result from several disease processes (Table 264-1). Distinction should be drawn between primary (degenerative) MR, in which the leaflets and/or chordae tendineae are primarily responsible for abnormal valve function, and secondary (functional) MR, in which the leaflets and chordae tendineae are usually normal but the regurgitation is caused by left ventricular (LV) remodeling, annular dilation, papillary muscle displacement, dyssynchrony, posterior leaflet tethering, or their combination. Patient assessment, treatment approach, and long-term prognosis differ significantly between primary and secondary MR. Mitral valve prolapse (MVP) is discussed more extensively

**TABLE 264-1 Major Causes of Mitral Regurgitation (MR)**

| Etiologies |
|---|
| **Acute** |
| IE |
| Papillary muscle rupture (post-MI) |
| Chordal rupture/leaflet flail (MVP, IE) |
| Blunt trauma |
| **Chronic** |
| **Primary (affecting leaflets, chordae)** |
| Myxomatous (MVP, Barlow's, *forme fruste*) |
| Rheumatic fever |
| IE (healed) |
| Congenital (cleft, AV canal) |
| Radiation |
| **Secondary (leaflets, chordae are "innocent bystanders")** |
| Ischemic cardiomyopathy |
| Dilated cardiomyopathy |
| HOCM (with SAM) |
| AF with LA enlargement and annular dilation (atrial functional MR) |
| **Mitral annular calcification[a]** |

[a]Mitral annular calcification may include elements of both primary and secondary MR (mixed) as the disease process may encroach on the leaflets, impair the normal sphincteric function of the annulus, or both. There are additional examples of "mixed" secondary MR such as the coexistence of MVP with an ischemic cardiomyopathy.

*Abbreviations:* AF, atrial fibrillation; AV, atrioventricular; HOCM, hypertrophic obstructive cardiomyopathy; IE, infective endocarditis; LA, left atrial; LV, left ventricular; MI, myocardial infarction; MVP, mitral valve prolapse; SAM, systolic anterior motion.

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in Chap. 265. The rheumatic process produces rigidity, deformity, and retraction of the valve cusps and commissural fusion, as well as shortening, contraction, and fusion of the chordae tendineae. MR can persist after resolution of the acute phase of infection and inflammation. MR may occur as a congenital anomaly (Chap. 269), most commonly as a defect of the endocardial cushions (atrioventricular cushion defects). A cleft anterior mitral valve leaflet accompanies ostium primum atrial septal defect. Radiation can result in leaflet thickening, retraction, and calcification, often in association with annular and chordal involvement and some degree of mitral stenosis. Chronic MR occurs frequently after prior MI(s) associated with changes in LV size, shape, and function. Similar mechanisms of annular dilation and ventricular remodeling contribute to the MR that occurs among patients with nonischemic forms of dilated cardiomyopathy once the LV end-diastolic dimension reaches 6 cm. The MR associated with hypertrophic obstructive cardiomyopathy (HOCM) is usually dynamic in nature and dependent on systolic anterior motion of the anterior mitral valve leaflet into a narrowed LV outflow tract. Patients with chronic persistent atrial fibrillation (AF) may develop atrial remodeling and annular dilation with inadequate leaflet lengthening and MR (atrial functional MR). Secondary MR due to LV remodeling is more frequently encountered in the community than secondary MR that occurs in association with AF and annular dilation. Annular calcification can result in MR when it encroaches on the leaflets or results in decreased sphincteric function and is especially prevalent among patients with advanced renal disease and is commonly observed in women >65 years of age with hypertension and diabetes mellitus. Irrespective of cause, chronic severe MR is often progressive because enlargement of the left atrium (LA) places tension on the posterior mitral leaflet, pulling it further away from the mitral orifice and thereby aggravating the valvular dysfunction. Similarly, LV dilation increases the regurgitation, which, in turn, enlarges the LA and LV further, resulting in a vicious circle; hence the aphorism, "MR begets MR."

## PATHOPHYSIOLOGY

The resistance to LV emptying (LV afterload) is reduced in patients with MR. As a consequence, the LV is decompressed into the LA during ejection, and with the reduction in LV size during systole, there is a rapid decline in LV tension. The initial compensation to MR is more complete LV emptying. However, LV volume increases progressively with time as the severity of the regurgitation increases and as LV contractile function deteriorates. This increase in LV volume is often accompanied by a reduced forward cardiac output (CO). LV compliance is often increased, and thus, LV diastolic pressure does not increase until late in the course. The regurgitant volume varies directly with the LV systolic pressure and the size of the regurgitant orifice; the latter, in turn, is influenced by the extent of LV and mitral annular dilation. Because ejection fraction (EF) rises in severe MR in the presence of normal LV function, even a modest reduction in this parameter (<60%) reflects significant contractile dysfunction.

During early diastole, as the distended LA empties, there is a particularly rapid *y* descent in the absence of accompanying MS. A brief, early diastolic LA-LV pressure gradient (often generating a rapid filling sound [$S_3$] and mid-diastolic murmur masquerading as MS) may occur in patients with pure, severe MR as a result of the very rapid flow of blood across a normal-sized mitral orifice.

Measurements of LV ejection fraction (LVEF), CO, pulmonary arterial (PA) systolic pressure, regurgitant volume, regurgitant fraction (RF), and the effective regurgitant orifice area can be obtained during a careful Doppler echocardiographic examination. These measurements can also be obtained accurately with cardiac magnetic resonance (CMR) imaging, although this technology is not widely available. Left and right heart catheterization with contrast ventriculography is used less frequently. Chronic, severe MR is defined by a regurgitant volume ≥60 mL/beat, regurgitant fraction (RF) ≥50%, and effective regurgitant orifice area ≥0.40 $cm^2$. In patients with secondary MR, in whom the severity of MR can be underappreciated using echocardiographic/Doppler techniques, lesser degrees of regurgitation may carry relatively greater prognostic weight. The adverse prognosis in secondary MR related to adverse LV remodeling is intimately related to the degree of myocardial dysfunction.

**LA Compliance** In acute severe MR, the regurgitant volume is delivered into a normal-sized LA having normal or reduced compliance. As a result, LA pressures rise markedly for any increase in LA volume. The *v* wave in the LA pressure pulse is usually prominent, LA and pulmonary venous pressures are markedly elevated, and pulmonary edema is common. Because of the rapid rise in LA pressures during ventricular systole, the murmur of acute MR is early in timing and decrescendo in configuration ending well before $S_2$, as a reflection of the progressive diminution in the LV-LA pressure gradient. LV systolic function in acute MR may be normal, hyperdynamic, or reduced, depending on the clinical context.

Patients with chronic severe MR, on the other hand, develop marked LA enlargement and *increased* LA compliance with little if any increase in LA and pulmonary venous pressures for any increase in LA volume. The LA *v* wave is relatively less prominent. The murmur of chronic MR is classically holosystolic in timing and plateau in configuration, as a reflection of the near-constant LV-LA pressure gradient. These patients usually complain of severe fatigue and exhaustion secondary to a low forward CO, whereas symptoms resulting from pulmonary congestion are less prominent initially; AF is almost invariably present once the LA dilates significantly.

## SYMPTOMS

Patients with chronic mild-to-moderate, isolated MR are usually asymptomatic. This form of LV volume overload is well tolerated. Fatigue, exertional dyspnea, and orthopnea are the most prominent complaints in patients with chronic severe MR. Palpitations are common and may signify the onset of AF. Late-onset right-sided heart failure, with painful hepatic congestion, ankle edema, distended neck veins, ascites, and secondary tricuspid regurgitation (TR), occurs in patients with MR who have associated pulmonary vascular disease and pulmonary hypertension. Acute pulmonary edema is common in patients with acute severe MR.

## PHYSICAL FINDINGS

In patients with chronic severe MR, the arterial pressure is usually normal, although the carotid arterial pulse may show a sharp, low-volume upstroke owing to the reduced forward CO. A systolic thrill is often palpable at the cardiac apex, the LV is hyperdynamic with a brisk systolic impulse and a palpable rapid-filling wave ($S_3$), and the apex beat is often displaced laterally.

In patients with acute severe MR, the arterial pressure may be reduced with a narrow pulse pressure, the jugular venous pressure and waveforms may be normal or increased and exaggerated, the apical impulse is not displaced, and signs of pulmonary congestion are prominent.

**Auscultation** $S_1$ is generally absent, soft, or buried in the holosystolic murmur of chronic, severe MR. In patients with severe MR, the aortic valve may close prematurely (due to the reduced forward cardiac output), resulting in wide but physiologic splitting of $S_2$. A low-pitched $S_3$ occurring 0.12–0.17 s after the aortic valve closure sound, i.e., at the completion of the rapid-filling phase of the LV, is believed to be caused by the sudden tensing of the papillary muscles, chordae tendineae, and valve leaflets. It may be followed by a short, rumbling, mid-diastolic murmur, even in the absence of structural MS. In patients with ischemic or dilated cardiomyopathy, however, a third sound ($S_3$) may also signify ventricular dysfunction. A fourth heart sound is often audible in patients with *acute* severe MR who are in sinus rhythm. A presystolic murmur is not ordinarily heard with isolated MR.

A systolic murmur of at least grade III/VI intensity is the most characteristic auscultatory finding in chronic severe MR. It is usually holosystolic (see Fig. 239-5*A*), but as previously noted, it is decrescendo and ceases in mid-to-late systole in patients with acute severe MR. The systolic murmur of chronic MR is usually most prominent at the apex and radiates to the axilla. However, in patients with ruptured chordae tendineae or primary involvement of the posterior mitral leaflet with prolapse or flail, the regurgitant jet is eccentric, directed anteriorly, and

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strikes the LA wall adjacent to the aortic root. In this situation, the systolic murmur is transmitted to the base of the heart and, therefore, may be confused with the murmur of AS. In patients with ruptured chordae tendineae, the systolic murmur may have a cooing or "seagull" quality, whereas a flail leaflet may produce a murmur with a musical quality. The systolic murmur of chronic MR not due to MVP is intensified by isometric exercise (handgrip) but is reduced during the strain phase of the Valsalva maneuver because of the associated decrease in LV preload.

## LABORATORY EXAMINATION

**ECG** In patients with sinus rhythm, there is evidence of LA enlargement, but right atrial (RA) enlargement also may be present when pulmonary hypertension is significant and affects RV function and size. Chronic severe MR is frequently associated with AF. In many patients, there is no clear-cut ECG evidence of enlargement of either ventricle. In others, the signs of eccentric LV hypertrophy are present.

**Echocardiogram** Transthoracic echocardiography (TTE) is indicated to assess the mechanism of the MR and its hemodynamic severity. LV function can be assessed from LV end-diastolic and end-systolic volumes and EF. Observations can be made regarding leaflet structure and function, chordal integrity, LA and LV size, annular calcification, and regional and global LV systolic function. Doppler imaging should demonstrate the width or area of the color flow MR jet within the LA, the duration and intensity of the continuous wave Doppler signal, the pulmonary venous flow contour, the early peak mitral inflow velocity, and quantitative measures of regurgitant volume, RF, and effective regurgitant orifice area. In addition, the PA pressures (PAPs) can be estimated from the TR jet velocity. TTE is also indicated to follow the course of patients with chronic MR and to provide rapid assessment for any clinical change. Transesophageal echocardiography (TEE) provides greater anatomic detail than TTE (see Fig. 241-5). Exercise testing with TTE can be useful to assess exercise capacity as well as any dynamic change in MR severity, PA systolic pressures, and biventricular function, for patients in whom there is a discrepancy between clinical findings and the results of other noninvasive testing.

**Chest X-Ray** The LA and LV are the dominant chambers in chronic MR. Late in the course of the disease, the LA may be massively enlarged and forms the right border of the cardiac silhouette. Pulmonary venous congestion, interstitial edema, and Kerley B lines are sometimes noted. Marked calcification of the mitral leaflets occurs commonly in patients with long-standing, combined rheumatic MR and MS, as well as in patients with radiation-induced mitral valve disease. Calcification of the mitral annulus may be visualized, particularly on the lateral view of the chest. Patients with acute severe MR may have asymmetric pulmonary edema if the regurgitant jet is directed predominantly to the orifice of an upper lobe pulmonary vein.

## TREATMENT (FIGS. 264-1 AND 264-2)

### Mitral Regurgitation

#### MEDICAL TREATMENT

The management of chronic severe MR depends to some degree on its cause. Anticoagulation with either warfarin or a direct oral agent (e.g., apixaban, rivaroxaban) should be provided if AF intervenes, as guided by the $CHA_2DS_2$-VASc risk score. The direct oral anticoagulants should not be used if moderate or severe rheumatic mitral

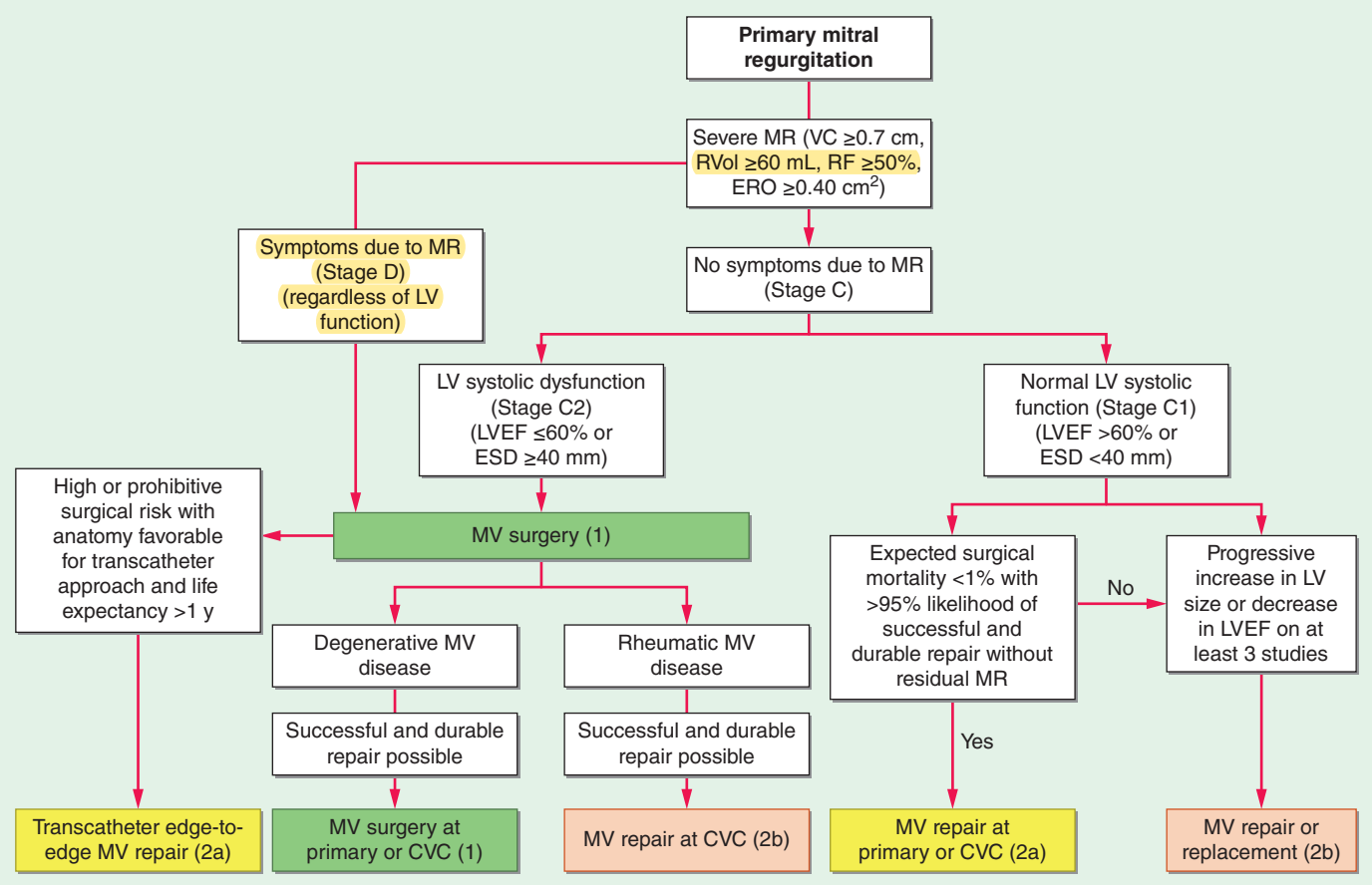



**FIGURE 264-1 Management of primary mitral regurgitation (MR).** See legend for Fig. 261-4 for explanation of treatment recommendations (Class I, IIa, IIb) and disease stages (B, C1, C2, D). Preoperative coronary angiography should be performed routinely as determined by age, symptoms, and coronary risk factors. Cardiac catheterization and angiography may also be helpful when there is a discrepancy between clinical and noninvasive findings. Mitral valve repair is strongly preferred over valve replacement whenever feasible for surgical treatment of primary MR. Transcatheter edge-to-edge repair (TEER) is reserved for high or prohibitive surgical risk patients with appropriate anatomy on transesophageal imaging. CVC, comprehensive valve center; EF, ejection fraction; ERO, effective regurgitant orifice; ESD, end-systolic dimension; LV, left ventricular; MV, mitral valve; RF, regurgitant fraction; RVol, regurgitant volume; VC, vena contracta. *(Reproduced with permission from CM Otto et al: ACC/AHA guideline for the management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 143:e72, 2021.)*

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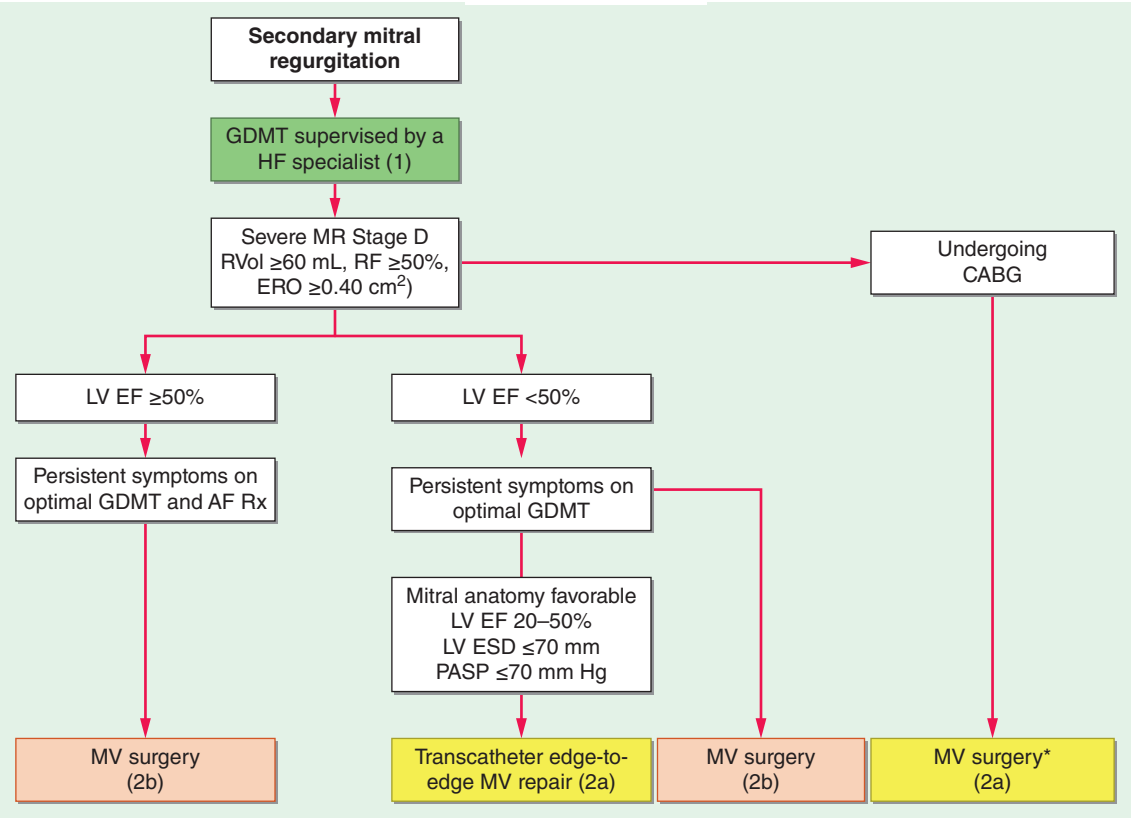



**FIGURE 264-2 Management of secondary mitral regurgitation.** See legend for Fig. 261-4 for explanation of treatment recommendations (Class I, IIa, IIb) and disease stages (B, C1, C2, D). Preoperative coronary angiography should be performed routinely as determined by age, symptoms, and coronary risk factors. Cardiac catheterization and angiography may also be helpful when there is a discrepancy between clinical and noninvasive findings. Surgery is recommended for patients with left ventricular ejection fraction (LVEF) >50%. Transcatheter edge-to-edge repair (TEER) is reasonable is selected patients after guideline-directed management and therapy (GDMT) has been optimized. *MV replacement may be preferred over MV repair for ischemic MR; AF, atrial fibrillation; CABG, coronary artery bypass grafting; EF, ejection fraction; ERO, effective regurgitant orifice; ESD, end-systolic dimension; HF, heart failure; LV, left ventricular; MR, mitral regurgitation, MV, mitral valve; PASP, pulmonary artery systolic pressure; RF, regurgitant fraction; RVol, regurgitant volume; Rx, treatment. *(Reproduced with permission from CM Otto et al: ACC/AHA guideline for the management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 143:e72, 2021.)*

stenosis is also present; they are also not approved for use in patients with mechanical prosthetic heart valves. Cardioversion should be considered depending on the clinical context, AF chronicity, and LA size. In contrast to the acute setting, there are no large, long-term prospective studies to substantiate the use of vasodilators for the treatment of chronic, isolated severe MR with preserved LV systolic function *in the absence of systemic hypertension.* The severity of secondary MR in the setting of an ischemic or dilated cardiomyopathy may diminish with aggressive guideline-directed therapy (GDMT) of heart failure including the use of diuretics for decongestion, beta blockers, angiotensin-converting enzyme (ACE) inhibitors/angiotensin receptor blockers, angiotensin-neprilysin inhibitors, mineralocorticoid receptor antagonists, sodium-glucose cotransporter-2 inhibitors and biventricular pacing (cardiac resynchronization therapy [CRT]) when indicated. Antibiotic prophylaxis for prevention of IE is indicated for MR patients with a prior history of IE. Asymptomatic patients with severe MR in sinus rhythm with normal LV size and systolic function should avoid isometric forms of exercise.

Patients with acute severe MR require urgent stabilization and preparation for surgery. Diuretics, intravenous vasodilators (particularly sodium nitroprusside), and even mechanical support may be needed for patients with post-MI papillary muscle rupture or other forms of acute severe MR.

## SURGICAL TREATMENT

In the selection of patients with chronic, severe, primary MR for surgical treatment, the often slowly progressive nature of the condition must be balanced against the immediate and long-term risks associated with operation. These risks are significantly lower for primary valve repair than for valve replacement (Table 264-2). Repair usually consists of valve reconstruction using a variety of valvuloplasty techniques and insertion of an annuloplasty ring. Repair spares the patient the long-term adverse consequences of valve replacement, including thromboembolic and hemorrhagic complications in the case of mechanical prostheses and late valve failure necessitating repeat valve replacement in the case of bioprostheses. In addition, by preserving the integrity of the papillary muscles, subvalvular apparatus, and chordae tendineae, mitral repair and valvuloplasty maintain LV function to a relatively greater degree than does valve replacement.

Surgery for chronic severe primary MR is indicated once symptoms occur, especially if valve repair is feasible (Fig. 264-1). Surgery should also be recommended for asymptomatic patients with LV dysfunction characterized by an EF ≤60% or an LV end-systolic dimension (LV ESD) ≥40 mm. Other indications for early consideration of mitral valve repair in asymptomatic patients include a progressive decrease in LVEF or increase in LV ESD on serial imaging

**TABLE 264-2 Mortality Rates after Mitral Valve Surgery[a]**

| OPERATION | NUMBER | UNADJUSTED OPERATIVE MORTALITY (%) |
|---|---|---|
| MVR (isolated) | 10,699 | 4.5 |
| MVR + CAB | 3509 | 9.6 |
| MVRp | 12,424 | 1.2 |
| MVRp + CAB | 4093 | 5.4 |

[a]Data are for calendar year 2018 during which 1088 participant groups reported a total of 287,872 procedures. Surgical mitral valve commissurotomy cases are included in the mitral valve repair procedures.

*Abbreviations:* CAB, coronary artery bypass; MVR, mitral valve replacement; MVRp, mitral valve repair.

*Source:* Adapted from ME Bowdish et al: Ann Thorac Surg 109:1646, 2020.

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as well as MV anatomy that would predict a >95% of a successful and durable repair in a low surgical risk patient. These aggressive recommendations for surgery are predicated on the adverse long-term consequences of waiting for LV function to decline further as well as the outstanding results achievable with mitral valve repair by reference surgeons at high-volume centers. Indeed, repair of myxomatous MR (e.g., prolapse, flail) in patients <75 years with normal LV systolic function and no coronary artery disease (CAD) can now be performed by experienced surgeons with <1% perioperative mortality risk. The risk of stroke, however, is also ~1%. Repair is feasible in up to 95% of patients with myxomatous disease operated on by a high-volume surgeon in a referral center of excellence. Repair techniques include chordal transfer, creation of neochords, limited leaflet resection, and insertion of an annuloplasty band. Long-term durability is excellent; the incidence of reoperative surgery for failed primary repair is ~1% per year for the first 10 years after surgery. For patients with AF, left or biatrial maze surgery, or radiofrequency isolation of the pulmonary veins, along with left atrial appendage amputation, is performed to reduce the risk of recurrent postoperative AF and associated thrombus formation.

The surgical management of patients with secondary MR is more complicated. Surgery for patients with ischemic MR most often involves simultaneous coronary artery revascularization. Current surgical practice includes either annuloplasty repair with an undersized, rigid ring or chord-sparing valve replacement for patients with moderate or greater degrees of MR. Valve repair for ischemic MR is associated with lower perioperative mortality rates than valve replacement but significantly higher rates of recurrent MR over time. Thus, replacement may be preferred over repair in this context. In patients with ischemic MR and significantly impaired LV systolic function (EF <30%), the risk of surgery is higher, recovery of LV performance is incomplete, and long-term survival is reduced. Referral for surgery must be individualized and made only after aggressive attempts to improve symptoms with GDMT and CRT, when indicated. The routine performance of surgical valve repair in patients with significant secondary MR due to a dilated cardiomyopathy has not been shown to improve long-term survival compared with optimal GDMT. Patients with acute severe MR can often be stabilized temporarily with appropriate medical therapy, but surgical correction will be necessary emergently in the case of papillary muscle rupture and within days to weeks in most other settings.

When surgical treatment is contemplated, left and right heart catheterization and left ventriculography *may* be helpful in confirming the presence of severe MR in patients in whom there is a discrepancy between the clinical and TTE findings that cannot be resolved with TEE or CMR. Coronary angiography identifies patients who require concomitant coronary revascularization.

## TRANSCATHETER MITRAL VALVE REPAIR AND REPLACEMENT

A transcatheter approach to the treatment of either primary or secondary MR may be feasible in selected patients with appropriate mitral valve anatomy. One approach involves the deployment of a clip delivered via transseptal puncture that grasps the leading edges of the mitral leaflets in their mid-portion (anterior scallop to posterior scallop or A2-P2; **Fig. 264-3**). The length and width of the gap between these leading edges, as well as other considerations such as leaflet thickening and calcification, dictate patient eligibility. The clip device for transcatheter edge-to-edge repair (TEER) is commercially available for treatment of both primary and secondary MR in appropriately selected patients (Figs. 264-1 and 264-2). The results of transthoracic and transesophageal echocardiographic imaging are critical to patient selection, along with a detailed assessment of surgical risk, comorbidities, and the adequacy of GDMT for heart failure. The use of TEER with a clip device in addition to medical therapy was shown to be superior to medical therapy alone in a trial involving symptomatic heart failure patients with reduced EF and at least moderately severe secondary MR. Patients treated with the clip device had fewer heart failure hospitalizations and longer survival than those treated medically. This was the first trial to show such benefit in patients with secondary MR and has impacted clinical practice. Other transcatheter approaches to mitral valve repair include the deployment of a device within the coronary sinus that can be adjusted to reduce mitral annular circumference and the effective orifice area of the valve much like a surgically implanted ring. Variations in the anatomic relationship of the coronary sinus to the mitral annulus and circumflex coronary artery have limited the applicability of this technique. Attempts to reduce the septal-lateral dimension of a dilated annulus using adjustable cords placed across the LV in a subvalvular location have also been investigated. Construction of neochords to the mitral leaflets under TEE guidance using a system delivered via the cardiac apex is also under study. Investigational experience to date with transcatheter mitral valve replacement systems is in early clinical stages, although the field is evolving rapidly.



**FIGURE 264-3 Clip used to grasp the free edges of the anterior and posterior leaflets in their midsections during transcatheter repair** of selected patients with mitral regurgitation. *(MitraClip is a trademark of Abbott or its related companies. Reproduced with permission from Abbott © 2021. All rights reserved.)*

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# 265 Mitral Valve Prolapse

**Patrick T. O'Gara, Joseph Loscalzo**

The role of the physical examination in the evaluation of patients with valvular heart disease is also considered in **Chaps. 42 and 239**; of electrocardiography (ECG) in **Chap. 240**; of echocardiography and other noninvasive imaging techniques in **Chap. 241**; and of cardiac catheterization and angiography in **Chap. 242**.

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# MITRAL VALVE PROLAPSE

Mitral valve prolapse (MVP), also variously termed the *systolic click-murmur syndrome, Barlow's syndrome* (Fig. 265-1), *floppy-valve syndrome,* and *billowing mitral leaflet syndrome*, is a relatively common but highly variable clinical syndrome resulting from diverse pathologic mechanisms affecting the mitral valve apparatus. Among these are excessive or redundant mitral leaflet tissue, which is commonly associated with myxomatous degeneration and greatly increased concentrations of certain glycosaminoglycans. MVP is the most common abnormality leading to primary mitral regurgitation (MR) (see Chap. 264).

In most patients with MVP, the cause is unknown, but in some, it appears to be genetically determined. A reduction in the production of type III collagen has been implicated, and electron microscopy has revealed fragmentation of collagen fibrils.

MVP is a frequent finding in patients with heritable disorders of connective tissue, including Marfan syndrome (Chap. 413), osteogenesis imperfecta, and Ehlers-Danlos syndrome. MVP may be associated with thoracic skeletal deformities similar to but not as severe as those in Marfan syndrome, such as a high-arched palate and alterations of the chest and thoracic spine, including the so-called straight back syndrome. Other associated features can include a history of inguinal hernias, joint dislocations, meniscal tears, and easy bruisability.

In most patients with MVP, myxomatous degeneration is confined to the mitral valve, although the tricuspid and aortic valves may also be affected. The posterior mitral leaflet is usually more affected than the anterior, and the mitral valve annulus is often dilated. In many patients, elongated, redundant, or ruptured chordae tendineae cause or contribute to the regurgitation.

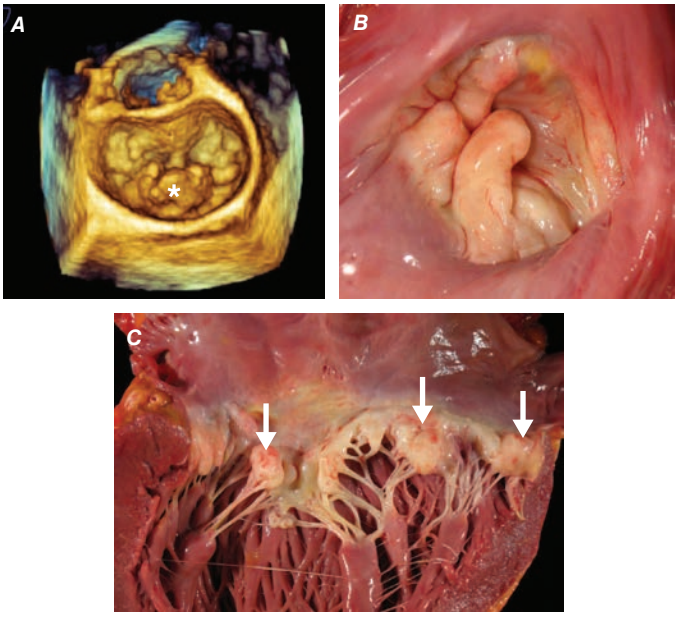



**FIGURE 265-1 Congenital or developmental mitral valve prolapse.** Myxomatous thickening and prolapse of the mitral valve can occur in isolation in 2–3% of the general population or may be associated with heritable collagen-vascular disorders and aortic root dilation, such as in Marfan syndrome. Myxomatous degeneration of the valve predisposes to severe regurgitation and chordal rupture and is a frequent indication for mitral valve repair or replacement. Prolapse can affect one or both leaflets, to varying degrees. ***A.*** Three-dimensional transesophageal echocardiogram showing a myxomatous mitral valve from the left atrial *en face* aspect. There is billowing and prolapse of the entire middle scallop of the posterior leaflet (*asterisk*). *(Figure courtesy of Douglas C. Shook, MD, Department of Anesthesiology, Perioperative and Pain Medicine, Brigham and Women's Hospital.)* ***B.*** The posterior leaflet of the mitral valve demonstrates marked prolapse and hooding in all segments and severe redundancy in this photograph taken from the vantage point of the left atrium. ***C.*** Opening the left heart reveals prominent mitral leaflet hooding (*arrows*). The chordae are focally thickened but are not fused as would be the case in rheumatic valve disease. *(Used with permission from JC Wu, RF Padera: Clinicopathologic correlates, in Atlas of Echocardiography, 2nd ed, SD Solomon [ed], E Braunwald [series ed]. Philadelphia, Current Medicine Group LLC, 2008. p 363.)*

MVP also may occur rarely as a sequel to acute rheumatic fever, in ischemic heart disease, and in various cardiomyopathies, as well as in 20% of patients with ostium secundum atrial septal defect.

MVP may lead to excessive stress on the papillary muscles, which, in turn, leads to dysfunction and ischemia of the papillary muscles and the subjacent ventricular myocardium. Rupture of chordae tendineae and progressive annular dilation and calcification contribute to valvular regurgitation, which then places more stress on the diseased mitral valve apparatus, thereby creating a vicious cycle. ECG changes (see below) and ventricular arrhythmias described in some patients with MVP appear to result from regional ventricular dysfunction and fibrosis related to the increased stress placed on the papillary muscles.

## CLINICAL FEATURES

MVP is more common in women than men and occurs most frequently between the ages of 15 and 30 years; the clinical course is most often benign. MVP may also be observed in older (>50 years) patients, often men, in whom MR is often more severe because of chordal rupture and requires surgical treatment. There is an increased familial incidence for some patients, suggesting an autosomal dominant form of inheritance with incomplete penetrance. MVP varies in its clinical expression, ranging from only a systolic click and murmur with mild prolapse of the posterior leaflet to severe MR due to chordal rupture and leaflet flail. The degree of myxomatous change of the leaflets can also vary widely. In many patients, the condition progresses over years or decades; in others, it worsens rapidly as a result of chordal rupture or endocarditis.

Most patients are asymptomatic and remain so for their entire lives. However, in North America, MVP is now the most common cause of isolated severe MR requiring surgical treatment. Arrhythmias, most commonly ventricular premature contractions and paroxysmal supraventricular and ventricular tachycardia, as well as atrial fibrillation (AF), have been reported and may cause palpitations, lightheadedness, and syncope. Sudden death is a very rare complication and occurs most often in patients with severe MR and depressed left ventricle (LV) systolic function, although it can occur in individuals with normal LV size and function. A small subset of MVP patients with high-grade ventricular ectopy has been identified with phenotypic features including electrocardiographic inferior-apical T-wave abnormalities, high-density premature ventricular complexes at rest, mitral annular disjunction (defined as abnormal atrial displacement of the mitral valve leaflet hinge point), and papillary muscle fibrosis on cardiac magnetic resonance imaging with late gadolinium enhancement. In addition, there may be an excess risk of sudden death among patients with a flail leaflet. Many patients have chest pain that is difficult to evaluate; it is often substernal, prolonged, and not related to exertion, but may rarely resemble angina pectoris. Transient cerebral ischemic attacks secondary to emboli from the mitral valve due to endothelial disruption have been reported. Infective endocarditis may occur in patients with MR and/or leaflet thickening.

**Auscultation** A frequent finding is the mid- or late (nonejection) systolic click, which occurs 0.14 s or more after $S_1$ and is thought to be generated by the sudden tensing of slack, elongated chordae tendineae or by the prolapsing mitral leaflet when it reaches its maximal excursion. Systolic clicks may be multiple and may be followed by a high-pitched, mid-late systolic crescendo–decrescendo murmur, which occasionally is "whooping" or "honking" and is heard best at the apex. Radiation of the murmur will depend on the involved leaflet. With posterior leaflet prolapse, the jet of MR is directed anteriorly and the murmur will radiate to the base of the heart. With anterior leaflet involvement, the jet of MR is directed posteriorly and the murmur will radiate to the axilla and back. The click and murmur occur earlier with standing, during the strain phase of the Valsalva maneuver and with any intervention that decreases LV volume (preload), exaggerating the propensity of the leaflet to prolapse. Conversely, squatting and isometric exercises, which increase LV volume, diminish MVP; the click-murmur complex is delayed, moves away from $S_1$, and may even disappear. Some patients have a mid-systolic click without a murmur; others have a murmur without a click. Still others have both sounds at different times.

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## LABORATORY EXAMINATION

The ECG most commonly is normal but may show biphasic or inverted T waves in leads II, III, and aVF and, occasionally, supraventricular or ventricular premature beats. Transthoracic echocardiography (TTE) is particularly effective in identifying the abnormal position and prolapse of the mitral valve leaflets. A useful echocardiographic definition of MVP is systolic displacement (in the parasternal long axis view) of the belly of the mitral valve leaflets by at least 2 mm into the left atrium (LA) superior to the plane of the mitral annulus. There can be prolapse of one or both leaflets (Fig. 265-2). Color flow and continuous wave Doppler imaging is helpful to evaluate the associated MR and provide estimates of severity. The jet lesion of MR due to MVP is most often eccentric, and assessment of the effective regurgitant orifice area and regurgitant volume can be difficult with standard techniques. Both three-dimensional echocardiography and cardiac magnetic resonance imaging can provide more precise determinations of LV volumes. Transesophageal echocardiography (TEE) is indicated when more accurate anatomic information is required and is performed routinely for intraoperative guidance during valve repair. Exercise testing can be performed when there is uncertainty regarding functional capacity. It is often combined with rest and immediate poststress TTE to assess LV and right ventricular (RV) function and the dynamic nature of MR and pulmonary artery pressures. Left ventriculography done at the time of right and left heart catheterization is rarely necessary but can also show prolapse of the posterior and sometimes of both mitral valve leaflets.

## TREATMENT

### Mitral Valve Prolapse

Infective endocarditis prophylaxis is indicated for patients with a prior history of endocarditis. Beta blockers sometimes relieve chest pain and control palpitations. Decisions regarding anticoagulation for stroke prevention in AF should be based on the CHA2DS2-VASc score and an assessment of bleeding risk. If the patient is symptomatic from severe MR, mitral valve repair is indicated (see Fig. 264-1). Other indications for surgery for MVP with severe primary MR include findings of established or progressive LV systolic dysfunction. Surgery can also be considered for low-risk asymptomatic patients in whom a successful and durable repair can be achieved with at least 95% likelihood by an expert surgeon. Mitral valve repair is preferred over replacement in patients with MVP or flail mitral leaflet (see Table 264-2); technical success is dependent not only on the anatomic findings but also on the skill and experience of the surgeon. Repair of isolated posterior leaflet prolapse is usually straightforward, but increasingly more complex pathologies (e.g., anterior leaflet prolapse, bileaflet prolapse, Barlow's deformity) require advanced skills. Careful pre- and intraoperative TEE imaging is an important component of patient evaluation and surgical planning. Transcatheter edge-to-edge repair (TEER) using a clip to grasp the anterior and posterior leaflets together can be considered for treatment of symptomatic patients at prohibitive or high surgical risk with severe primary MR due to MVP (see Fig. 264-3). Most often, the MR will be reduced in severity but not eliminated. Nevertheless, symptom status and indices of LV size and function can be improved with this approach, which is now offered at >475 specialized sites in the United States. Reported hospital mortality rates following the procedure are ~2%. Other transcatheter repair and replacement devices are not yet approved for clinical use in the United States (see Chap. 264).

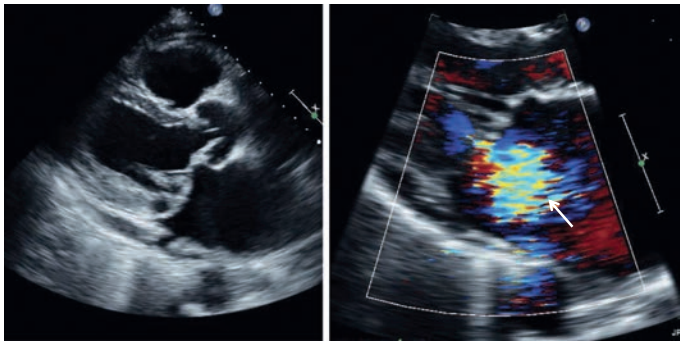


**FIGURE 265-2. Barlow's valve with classic mitral valve prolapse**, as seen on transthoracic echocardiogram in parasternal long-axis windows. Left: parasternal long-axis window, showing both myxomatous leaflets billowing into the left atrium in late systole. Right: same window with color Doppler showing significant mitral regurgitation (*arrow*) in systole. *(Courtesy of Justina Wu, MD, PhD.)*

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# 266 Tricuspid Valve Disease

**Patrick T. O'Gara, Joseph Loscalzo**

## TRICUSPID STENOSIS

Tricuspid stenosis (TS), which is much less prevalent than mitral stenosis (MS) in North America and Western Europe, is generally rheumatic in origin and is more common in women than men (Table 266-1). It does not occur as an isolated lesion and is usually associated with MS. Hemodynamically significant TS occurs in 5–10% of patients with severe MS; rheumatic TS is commonly associated with some degree of tricuspid regurgitation (TR). Nonrheumatic causes of TS are rare.

### PATHOPHYSIOLOGY

A diastolic pressure gradient between the right atrium (RA) and right ventricle (RV) defines TS. It is augmented when the transvalvular blood flow increases during inspiration and declines during expiration. A mean diastolic pressure gradient of 4 mmHg is usually sufficient to elevate the mean RA pressure to levels that result in systemic venous congestion. Unless sodium intake has been restricted and diuretics administered, this venous congestion is associated with hepatomegaly, ascites, and edema, sometimes severe. In patients with sinus rhythm, the RA *a* wave may be extremely tall and may even approach the level of the RV systolic pressure. The *y* descent is prolonged. The cardiac output (CO) at rest is usually depressed, and it fails to rise during exercise. The low CO is responsible for the normal or only slightly elevated left atrial (LA), pulmonary artery (PA), and RV systolic pressures despite the presence of MS. Thus, the presence of TS can mask the hemodynamic and clinical features of any associated MS.

### SYMPTOMS

Because the development of MS generally precedes that of TS, many patients initially have symptoms of pulmonary congestion and fatigue. Characteristically, patients with severe TS complain of relatively little dyspnea for the degree of hepatomegaly, ascites, and edema that they have. However, fatigue secondary to a low CO and discomfort due to refractory edema, ascites, and marked hepatomegaly are common in patients with advanced TS and/or TR. In some patients, TS may be suspected for the first time when symptoms of right-sided failure persist after an adequate mitral commissurotomy.

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**TABLE 266-1 Causes of Tricuspid Valve Diseases**

| VALVE LESION | ETIOLOGIES |
|---|---|
| Tricuspid stenosis | Rheumatic |
| | Congenital |
| Tricuspid regurgitation | Primary (organic) |
| | Rheumatic |
| | Endocarditis |
| | Myxomatous (TVP) |
| | Carcinoid |
| | Radiation |
| | Congenital (Ebstein's) |
| | Trauma (including that due to intracardiac leads and RV endomyocardial biopsy) |
| | Papillary muscle injury (post-MI) |
| | Secondary (functional) |
| | RV and tricuspid annular dilation due to multiple causes (e.g., long-standing pulmonary HTN, remodeling post-RV MI, left-sided heart disease, cardiomyopathy, AF (atrial functional tricuspid regurgitation), chronic RV apical pacing (dyssynchrony) |

*Abbreviations:* AF, atrial fibrillation; HTN, hypertension; MI, myocardial infarction; RV, right ventricular; TVP, tricuspid valve prolapse.

### PHYSICAL FINDINGS

Because TS usually occurs in the presence of other obvious valvular disease, the diagnosis may be missed unless it is considered. Severe TS is associated with marked hepatic congestion, often resulting in cirrhosis, jaundice, serious malnutrition, anasarca, and ascites. Congestive hepatomegaly and, in cases of severe tricuspid valve disease, splenomegaly are present. The jugular veins are distended, and in patients with sinus rhythm, there may be giant *a* waves. The *v* waves are less conspicuous, and because tricuspid obstruction impedes RA emptying during diastole, there is a slow *y* descent. In patients with sinus rhythm, there may be prominent presystolic pulsations of the enlarged liver as well.

On auscultation, an opening snap (OS) of the tricuspid valve may rarely be heard ~0.06 s after pulmonic valve closure. The diastolic murmur of TS has many of the qualities of the diastolic murmur of MS, and because TS almost always occurs in the presence of MS, it may be missed. However, the tricuspid murmur is generally heard best along the left lower sternal border and over the xiphoid process and is most prominent during presystole in patients with sinus rhythm. The murmur of TS is augmented during inspiration, and it is reduced during expiration and particularly during the strain phase of the Valsalva maneuver, when tricuspid transvalvular flow is reduced.

### LABORATORY EXAMINATION

The electrocardiogram (ECG) features of RA enlargement **(see Fig. 240-8)** include tall, peaked P waves in lead II, as well as prominent, upright P waves in lead $V_1$. The *absence* of ECG evidence of RV hypertrophy (RVH) in a patient with right-sided heart failure who is believed to have MS should suggest associated tricuspid valve disease. The chest x-ray in patients with combined TS and MS shows particular prominence of the RA and superior vena cava without much enlargement of the PA and with less evidence of pulmonary vascular congestion than occurs in patients with isolated MS; engorgement of the azygos vein can often be appreciated. On transthoracic echocardiographic (TTE) examination, the tricuspid valve is usually thickened and domes in diastole; the transvalvular gradient can be estimated by continuous wave Doppler echocardiography. Severe TS is characterized by a valve area ≤1 $cm^2$ or pressure half-time of ≥190 ms. The RA and inferior vena cava (IVC) are enlarged. TTE provides additional information regarding the severity of any associated TR, mitral valve structure and function, left ventricular (LV) and RV size and function, and PA pressure. Cardiac catheterization is not routinely necessary for assessment of TS.

## TREATMENT

### Tricuspid Stenosis

Patients with TS generally exhibit marked systemic venous congestion; salt restriction, bed rest, and diuretic therapy are required during the preoperative period. Such a preparatory period may diminish hepatic congestion and thereby improve hepatic function sufficiently so that the risks of operation, particularly bleeding, are diminished. Surgical relief of the TS should be carried out, preferably at the time of surgical mitral commissurotomy or mitral valve replacement (MVR) for mitral valve disease, in patients with moderate or severe TS who have mean diastolic pressure gradients exceeding ~4 mmHg and tricuspid orifice areas <1.5–2 $cm^2$. TS is almost always accompanied by significant TR. Operative repair may permit substantial improvement of tricuspid valve function. If repair cannot be accomplished, the tricuspid valve may have to be replaced. Meta-analysis has shown no difference in overall survival between mechanical and tissue valve replacement. Mechanical valves in the tricuspid position are more prone to thromboembolic complications than in other positions. Percutaneous tricuspid balloon commissurotomy for isolated severe TS without significant TR is very rarely performed.

## TRICUSPID REGURGITATION

More than 85% of TR cases encountered in clinical practice are secondary (functional) in nature and related to tricuspid annular dilation and leaflet tethering in the setting of RV remodeling caused by pressure or volume overload (or both), myocardial infarction (MI), or trauma (Table 266-1). Secondary TR is commonly seen in the late stages of heart failure due to rheumatic or congenital heart disease with severe PA hypertension (PA systolic pressure >55 mmHg), as well as in other types of left-sided valvular (e.g., mitral regurgitation) or myocardial diseases (e.g., ischemic and idiopathic dilated cardiomyopathies). Secondary TR can also develop from chronic RV apical pacing and dyssynchronous contraction; in some patients, the RV leads may also perforate or entrap the TV leaflets. TR can often emerge in the setting of new-onset atrial fibrillation (AF), particularly in older patients (atrial functional TR). Rheumatic fever may produce primary TR, often associated with TS. Tricuspid valve prolapse, carcinoid heart disease, endomyocardial fibrosis, radiation, infective endocarditis, and leaflet trauma can also produce primary TR. Less commonly, primary TR results from congenitally deformed tricuspid valves and can occur with defects of the atrioventricular canal, as well as with Ebstein's malformation of the tricuspid valve **(Chap. 269)**.

### PATHOPHYSIOLOGY

The incompetent tricuspid valve allows blood to flow backward from the RV into the RA, the volume of which is dependent on the driving pressure (i.e., RV systolic pressure) and the size of the regurgitant orifice. The severity and physical signs of TR can vary as a function of PA systolic pressure (in the absence of RV outflow tract stenosis), the dimension of the tricuspid valve annulus, the respiratory cycle-dependent changes in RV preload, and RA compliance. RV filling is increased during inspiration. Forward CO is reduced and does not augment with exercise. Significant degrees of TR will lead to RA enlargement and elevation of the RA and jugular venous pressures with prominent *c-v* waves in the pulse tracings. Progressively severe TR can lead to "ventricularization" of the RA wave form **(see Fig. 239-1*B*)**. Severe TR is also characterized by RV dilation (RV volume overload) and eventual systolic dysfunction, the progression of which can be accelerated by a concomitant pressure load from PA hypertension or by myocardial fibrosis from previous injury.

### SYMPTOMS

Mild or moderate degrees of TR are usually well tolerated in the absence of other hemodynamic disturbances. Because TR most often coexists with left-sided valve lesions, LV dysfunction, and/or PA hypertension, symptoms related to these lesions may dominate the clinical picture. Fatigue and exertional dyspnea owing to reduced forward CO are early symptoms of isolated, severe TR. As the disease progresses and RV

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function declines, patients may report cervical pulsations, abdominal fullness/bloating, diminished appetite, and muscle wasting, although with progressive weight gain and painful swelling of the lower extremities.

### PHYSICAL FINDINGS

The neck veins in patients with severe TR are distended with prominent *c-v* waves and rapid *y* descents (in the absence of TS). TR is more often diagnosed by examination of the neck veins than by auscultation of the heart sounds. Other findings may include marked hepatomegaly with systolic pulsations, ascites, pleural effusions, edema, and a positive hepatojugular reflux sign. A prominent RV pulsation in the left parasternal region and a blowing holosystolic murmur along the lower left sternal margin, which may be intensified during inspiration (Carvallo's sign) and reduced during expiration or the strain phase of the Valsalva maneuver, are characteristic findings. The murmur of TR may sometimes be confused with that of mitral regurgitation (MR) unless attention is paid to its variation during the respiratory cycle and the extent of RV enlargement is appreciated. AF is usually present in the chronic phase of the disease.

### LABORATORY EXAMINATION

The ECG may show changes characteristic of the lesion responsible for the TR, e.g., an inferior Q-wave MI suggestive of a prior RV MI, RVH, or a bizarre right bundle branch block–type pattern with preexcitation in patients with Ebstein's anomaly. ECG signs of RA enlargement may be present in patients with sinus rhythm; AF is frequently noted. The chest x-ray may show RA and RV enlargement, depending on the chronicity and severity of TR. TTE is usually definitive with demonstration of RA dilation and RV volume overload and prolapsing, flail, scarred, or displaced/tethered tricuspid leaflets with annular dilatation; the diagnosis and assessment of TR can be made by color flow Doppler imaging (see Fig. 241-8). Severe TR is accompanied by hepatic vein systolic flow reversal. Continuous wave Doppler of the TR velocity profile is useful in estimating PA systolic pressure, except when the TR is very severe and the jet velocity is blunted by rapidly increasing RA pressure. Accurate assessment of TR severity, PA pressures, and RV size and systolic function with TTE can be quite challenging in many patients. Real-time three-dimensional echocardiography and cardiac magnetic resonance (CMR) imaging provide alternative imaging modalities, although they are not widely available. In patients with severe TR, the CO is usually markedly reduced, and the RA pressure pulse may not exhibit an *x* descent during early systole but rather show a prominent *c-v* wave with a rapid *y* descent. The mean RA and RV end-diastolic pressures are often elevated. Exercise testing can be used to assess functional capacity in patients with asymptomatic severe TR. The prognostic significance of exercise-induced changes in TR severity and RV function has not been well studied.

## TREATMENT

### Tricuspid Regurgitation (Fig. 266-1)

Diuretics can be useful for patients with severe TR and signs of right heart failure. An aldosterone antagonist may be particularly

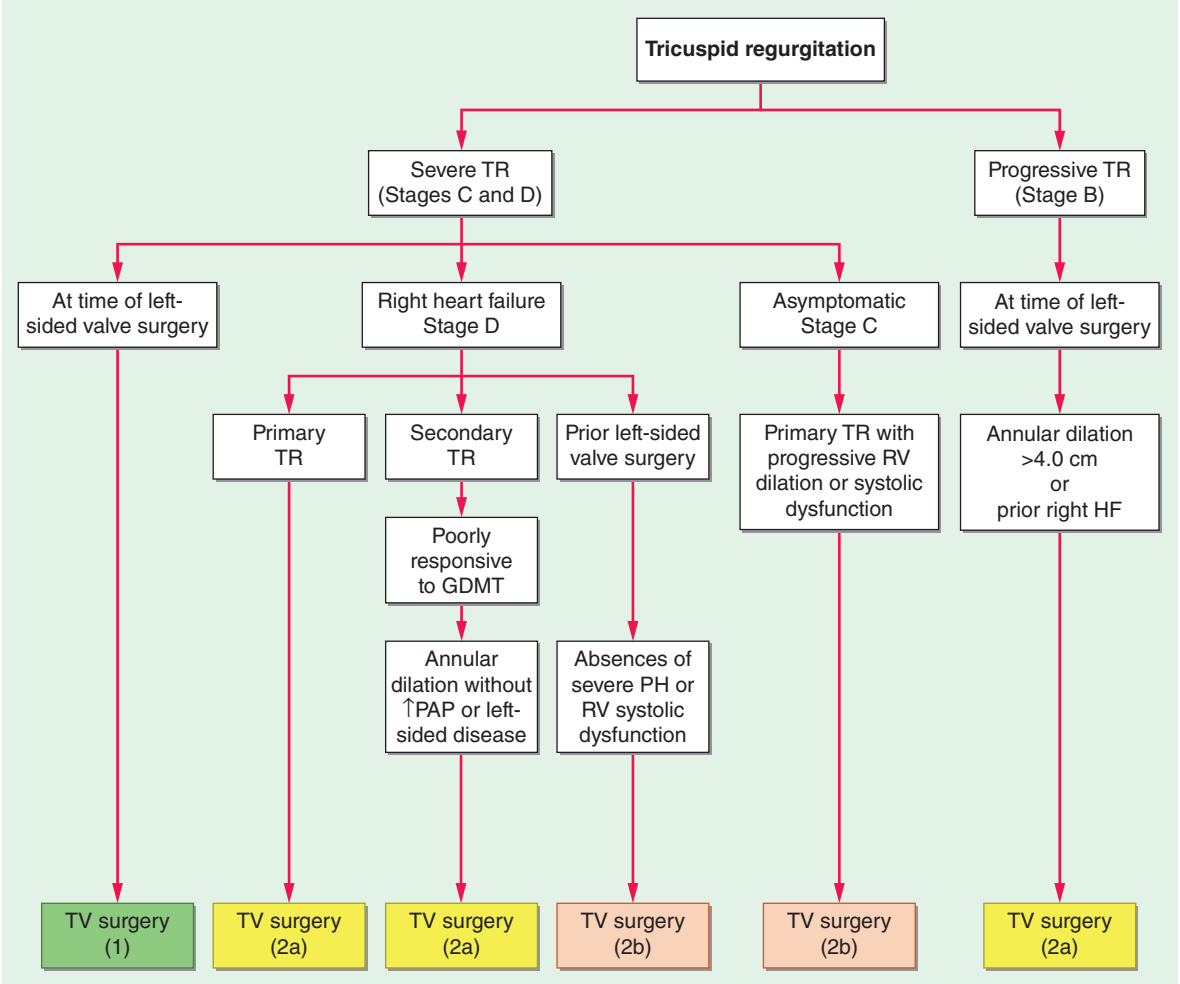



**FIGURE 266-1 Management of tricuspid regurgitation.** See legend for Fig. 261-4 for explanation of treatment recommendations (Class I, IIa, IIb) and disease stages (B, C, D). Preoperative coronary angiography should be performed routinely as determined by age, symptoms, and coronary risk factors. Cardiac catheterization and angiography may also be helpful when there is a discrepancy between clinical and noninvasive findings. GDMT, guideline-directed management and therapy; HF, heart failure; PAP, pulmonary artery pressure; PH, pulmonary hypertension; RV, right ventricular; TR, tricuspid regurgitation; TV, tricuspid valve. Annular dilation is defined by >40 mm on transthoracic echocardiography (>21 mm/m$^2$) or >70 mm on direct intraoperative measurement. *(Reproduced with permission from CM Otto et al: 2020 AHA/ACC Guideline for management of patients with valvular heart disease: A report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. Circulation 143:e72, 2021.)*

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helpful because many patients have secondary hyperaldosteronism from marked hepatic congestion. Therapies to reduce elevated PA pressures and/or pulmonary vascular resistance, including those targeted at left-sided heart disease, can also be considered for patients with PA hypertension and severe secondary TR. Tricuspid valve surgery is recommended for patients with severe TR who are undergoing left-sided valve surgery and is also undertaken frequently for treatment of even moderate TR in patients undergoing left-sided valve surgery who have tricuspid annular dilation (>40 mm), a history of right heart failure, or PA hypertension. Operation most often comprises repair rather than replacement in these settings and has become routine in most major surgical centers. Surgery may also infrequently be required for treatment of severe, primary TR with right heart failure not responsive to standard medical therapy or because of progressively declining RV systolic function. Reported perioperative mortality rates for isolated tricuspid valve surgery (repair and replacement) are high (~8–9%) and likely are influenced by the hazards encountered during reoperation on patients who have undergone previous left-sided valve surgery and have reduced RV function. Indwelling pacemaker or defibrillator leads can also pose technical challenges. Investigation of transcatheter tricuspid valve repair and replacement systems is ongoing.

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# 267 Pulmonic Valve Disease

**Patrick T. O'Gara, Joseph Loscalzo**

## PULMONIC STENOSIS

Pulmonic valve stenosis (PS) is essentially a congenital disorder (Table 267-1). With isolated PS, the valve is typically domed. Dysplastic pulmonic valves are seen as part of the Noonan syndrome (Chap. 281), which maps to chromosome 12. Mutations in the *PTPN1* gene are associated with about half of all cases of Noonan syndrome. Much less common etiologies include carcinoid and obstructing tumors or bulky vegetations. The pulmonic valve is only very rarely affected by the rheumatic process.

### PATHOPHYSIOLOGY

PS is defined hemodynamically by a systolic pressure gradient between the right ventricle (RV) and the main pulmonary artery (PA). RV hypertrophy (RVH) develops as a consequence of sustained obstruction to RV outflow, and systolic ejection is prolonged. Compared with the ability of the LV to compensate for the pressure overload imposed by aortic stenosis (AS), RV dysfunction from afterload mismatch occurs earlier in the course of PS and at lower peak systolic pressures, because the RV adapts less well to this type of hemodynamic burden. With normal systolic function and cardiac output (CO), severe PS is defined by a peak systolic gradient across the pulmonic valve of >64 mmHg (mean gradient >35 mm Hg, Doppler jet velocity >4 m/s); moderate PS correlates with a peak gradient of 36–64 mmHg (Doppler jet velocity 3-4 m/s). Mild PS is characterized by a jet velocity <3 m/s (peak gradient <36 mmHg). PS rarely progresses in patients with mild PS mmHg but may worsen in those with moderate disease due to valve thickening and calcification with age. The right atrial (RA) *a* wave elevates in relation to the higher pressures needed to fill a noncompliant, hypertrophied RV. A prominent RA *v* wave signifies functional tricuspid regurgitation (TR) from RV and annular dilation. The CO is maintained until late in the course of the disease.

**TABLE 267-1 Causes of Pulmonic Valve Disease**

| VALVE LESION | ETIOLOGIES |
|---|---|
| Pulmonic stenosis | Congenital |
| | Carcinoid |
| | Tumor |
| | Endocarditis |
| Pulmonic regurgitation | Primary valve disease |
| | Congenital |
| | Post-valvotomy |
| | Endocarditis |
| | Carcinoid |
| | Annular enlargement |
| | Pulmonary hypertension |
| | Idiopathic dilation |
| | Marfan syndrome |

### SYMPTOMS

Patients with mild or even moderate PS are usually asymptomatic and first come to medical attention because of a heart murmur (or early systolic click) that leads to echocardiography. With severe PS, patients may report exertional dyspnea or early-onset fatigue. Anginal chest pain from RV oxygen supply-demand mismatch and syncope may occur with very severe forms of obstruction, particularly in the presence of a destabilizing trigger such as atrial fibrillation, fever, infection, anemia, or pregnancy.

### PHYSICAL FINDINGS

The murmur of mild or moderate PS is mid-systolic in timing, crescendo–decrescendo in configuration, heard best in the left second interspace, and usually introduced by an ejection sound (click) in younger adults whose valves are still pliable. The ejection sound is the only right-sided acoustic event that decreases in intensity with inspiration. This phenomenon reflects premature opening of the pulmonic valve by the elevated RV end-diastolic (post-atrial *a* wave) pressure. The systolic murmur increases in intensity during inspiration. With progressively severe PS, the ejection sound moves closer to the first heart sound and eventually becomes inaudible. A right-sided fourth heart sound may emerge. The systolic murmur peaks later and may persist through the aortic component of the second heart sound ($A_2$). Pulmonic valve closure is delayed, and the pulmonic component of the second heart sound ($P_2$) is reduced or absent. A prominent *a* wave, indicative of the higher atrial pressure necessary to fill the noncompliant RV, may be seen in the jugular venous pulse. A parasternal or RV lift can be felt with significant pressure overload. Signs of right heart failure, such as hepatomegaly, ascites, and edema, are uncommon but may appear very late in the disease.

### LABORATORY EXAMINATION

The electrocardiogram (ECG) will show right axis deviation, RVH, and RA enlargement in adult patients with severe PS. Chest x-ray findings include poststenotic dilation of the main PA in the frontal plane projection and filling of the retrosternal airspace due to RV enlargement on

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the lateral film. In some patients with RVH, the cardiac apex appears to be lifted off the left hemidiaphragm. The RA may also be enlarged. Transthoracic echocardiography (TTE) allows definitive diagnosis and characterization in most cases, with depiction of the valve and assessment of the jet velocity, gradient, RV function, PA pressures (which should be low), and any associated cardiac lesions. Transesophageal echocardiography (TEE) may be useful in some patients for improved delineation of the RV outflow tract (RVOT) and assessment of infundibular hypertrophy. Cardiac catheterization is not usually necessary for diagnostic purposes, but if performed, pressures should be obtained from just below and above the pulmonic valve with attention to the possibility that a dynamic component to the gradient may exist. The correlation between Doppler assessment of peak instantaneous gradient and catheterization-measured peak-to-peak gradient is weak. The latter may correlate better with the Doppler mean gradient.

## TREATMENT

### Pulmonic Stenosis

Diuretics can be used to treat symptoms and signs of right heart failure. Provided there is less than moderate pulmonic regurgitation (PR), percutaneous pulmonic balloon valvuloplasty is recommended for symptomatic patients with moderate or severe PS and for asymptomatic patients with a peak gradient >64 mmHg (or mean gradient >35 mmHg). Surgery may be required when the valve is dysplastic (as seen in patients with Noonan's syndrome and other disorders). A multidisciplinary heart team is best positioned to make treatment decisions of this nature.

## PULMONIC REGURGITATION

PR may develop as a consequence of primary valve pathology, annular enlargement, or their combination; after surgical treatment of RVOT obstruction in children with such disorders as tetralogy of Fallot; or after percutaneous pulmonic balloon valvotomy (Table 267-1). Carcinoid usually causes mixed pulmonic valve disease with PR and PS. Long-standing severe PA hypertension from any cause can result in dilation of the pulmonic valve ring and PR.

### PATHOPHYSIOLOGY

Severe PR results in RV chamber enlargement and eccentric hypertrophy. As is the case for aortic regurgitation (AR), PR is a state of increased preload and afterload. The reverse pressure gradient from the PA to the RV, which drives the PR, progressively decreases throughout diastole and accounts for the decrescendo nature of the diastolic murmur. As RV diastolic pressure increases, the murmur becomes shorter in duration. The forward CO is preserved during the early stages of the disease but may not increase normally with exercise and declines over time. A reduction in RV ejection fraction may be an early indicator of hemodynamic compromise. In advanced stages, there is significant enlargement of the RV and RA with marked elevation of the jugular venous pressure.

### SYMPTOMS

Mild or moderate degrees of PR do not, by themselves, result in symptoms. Other issues, such as PA hypertension, may dominant the clinical picture. With progressively severe PR and RV dysfunction, fatigue, exertional dyspnea, abdominal fullness/bloating, and lower extremity swelling may be reported.

### PHYSICAL FINDINGS

The physical examination hallmark of PR is a high-pitched, decrescendo diastolic murmur (Graham Steell murmur) heard along the left sternal border that can be difficult to distinguish from the more frequently appreciated murmur of AR. The Graham Steell murmur may become louder with inspiration and is usually associated with a loud and sometimes palpable $P_2$ and an RV lift, as would be expected in patients with significant PA hypertension of any cause. Survivors of childhood surgery for tetralogy of Fallot or PS/pulmonary atresia may have an RV-PA conduit that is freely regurgitant because it does not contain a valve. PA pressures in these individuals are not elevated, and the diastolic murmur can be misleadingly low pitched and of short duration despite significant degrees of PR and RV volume overload.

### LABORATORY EXAMINATION

Depending on both the etiology and severity of PR, the ECG may show findings of RVH and RA enlargement. On chest x-ray, the RV and RA may be enlarged. Pulmonic valve morphology and function can be assessed with transthoracic Doppler echocardiography. RV systolic pressure can be estimated from the tricuspid valve systolic jet velocity. Cardiac magnetic resonance (CMR) imaging can provide greater anatomic detail, particularly in patients with repaired congenital heart disease, and more precise assessment of RV volumes and function. Cardiac catheterization is not routinely necessary but would be performed as part of a planned transcatheter PV procedure.

## TREATMENT

### Pulmonic Regurgitation

In patients with functional PR due to PA hypertension and annular dilation, efforts to reduce PA vascular resistance and pressure should be pursued. Such efforts may include pharmacologic/vasodilator and/or surgical/interventional strategies, depending on the cause of the PA hypertension (e.g., idiopathic PA hypertension, left-sided heart valve disease). Diuretics can be used to treat the manifestations of right heart failure. Surgical valve replacement for primary, severe, pulmonic valve disease, such as carcinoid or endocarditis, is rarely undertaken. Transcatheter pulmonic valve replacement has been successfully performed in many patients with severe PR after childhood repair of tetralogy of Fallot or pulmonic valve stenosis or atresia. This procedure was introduced clinically prior to transcatheter aortic valve replacement.

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# 268 Multiple and Mixed Valvular Heart Disease

**Patrick T. O'Gara, Joseph Loscalzo**

Many acquired and congenital cardiac lesions may result in stenosis and/or regurgitation of one or more heart valves. For example, rheumatic heart disease can involve the mitral (mitral stenosis [MS], mitral regurgitation [MR], or MS and MR), aortic (aortic stenosis [AS], aortic regurgitation [AR], or AS and AR), and tricuspid (tricuspid stenosis [TS], tricuspid regurgitation [TR], or TS and TR) valve, alone or in combination. The common association of functional TR with significant mitral valve disease is discussed in **Chap. 266**. Severe mitral annular calcification can result in regurgitation (due to decreased annular shortening during systole) and mild or moderate stenosis (caused by extension of the calcification onto the leaflets resulting in restricted valve opening). Patients with severe AS and LV remodeling may develop functional MR that may not improve after isolated aortic

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valve replacement (AVR). Primary MR due to mitral valve prolapse or chordal rupture has been noted in patients with severe AS. Aortic valve infective endocarditis (IE) may secondarily involve the mitral apparatus either by abscess formation and contiguous spread via the intervalvular fibrosa or by "drop metastases" from the aortic leaflets onto the anterior leaflet of the mitral valve. Mediastinal radiation may result in aortic, mitral, and even tricuspid valve disease, most often with mixed stenosis and regurgitation. Carcinoid heart disease may cause mixed lesions of either or both the tricuspid and pulmonic valves. Ergotamines, and the previously used combination of fenfluramine and phentermine, can rarely result in mixed lesions of the aortic and/or mitral valve. Patients with Marfan syndrome may have both AR from aortic root dilation and MR due to mitral valve prolapse (MVP). Myxomatous degeneration causing prolapse of multiple valves (mitral, aortic, tricuspid) can also occur in the absence of an identifiable connective tissue disorder. Bicuspid aortic or pulmonic valve disease can result in mixed stenosis and regurgitation. The former is also associated with aortic aneurysm disease and a predisposition to aortic dissection.

## PATHOPHYSIOLOGY

In patients with multivalvular heart disease, the pathophysiologic derangements associated with the more proximal valve disease can mask the full expression of the attributes of the more distal valve lesion. For example, in patients with rheumatic mitral and aortic valve disease, the reduction in cardiac output (CO) imposed by the mitral valve disease will decrease the magnitude of the hemodynamic derangements related to the severity of the aortic valve lesion (stenotic, regurgitant, or both). Alternatively, the development of atrial fibrillation (AF) during the course of MS can lead to sudden worsening in a patient whose aortic valve disease was not previously felt to be significant. The development of reactive pulmonary vascular disease, sometimes referred to as a "secondary obstructive lesion in series," can impose an additional challenge in these settings. As CO falls with progressive tricuspid valve disease, the severity of any associated mitral or aortic disease can be underestimated.

One of the most common examples of multivalve disease is that of functional TR in the setting of significant mitral valve disease. Functional TR occurs as a consequence of right ventricular and annular dilation; pulmonary artery (PA) hypertension may be present. The tricuspid leaflets are morphologically normal. Progressive degrees of TR lead to right ventricular volume overload and continued chamber and annular dilation. The TR is usually central in origin; reflux into the right atrium (RA) is expressed as large, systolic *c-v* waves in the RA pressure pulse. The height of the *c-v* wave is dependent on RA compliance and the volume of regurgitant flow. The RA waveform may become "ventricularized" in advanced stages of chronic, severe TR. CO falls and the severity of the associated mitral valve disease may become more difficult to appreciate. Findings related to advanced right heart failure (e.g., ascites, edema) predominate. Primary rheumatic tricuspid valve disease may occur with rheumatic mitral disease and cause hemodynamic changes reflective of TR, TS, or their combination. With TS, the *y* descent in the RA pressure pulse is prolonged. Typically, findings related to the mitral valve disease predominate over those related to the tricuspid valve disease.

Another example of rheumatic, multivalve disease involves the combination of mitral and aortic valve pathology, frequently characterized by MS and AR. In isolated MS, left ventricular (LV) preload and diastolic pressure are reduced as a function of the severity of inflow obstruction. With concomitant AR, however, LV filling is enhanced and diastolic pressure may rise depending on the compliance characteristics of the chamber. Because the CO falls with progressive degrees of MS, transaortic valve flows will decline, masking the potential severity of the aortic valve lesion (AR, AS, or its combination). As noted above, onset of AF in such patients can be especially deleterious. The loss of atrial systole with AF may result in a critical reduction in CO, a rise in LA and LV diastolic pressures, and a further deleterious increase in heart rate.

Secondary (functional) MR may complicate the course of some patients with severe AS. The mitral valve leaflets and chordae tendineae are usually normal. Incompetence is related to changes in LV geometry (remodeling) and abnormal systolic tethering of the leaflets in the context of markedly elevated LV systolic pressures. Relief of the excess afterload with surgical or transcatheter AVR may result in reduction of the secondary MR. Persistence of significant, secondary MR following AVR is associated with impaired functional outcomes and reduced survival. Identification of patients who would benefit from concomitant treatment of their secondary MR at time of AVR is quite challenging. Most surgeons advocate for repair of moderate-to-severe or severe secondary MR at time of surgical AVR. Significant primary MR may also coexist with AS and is routinely managed with repair or replacement at the time of AVR. There is increasing experience with the combination of transcatheter aortic valve implantation (TAVI) and transcatheter edge-to-edge mitral valve repair (TEER) in high surgical risk patients with severe AS and moderate-severe primary or secondary MR.

In patients with mixed AS and AR, assessment of valve stenosis can be influenced by the magnitude of the regurgitant valve flow. Because transvalvular systolic flow velocities are augmented in patients with AR and preserved LV systolic function, the LV-aortic Doppler-derived pressure gradient and the intensity of the systolic murmur will be elevated to values higher than expected for the true systolic valve orifice size as measured by planimetry. Uncorrected, the Gorlin formula, which relies on forward CO (systolic transvalvular flow) and the mean pressure gradient for calculation of valve area, is not accurate in the setting of mixed aortic valve disease. Similar considerations apply to patients with mixed mitral valve disease. The peak mitral valve Doppler E wave velocity ($v_0$) is increased in the setting of severe MR because of enhanced early diastolic flow and may not accurately reflect the contribution to left atrial (LA) hypertension from any associated MS. When either AR or MR is the dominant lesion in patients with mixed aortic or mitral valve disease, respectively, the LV is dilated. When AS or MS predominates, LV chamber size will be normal or small. It can sometimes be difficult to ascertain whether stenosis or regurgitation is the dominant lesion in patients with mixed valve disease, although an integrated clinical and noninvasive assessment can usually provide clarification for purposes of patient management. For patients with moderate, mixed AS and AR in whom stenosis is the dominant lesion, the natural history tends to parallel what might be expected for isolated severe AS, and the treatment approach should be accordingly aligned.

Patients with significant AS, a nondilated LV chamber, and concentric hypertrophy will poorly tolerate the abrupt development of aortic regurgitation, as may occur, for example, with IE or after surgical AVR or TAVI complicated by paravalvular leakage. The noncompliant LV is not prepared to accommodate the sudden volume load, and as a result, LV diastolic pressure rises rapidly and severe heart failure develops. Indeed, paravalvular regurgitation is a significant risk factor for short- to intermediate-term death following transcatheter AVR. Conditions in which the LV may not be able to dilate in response to chronic AR (or MR) include radiation heart disease and, in some patients, the cardiomyopathy associated with obesity and diabetes. Noncompliant ventricles of small chamber size predispose to earlier onset diastolic dysfunction and heart failure in response to any further perturbation in valve function.

## SYMPTOMS

Compared with patients with isolated, single-lesion valve disease, patients with multiple or mixed valve disease may develop symptoms at a relatively earlier stage in the natural history of their disease. Symptoms such as exertional dyspnea and fatigue are usually related to elevated filling pressures, reduced CO, or their combination. Palpitations may signify AF and identify mitral valve disease as an important component of the clinical presentation, even when not previously suspected. Chest pain compatible with angina could reflect left or right ventricular oxygen supply/demand mismatch on a substrate of hypertrophy and pressure/volume overload with or without superimposed coronary artery disease. Symptoms related to right heart failure (abdominal fullness/bloating, edema) are late manifestations of advanced disease.

## PHYSICAL FINDINGS

Mixed disease of a single valve is most often manifested by systolic and diastolic murmurs, each with the attributes expected for the valve in

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question. Thus, patients with AS and AR will have characteristic mid-systolic, crescendo-decrescendo and blowing, decrescendo diastolic murmurs at the base of the heart in the second right interspace and along the left sternal edge, respectively. Many patients with significant AR have mid-systolic outflow murmurs even in the absence of valve sclerosis/stenosis, and other findings of AS must be sought. The separate murmurs of AS and AR can occasionally be difficult to distinguish from the continuous murmurs associated with either a patent ductus arteriosus (PDA) or ruptured sinus of Valsalva aneurysm. With mixed aortic valve disease, the systolic murmur should end before, and not envelope or extend through, the second heart sound ($S_2$). The murmur associated with a PDA is heard best to the left of the upper sternum. The continuous murmur heard with a ruptured sinus of Valsalva aneurysm is often first appreciated after an episode of acute chest pain. An early ejection click, which usually defines bicuspid aortic valve disease in young adults, is often not present in patients with congenital mixed AS and AR. As noted above, both the intensity and duration of these separate murmurs can be influenced by a reduction in CO and transvalvular flow due to coexistent mitral valve disease. In patients with isolated MS and MR, expected findings would include a blowing, holosystolic murmur and a mid-diastolic rumble (with or without an opening snap) best heard at the cardiac apex. An irregularly irregular heart rhythm in such patients would likely signify AF. Findings with TS and TR would mimic those of left-sided MS and MR, save for the expected changes in the murmurs with respiration. The murmurs of pulmonic stenosis and regurgitation behave in a fashion directionally similar to AS and AR; dynamic changes during respiration should be noted. Specific attributes of these cardiac murmurs are reviewed in **Chaps. 42 and 266**.

### LABORATORY EXAMINATION

The electrocardiogram (ECG) may show evidence of ventricular hypertrophy and/or atrial enlargement. ECG signs indicative of right-sided cardiac abnormalities in patients with left-sided valve lesions should prompt additional assessment for PA hypertension and/or right-sided valve disease. The presence of AF in patients with aortic valve disease may be a clue to the presence of previously unsuspected mitral valve disease in the appropriate context. The chest x-ray can be reviewed for evidence of cardiac chamber enlargement, valve and/or annular calcification, and any abnormalities in the appearance of the pulmonary vasculature. The latter could include enlargement of the main and proximal pulmonary arteries with PA hypertension and pulmonary venous redistribution/engorgement or Kerley B lines with increasing degrees of LA hypertension. An enlarged azygos vein in the frontal projection indicates RA hypertension. Roentgenographic findings not expected based on a single or mixed valve lesion may reflect other valve disease.

Transthoracic echocardiography (TTE) is the most commonly used imaging modality for the diagnosis and characterization of multiple and/or mixed valvular heart disease and may often demonstrate findings not clinically suspected. Transesophageal echocardiography (TEE) may sometimes be required for more accurate assessment of valve anatomy (specifically, the mitral valve) and when IE is considered responsible for the clinical presentation. TTE findings of particular interest include those related to valve morphology and function, calcification, chamber size, ventricular wall thickness, biventricular function estimated PA systolic pressure, and the dimensions of the great vessels, including the root and ascending aorta, PA, and inferior vena cava. Exercise testing (with or without echocardiography) can be useful when the degree of functional limitation reported by the patient is not adequately explained by the findings on TTE performed at rest. An integrated assessment of the clinical and TTE findings is needed to help determine the dominant valve lesion(s) and establish an appropriate plan for treatment and follow-up. Natural history is usually influenced to a relatively greater degree by the dominant lesion.

Cardiac magnetic resonance (CMR) imaging can be used to provide additional anatomic and physiologic information when echocardiography proves suboptimal but is less well suited to the evaluation of valve morphology. Cardiac computed tomography (CT) has been used to assess intracardiac structures in patients with complicated IE. It is invaluable in planning for transcatheter valve implantation. Coronary CT angiography provides a noninvasive alternative for the assessment of coronary artery anatomy prior to surgery or transcatheter intervention.

Invasive hemodynamic evaluation with right and left heart catheterization may be required to characterize more completely the individual contributions of each lesion in patients with either multiple or mixed valvular heart disease. It is strongly recommended when there is a discrepancy between the clinical and noninvasive findings in a symptomatic patient. Measurement of PA pressures and calculation of pulmonary vascular resistance (PVR) can help inform clinical decision-making in certain patient subsets, such as those with advanced mitral and tricuspid valve disease. It is important to identify any potential contribution to the clinical picture from pulmonary vascular disease. Attention to the accurate assessment of CO is essential. Coronary angiography (if indicated) can be performed as part of the procedure. Contrast ventriculography and great vessel angiography are performed infrequently.

## TREATMENT

## Multiple and Mixed Valve Disease

Management of patients with multiple or mixed valve disease can be challenging. As noted above, it is helpful to determine the dominant valve lesion and proceed according to the treatment and follow-up recommendations for it **(Chaps. 261–267)**, being mindful of deviations from the expected course due to the contributions of more than one valve lesion. For example, AF that emerges in the course of moderate mitral valve disease may precipitate heart failure in patients with concomitant, severe aortic valve disease that was previously asymptomatic.

Medical therapies are limited and include diuretics when indicated for relief of congestion and anticoagulation to prevent stroke and thromboembolism in patients with AF. Blood pressure–lowering medications may be needed to treat systemic hypertension, which may aggravate left-sided regurgitant valve lesions, but should be initiated and titrated carefully. Pulmonary vasodilators to lower PVR are not generally effective in this context.

There is a paucity of evidence to inform practice guidelines for surgical and/or transcatheter valve intervention in patients with multiple or mixed valve disease. When there is a clear, dominant lesion, as for example in a patient with severe AS and mild AR, indications for intervention are straightforward and follow those recommended for patients with AS **(Chap. 261)**. In other patients, however, there is less clarity, and decisions regarding intervention should be based on several considerations, including those related to lesion severity, ventricular remodeling, functional capacity, and PA pressures. In this regard, it is important to realize that patients with multiple and/or mixed valve disease may develop limiting symptoms or signs of physiologic impairment even with moderate valve lesions.

Concomitant aortic and mitral valve replacement surgery is associated with a significantly higher perioperative mortality risk than replacement of either valve alone, and operation should be carefully considered. Double valve replacement surgery is usually performed for treatment of severe (unrepairable) valve disease at both locations and for the combination of severe disease at one location with moderate disease at the other to avoid the hazards of reoperation in the intermediate to late term for progressive disease of the unoperated valve. In addition, the presence of a prosthesis in the aortic position significantly restricts surgical exposure of the native mitral valve. The need for double valve replacement may also impact the decision regarding the type of prosthesis (i.e., mechanical vs tissue).

Tricuspid valve repair for moderate or severe secondary (functional) TR at the time of left-sided valve surgery is now commonplace, particularly if there is dilation of the tricuspid annulus

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(>40 mm). The addition of tricuspid valve repair, consisting usually of insertion of an annuloplasty ring, adds little time or complexity to the procedure and is well tolerated. Reoperation for repair (or replacement) of progressive TR years after initial surgery for left-sided valve disease, on the other hand, is associated with a relatively high perioperative mortality risk. Mitral valve repair or replacement for moderate or severe secondary MR at time of AVR for AS can usually be undertaken with acceptable risk for perioperative death or major complication.

The presence of moderate or severe MR in patients with rheumatic MS is a contraindication to percutaneous mitral balloon commissurotomy (PMBC). TAVI can be performed for mixed AS and AR when the anatomic findings related to annulus size, coronary height, and the distribution of calcium are favorable. Transcatheter management of both severe AS and severe primary or secondary MR (with deployment of an edge-to-edge clip) has been undertaken with increasing frequency in appropriately selected patients with prohibitive or high surgical risk. Further advances in transcatheter treatments for multiple and mixed valve disease are anticipated.

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# 269 Congenital Heart Disease in the Adult

**Anne Marie Valente, Michael J. Landzberg**

## PREVALENCE

The number of adults with congenital heart disease (CHD) living in the United States is estimated to be at least 1.4 million, with just over one in five having a complex form of CHD. The majority of adults with CHD were diagnosed in childhood, although a substantial percentage may have CHD first recognized as adults. Lifelong follow-up in coordination with, or directly by, clinicians with expertise in adult congenital heart disease (ACHD) is recommended. In this chapter, we will review the current field of ACHD, with an introduction to CHD nomenclature and cardiac development. This is followed by a summary of the more common CHD lesions that may be diagnosed in adulthood. Lastly, some of the common repaired CHD lesions that are encountered in adults are discussed. Throughout the chapter, to aid in the understanding of congenital cardiac anatomy and physiology, we include figures displaying the passage of blood flow between blood vessels and cardiac chambers in various disorders **(Fig. 269-1)**.

## THE CHANGING LANDSCAPE OF ADULT CHD

**A Relatively New Subspecialty in Cardiovascular Disease** Over the past decade, the field of caring for adults with CHD (ACHD) has blossomed, and several nationwide initiatives have been initiated in an attempt to standardize care. The American College of Cardiology and American Heart Association developed guidelines for the care of adults with CHD, first published in 2008 and revised in 2018. These guidelines emphasize the need for collaboration among primary care practitioners, cardiologists, and ACHD subspecialty cardiologists. The body of medical knowledge and competencies attendant with ACHD combined with skill acquisition in coordination of complex care over a patient's medical lifetime led in 2015 to ACHD board certification examinations by the American Board of Medical Subspecialties, as well as the establishment of requirements for advanced fellowship training in ACHD care by the Accreditation Council for Graduate Medical Education. In temporal association, the Adult Congenital Heart Association (ACHA) developed a process for ACHD care program accreditation based on standardization of infrastructural components felt requisite to achieve quality outcomes for ACHD.

## SPECIAL CONSIDERATIONS FOR THE ACHD PATIENT

Adults with CHD may not recognize subtle changes in their exercise capacity, some of which are associated with worse survival; by the time symptoms are recognized, irreversible physiologic changes may have occurred. ACHD patients are, therefore, advised to undergo regular evaluations for surveillance of anatomic, hemodynamic, and electrophysiologic sequelae that may be present. In addition, specific situations may arise in which it is prudent to review care in consultation with an ACHD specialist, several of which are outlined below.

**Non-Cardiac Surgery** Nearly all adults with CHD can be classified with stage A (harboring risk) or greater degrees of heart failure. As such, adults with CHD may demonstrate limited hemodynamic reserve to altered myocardial perfusion or loading conditions and may have subclinical organ dysfunction that is not recognized by standard laboratory assessment. Comprehensive, multispecialty assessment and care strategy review are recommended in advance of invasive or operative procedures for adults with CHD. **Table 269-1** lists the multiorgan considerations that should be considered in adults with CHD during perioperative resuscitation and convalescence. Anesthetic management requires particular knowledge of anatomy, physiologic consequence of underlying defects, myocardial and vascular performance, presence and nature of previous palliative procedures and residual shunts, alteration of venous or arterial pathways within the circulation, and status of noncardiovascular organ physiology.

**Pregnancy** Women with CHD should receive counseling regarding both maternal and fetal risks prior to conceiving a pregnancy and should be cared for in institutions with experience in treating CHD during pregnancy. Preconception evaluation includes detailed medical history, with particular attention to the women's functional capacity, which is closely linked to maternal and fetal outcomes. **Table 269-2** lists the World Health Organization classification of risk during pregnancy in women with heart disease; women at risk should be strongly counseled about the significant risks of morbidity and mortality during pregnancy and the postpartum period. Normal physiologic hemodynamic changes of pregnancy are significant, occur over a relatively condensed period of time, and may be compounded in adults with CHD. Silversides and colleagues have developed a weighted-risk score for pregnant women with heart disease, based on a large registry known as CARPREG 2. The highest-weighted risk factors (weight of 3 points) include a prior history of cardiac events or arrhythmias, decreased functional status (New York Heart Association class ≥III), and presence of a mechanical heart valve. Risk factors that account for 2 points include ventricular dysfunction, high-risk left-sided valve disease/left ventricular outflow tract obstruction, pulmonary hypertension, coronary artery disease, and high-risk aortopathy. One point is assigned for late pregnancy assessment or no prior cardiac intervention. In this cohort, 16% of women experienced an adverse cardiac outcome, primarily heart failure and arrhythmia related. The predicted risks for cardiac events stratified according to point score were as follows: ≤1 point, 5%; 2 points, 10%; 3 points, 15%; 4 points, 22%; and >4 points, 41%.

Prepregnancy medications should be reviewed to ensure their safety in pregnancy. Alternatives to angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers, and endothelin receptor blockers should be considered, as these agents are teratogenic and contraindicated during pregnancy and should be discontinued. Women requiring anticoagulation must be advised of the challenges of